

Functional MRI: fMRI Report

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Abstract

Understanding how the brain works has long been one of the most exciting and challenging goals in neuroscience. Functional Magnetic Resonance Imaging (fMRI) offers a powerful, non-invasive way to observe brain activity during different mental tasks. This report presents a comprehensive functional Magnetic Resonance Imaging (fMRI) analysis of the NYU Slow Flanker dataset using the FMRIB Software Library (FSL). Our primary objective is to investigate specifically the neural mechanisms underlying cognitive control. The brain's ability to suppress irrelevant information. Employing the General Linear Model (GLM) to evaluate task-evoked brain activity, we contrast Blood-Oxygen-Level-Dependent (BOLD) responses to congruent versus incongruent visual stimuli. Ultimately, this report outlines the complete analytical pipeline, documenting the experimental paradigm, structural and functional metadata, preprocessing steps, and statistical modeling used to evaluate these cognitive processes.

1 Introduction

The human brain is constantly bombarded with sensory information. The ability to filter out noise and focus on goal-relevant stimuli "broadly defined as cognitive control" is a fundamental part of how we manage our thoughts and actions. The primary objective of this functional Magnetic Resonance Imaging (fMRI) study is to isolate and map the neural networks responsible for this interference resolution.

To map the brain activity associated with this capacity, cognitive control is frequently probed using the Flanker task. During this behavioral paradigm, subjects are presented with a row of arrows and instructed to indicate the direction of the central target arrow (left or right) while ignoring the surrounding "flanking" arrows. The task comprises two primary conditions:

- **Congruent Condition:** The target and flanking arrows point in the identical direction (e.g., > > > > > or < < < < <). Because there is no competing visual information, this condition serves as a low-demand baseline that requires minimal cognitive control.
- **Incongruent Condition:** The flanking arrows point in the opposite direction of the target (e.g., > > < > > or < < > < <). This perceptual mismatch induces a strong response conflict, forcing the brain to actively suppress the prepotent—but incorrect—motor response to successfully complete the trial.

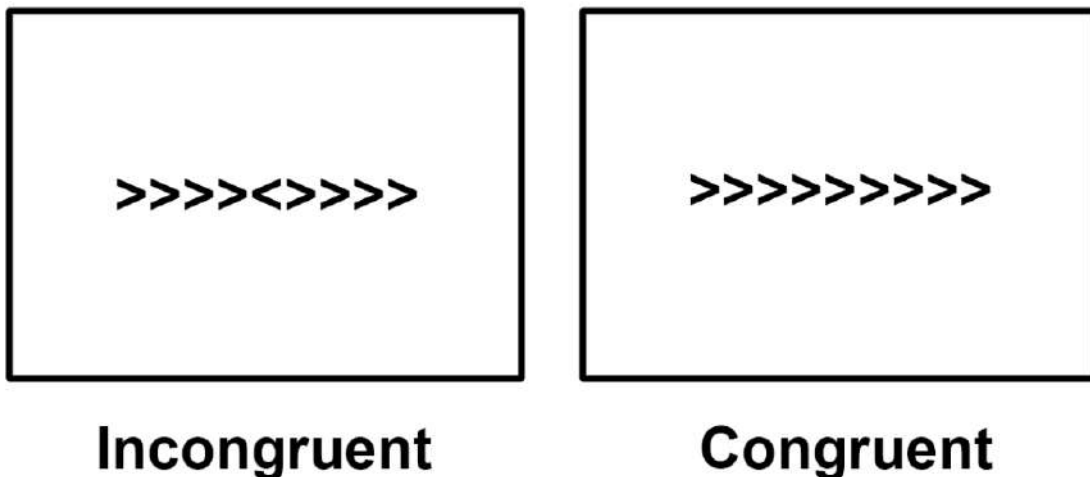


Figure 1: An example of the two conditions in the Flanker task: Congruent vs Incongruent

This visual conflict reliably carries a behavioral cost, manifesting as consistently slower reaction times and decreased accuracy during incongruent trials. The overarching goal of our fMRI analysis is to leverage this behavioral effect to

identify its neurobiological source. By estimating the magnitude of the Blood-Oxygen-Level-Dependent (BOLD) signal for both trial types, we will compute a direct statistical contrast (Incongruent > Congruent).

2 The Physics of fMRI and BOLD Contrast

fMRI relies on T2*-weighted images to detect the BOLD (Blood Oxygenation Level Dependent) signal. Understanding the physical principles behind this signal is essential for interpreting the neural activity maps generated in subsequent analyses.

2.1 The Physics of T2*

In MRI physics, T2* refers to the decay of the transverse magnetization signal. While standard T2 relaxation is based purely on the properties of the tissue itself, T2* takes into account local magnetic field inhomogeneities (distortions in the magnetic field). [2].

- When the magnetic field is uneven, the magnetic spins of protons dephase (fall out of sync) much faster.
- Faster dephasing leads to a shorter T2* and a darker image signal.
- A more uniform magnetic field leads to a longer T2* and a brighter image signal.

2.2 The Magnetic Properties of Blood

- **Deoxyhemoglobin (without oxygen):** This is paramagnetic. It acts like a tiny magnet, disrupting the local magnetic field around the blood vessel. This disruption causes protons to dephase faster, decreasing the T2* signal.
- **Oxyhemoglobin (with oxygen):** This is diamagnetic. It is magnetically neutral relative to brain tissue and does not disrupt the magnetic field. [3].

2.3 The BOLD Response (The fMRI Connection)

When a specific region of the brain becomes active (e.g., when you tap your finger or solve a math problem), it consumes oxygen. Here is the chain of events that links this to T2*:

1. **The Overcompensation:** The brain responds to this oxygen demand by rushing an oversupply of oxygenated blood to the active area (the hemodynamic response).
2. **The Washout:** This massive influx of fresh, oxygenated blood effectively flushes out the deoxygenated blood in that specific localized network of capillaries.
3. **The T2* Change:** Because the paramagnetic deoxyhemoglobin is washed away and replaced by diamagnetic oxyhemoglobin, the local magnetic field suddenly becomes more uniform.
4. **The Brighter Signal:** With fewer magnetic distortions, the protons dephase slower. T2* increases, resulting in a slightly stronger (brighter) signal on the MRI scanner. [4].

3 The Hemodynamic Response

The Hemodynamic Response (HDR) is the brain's localized, automatic system for managing the supply and demand of energy. Because neurons do not have their own internal reserves of energy or oxygen, they must rely on the vascular system to deliver them exactly when and where they are needed.

The physiology behind the HDR comes down to two main concepts: Neurovascular Coupling (how neurons talk to blood vessels) and the Phases of Blood Flow (the mechanical response of the vessels).

3.1 The Trigger: Neurovascular Coupling

When a specific region of the brain becomes active, it needs immediate energy (ATP) to restore its electrical gradients. This kicks off a chain reaction:

- **Glutamate Release:** Firing neurons release neurotransmitters, primarily glutamate.
- **Astrocyte Activation:** Star-shaped glial cells called astrocytes detect this glutamate. Astrocytes have "end-feet" that physically wrap around both synapses and nearby blood vessels, acting as the perfect middleman [5].
- **Vasoactive Signals:** The astrocytes experience a surge of calcium, causing them to release vasoactive chemical messengers—such as nitric oxide (NO) and prostaglandins—directly onto the smooth muscle cells of nearby arterioles.
- **Vasodilation:** These chemicals cause the smooth muscles around the arterioles to relax. The blood vessels widen (dilate), instantly dropping local vascular resistance and allowing blood to rush in.

3.2 The Core Paradox: Supply Exceeds Demand

To understand fMRI, you have to understand the defining quirk of the HDR: the brain overcompensates [6].

When neurons fire, the Cerebral Metabolic Rate of Oxygen (CMRO₂) increases slightly. However, the resulting increase in Cerebral Blood Flow (CBF) and Cerebral Blood Volume (CBV) is out of proportion—often increasing by 30% to 50%. The brain sends far more oxygenated blood than the neurons actually consume. This massive influx is what washes out the deoxygenated blood and triggers the T2* signal increase we discussed earlier.

3.3 The Three Phases of the HDR

If you plot this physiological response on a graph over time, it follows a very specific, predictable curve containing three distinct phases:

- **Phase 1: The Initial Dip (0 to 2 seconds)**
 - As soon as neurons fire, they consume the immediately available oxygen in the local capillaries.
 - Because the blood vessels haven't dilated yet, there is a tiny, temporary spike in deoxygenated blood.
 - *Note:* This dip is very small and fast, making it hard to capture on a standard fMRI, but it represents the truest spatial localization of neural firing.
- **Phase 2: The Peak (4 to 6 seconds)**
 - The vasoactive signals take effect, and arterioles dilate.
 - A flood of fresh, oxygenated arterial blood surges into the capillary bed.
 - This is the massive overcompensation phase that dramatically flushes out the deoxygenated blood, causing the classic peak in the BOLD signal.
- **Phase 3: The Post-Stimulus Undershoot (10 to 30 seconds)**
 - Neural activity stops, and arterial blood flow quickly drops back to baseline.
 - However, the veins that expanded to accommodate the rushing blood (venous ballooning) are slower to deflate back to their original size.
 - Because the veins are still slightly enlarged and blood flow has slowed down, deoxygenated blood temporarily pools in the area. This causes the signal to dip below the baseline before slowly returning to normal.

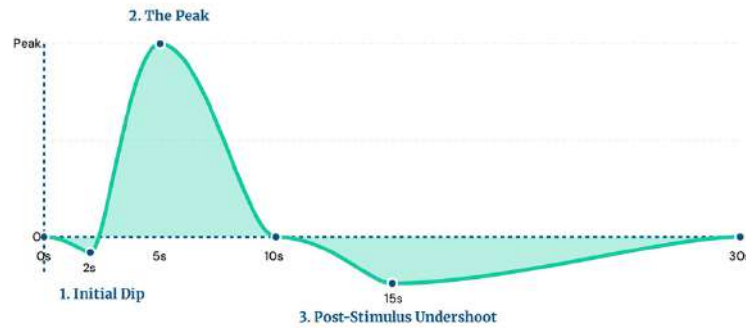


Figure 2: The predictable BOLD curve: A tiny initial deoxygenation dip, followed by a massive arterial flood (Peak), and a slow venous deflation (Undershoot).

4 Experimental Design

The Flanker task was administered to participants while fMRI data were acquired [1]. The task consisted of multiple runs, each containing a balanced number of congruent and incongruent trials presented in a randomized order. A single trial proceeds sequence as follows:

- **Fixation:** Subjects focus on a central cross to establish a neurophysiological baseline.
- **Stimulus Presentation (2000 ms):** A congruent or incongruent Flanker array is displayed, during which the subject registers a button-press response.
- **Jittered Inter-Trial Interval (8,000–14,000 ms):** A variable rest period featuring the fixation cross. This interval increments in whole seconds, with a mean duration of 12 seconds.

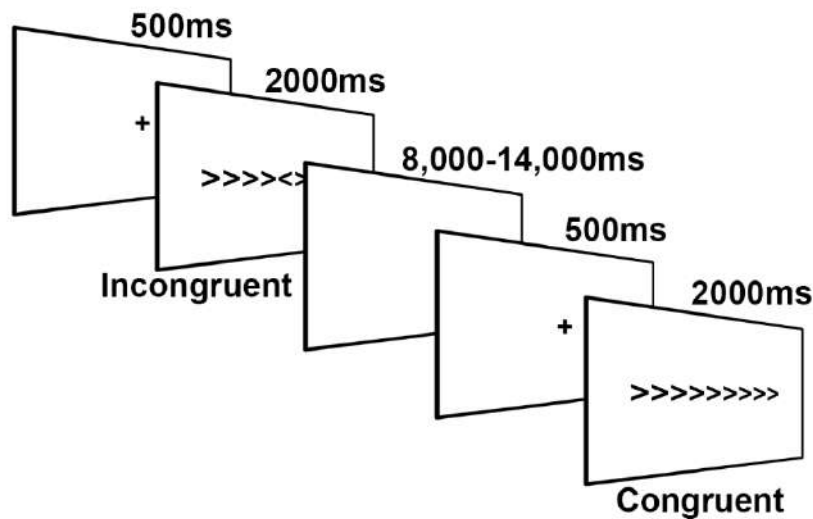


Figure 3: Illustration of the Flanker task for this study, adapted from Kelly et al. (2008). [1]

5 Dataset Description

The dataset includes scans from 26 participants, each with:

- A high-resolution anatomical scan (T1-weighted image)
- Two functional sessions (146 volumes each), with a 2-second repetition time (TR)

All data were collected using a Siemens Allegra 3.0 T scanner with a standard head coil, at the **NYU Center for Brain Imaging**. The data is organized according to the **Brain Imaging Data Structure (BIDS)** standard:

- Each participant's folder contains an **anat** (anatomical) and a **func** (functional) subfolder
- Functional scans are accompanied by onset files, which record the timing of Congruent and Incongruent trials during the Flanker task.

5.1 Anatomical Metadata (T1w.json)

Located in the root directory, this file dictates the hardware parameters for the high-resolution structural images acquired via a Siemens Allegra 3.0 Tesla scanner. Key parameters derived from the metadata include:

- **Sequence:** 3D MPRAGE sequence acquiring 176 contiguous slices to provide a detailed anatomical reference space.
- **Repetition Time (TR):** 2.5 s (2500 ms).
- **Echo Time (TE):** 0.00393 s (3.93 ms).
- **Inversion Time (TI):** 0.90 s (900 ms).
- **Flip Angle:** 8.0°.



Figure 4: Sub. 8 anatomical image

5.2 Functional Metadata (`task-flanker_bold.json`)

This file contains the physical scanning parameters for the functional runs. The variables crucial for preprocessing and future modeling are:

- **Sequence:** 2D Echo Planar Imaging (EPI).
- **Repetition Time (TR):** 2.0 s (2000 ms). *(This is the critical parameter for temporal modeling, indicating that one complete 3D brain volume was acquired every 2 seconds).*
- **Echo Time (TE):** 0.030 s (30 ms).
- **Flip Angle:** 80.0°.
- **Voxel Size & Coverage:** 3×3×4 mm across 40 slices, capturing 146 total volumes per experimental block.

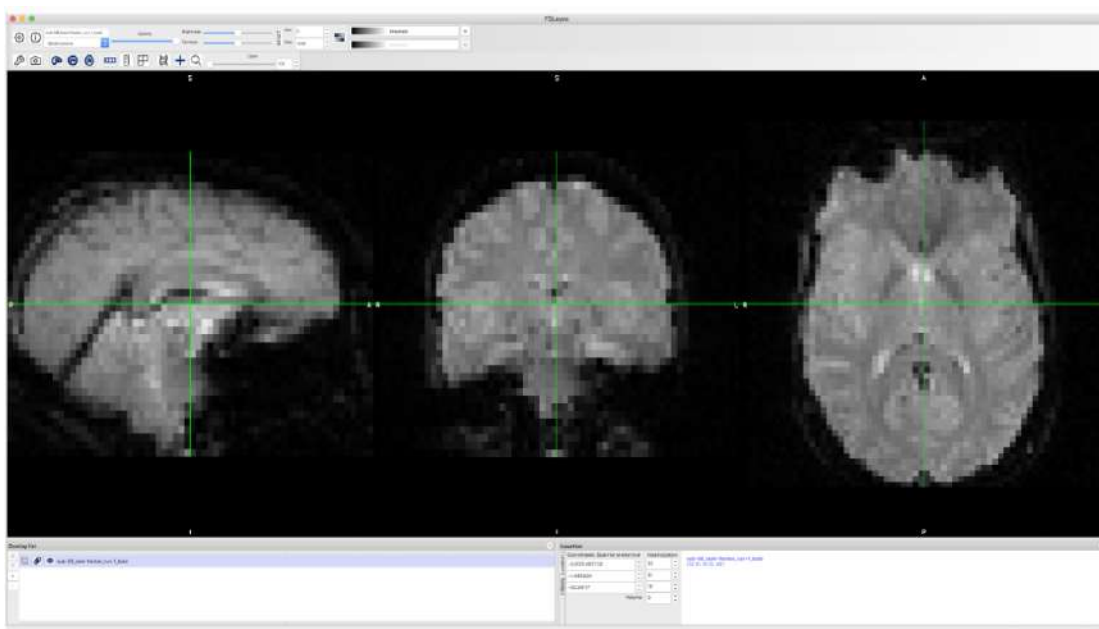


Figure 5: Sub. 8 functional image

5.3 Behavioral Task Logs (`events.tsv`)

Subjects performed two separate 5-minute functional blocks, yielding two `events.tsv` files inside each subject's `func` directory. These logs serve as the experimental timeline:

- **Trial Distribution:** 24 trials per block (12 congruent, 12 incongruent) presented in a pseudorandom order.
- **Recorded Variables:** The exact onset time, duration, and trial type (Congruent vs. Incongruent) for every stimulus.

Role in Statistical Modeling: Because the MRI scanner solely acquires images without recognizing the cognitive task, these behavioral logs are foundational. In subsequent analytical phases, these recorded onset times will be convoluted with a theoretical BOLD response curve to perform accurate model fitting.

6 Data Organization and BIDS Compliance

6.1 Introduction to BIDS and Its Purpose

Neuroimaging experiments inherently produce complex, multi-dimensional data that can be arranged in countless ways. Historically, the lack of a standardized organizational consensus led to misunderstandings, reduced reproducibility, and significant time wasted rewriting scripts to accommodate unique laboratory file structures.

To solve this, the field adopted the Brain Imaging Data Structure (BIDS). BIDS provides a simple, easily adoptable standard for organizing and describing neuroimaging and behavioral data. Utilizing BIDS offers several critical advantages:

- **Collaboration and Reproducibility:** It makes it exceptionally easy for other researchers to understand and work with the data without needing extensive guidance.
- **Software Compatibility:** A growing ecosystem of automated data analysis pipelines and software packages are designed to directly read and process BIDS-formatted data.
- **Public Sharing:** Major repositories, such as OpenNeuro, require datasets to be organized according to BIDS prior to publication.
- **Automated Quality Assurance:** Validation tools, such as the BIDS Validator, can automatically check the dataset's integrity and flag missing values or structural errors.

6.2 Application to the NYU Slow Flanker Dataset

The NYU Slow Flanker dataset (Accession `ds000102` on OpenNeuro) utilizes the BIDS framework to structure the neuroimaging data of its 26 participants. The presence of a BIDS Validation report at the top of the dataset's OpenNeuro page confirms its structural evaluation against these standardized rules.

The BIDS standard directly dictates the dataset's file hierarchy and naming conventions:

```
ds000102/
|-- sub-01/
|   |-- anat/
|   |   |-- sub-01_T1w.nii.gz
|   |   '-- sub-01_T1w.json
|   '-- func/
|       |-- sub-01_task-flanker_bold.nii.gz
|       |-- sub-01_task-flanker_bold.json
|       |-- sub-01_task-flanker_events.tsv
|       '-- sub-01_task-flanker_events.json
|-- sub-02/
|   |-- anat/
|   '-- func/
'-- ... (sub-03 through sub-26)
```

- **Subject-Level Directories:** All data is strictly separated into subject-specific folders (e.g., `sub-01`, `sub-02`), ensuring that each participant's data is compartmentalized.
- **Modality Subfolders:** Within each subject folder, the data is further divided by imaging modality. The high-resolution T1-weighted anatomical scans are housed in an `anat/` folder, while the Echo Planar Imaging (EPI) task-based sequences are stored in a `func/` folder.

- **Paired Metadata (.json sidecars):** BIDS mandates that compressed NIfTI image files (.nii.gz) are accompanied by JSON sidecar files. For example, the functional images are paired with JSON files that store essential scanner metadata (such as the 2000 ms Repetition Time and 80-degree flip angle) directly next to the image file.
- **Event Logging (_events.tsv):** Behavioral data corresponding to the event-related Flanker task is stored in tab-separated values (TSV) files. These files are essential for the General Linear Model (GLM), as they record the exact onset times, durations, and trial conditions (Congruent vs. Incongruent).

By strictly adhering to BIDS, the NYU Slow Flanker dataset ensures that its complex fMRI data can be seamlessly ingested by analytical software like FSL, drastically reducing preprocessing friction and ensuring a reproducible analytical pipeline.

7 Data Inspection & Quality Control

To ensure the reliability of the neuroimaging analysis, comprehensive quality control was performed on both the structural (anatomical) and functional data. The metrics below, extracted from the `aMRIQC.csv` and `fMRIQC.csv` derivatives, provide quantifiable measures of image clarity, artifact presence, and subject motion.

7.1 Anatomical MRI Quality Metrics (aMRIQC)

The anatomical metrics evaluate the clarity and tissue contrast of the high-resolution T1-weighted structural images. Good structural quality is critical for accurate brain extraction and spatial normalization.

Table 1: Anatomical MRI Quality Metrics

Metric	Description
CNR & SNR	Contrast-to-Noise Ratio and Signal-to-Noise Ratio: Foundational measures of image quality. SNR measures the strength of the actual brain signal compared to random background noise. CNR specifically measures how easily we can visually distinguish between different brain tissues (e.g., Gray Matter vs. White Matter). Higher values indicate clearer, more reliable images.
CJV	Coefficient of Joint Variation: Evaluates the overlap in intensity distributions between White Matter (WM) and Gray Matter (GM). Highly sensitive to head motion and intensity non-uniformity. Lower CJV indicates crisp, distinct boundaries between tissue types.
EFC	Entropy Focus Criterion: Acts as a mathematical proxy for image blurring and ghosting artifacts caused by head movement. Lower values (closer to zero) indicate a sharper image free of motion blurring.
FWHM	Full-Width Half Maximum: Measures the spatial smoothness or resolution of the image. Lower FWHM values indicate a sharper, higher-resolution image; higher values suggest excessive smoothing or blurring.
FBER	Foreground-to-Background Energy Ratio: Compares the variance of the signal inside the brain (foreground) to the variance of the noise outside the head (background). Higher values indicate a strong brain signal with minimal scanner background noise.

7.2 Functional MRI Quality Metrics (fMRIQC)

Functional metrics evaluate the stability of the BOLD signal over time. Because fMRI relies on detecting very small changes in blood oxygenation over hundreds of sequential images, temporal stability and motion tracking are paramount.

Table 2: Functional MRI Quality Metrics

Metric	Description
FD	Framewise Displacement: Direct measure of subject head motion. Calculates how much the subject’s head moved from one volume to the next. <code>mean_fd</code> provides average movement across the scan; <code>perc_fd</code> flags percentage of the scan with movement beyond threshold. Lower values indicate a highly compliant subject.
DVARS	Derivative of RMS Variance over Voxels: Measures signal disruption. Calculates how much image intensity changes across the entire brain from one time point to the next. Spikes correlate with sudden head twitches or scanner artifacts. Lower, stable values indicate a clean time-series.
tSNR	Temporal Signal-to-Noise Ratio: The most critical metric for fMRI. Measures how stable voxel signals are over the entire scan duration. High tSNR means stable baseline signal, making it easier to detect BOLD signal changes. Higher values indicate robust functional data.
GSR	Ghost-to-Signal Ratio: Measures presence of "ghosting" artifacts—a common issue in Echo-Planar Imaging (EPI) where faint, misplaced copies of the brain appear in the background. Lower values indicate clean image free of ghosting interference.
GCOR	Global Correlation: Measures average correlation across all voxels in the brain. High global correlation usually indicates systemic noise (breathing/heartbeat or scanner drift) rather than true task-based neural activity.

7.3 Evaluation of the Flanker Dataset Quality

The overall quality of the NYU Slow Flanker dataset was rigorously evaluated to ensure the integrity of downstream statistical modeling. Using the extracted Quality Control metrics (MRIQC) for both structural and functional scans, we assessed the cohort for image clarity, temporal stability, and motion-related artifacts. Overall, the dataset demonstrates a high standard of image quality typical of controlled cognitive neuroscience studies.

7.3.1 Anatomical Quality Assessment

The T1-weighted structural images display exceptional tissue contrast between gray and white matter. Across the entire cohort of 26 participants, the Signal-to-Noise Ratio (SNR) demonstrates robust signal integrity, ranging from 13.7 to over 18.7. Additionally, the highest Entropy Focus Criterion (EFC) recorded within the sample is a mere 0.56. Because this metric falls well below standard exclusion thresholds, it provides quantitative confirmation of minimal subject head motion, effectively ruling out any substantial image blurring or ghosting artifacts. Consequently, all 26 structural images comfortably exceed quality assurance standards and are approved for spatial normalization.

Table 3: Face Cropping Quality Assessment

Subject	Face Well Cropped
Sub 1	Yes
Sub 2	Yes
Sub 3	No
Sub 4	Yes
Sub 5	Yes
Sub 6	Yes
Sub 7	Yes
Sub 8	Yes
Sub 9	Yes
Sub 10	No
Sub 11	Yes
Sub 12	No
Sub 13	Yes

7.3.2 Functional Time-Series Stability

The four-dimensional BOLD data demonstrate outstanding temporal stability. While a mean Framewise Displacement (FD) exceeding 0.20 mm is conventionally used to identify problematic, high-motion fMRI runs, the absolute highest mean FD observed in this dataset is merely 0.133 mm. Moreover, an analysis of the `num_fd` parameter indicates a near-total absence of sudden motion spikes (such as swallowing or abrupt head jerks) across the vast majority of scans. Even the most motion-prone participant exhibited a maximum of just three minor movements during a five-minute acquisition block. As a direct result of this excellent participant compliance, both the DVARS metrics and outlier volume percentages remain exceptionally low.

7.3.3 Functional Data Inspection

Functional MRI data are collected as a time series sequence of volumes acquired over time. It is crucial to check for any motion artifacts, as this can degrade the quality of the data. To inspect the functional scans, we use “FSLeyes” to rapidly flip through all the volumes by clicking on the movie reel icon.

Table 4: Functional Run Quality Assessment by Subject

Subject ID	Func Run 1	Func Run 2
sub-01	pass	pass
sub-02	pass	pass
sub-03	warning	warning
sub-04	warning	pass
sub-05	pass	pass
sub-06	pass	pass
sub-07	pass	pass
sub-08	pass	pass
sub-09	pass	pass
sub-10	pass	pass
sub-11	pass	pass
sub-12	pass	pass
sub-13	pass	pass
sub-14	pass	warning
sub-15	pass	pass
sub-16	pass	pass
sub-17	pass	pass
sub-18	pass	pass
sub-19	warning	warning
sub-20	pass	pass
sub-21	pass	pass
sub-22	pass	pass
sub-23	pass	pass
sub-24	pass	pass
sub-25	warning	warning
sub-26	pass	warning

8 fMRI Data Preprocessing

8.1 Anatomical Brain Extraction (Skullstripping)

The first critical step in our preprocessing pipeline is the isolation of brain tissue from non-brain structures (such as the skull, scalp, face, and dura mater) in the high-resolution T1-weighted anatomical images. This process, commonly referred to as "skull-stripping," is essential; failing to remove the skull can severely disrupt the algorithms used later for spatial normalization and functional-to-anatomical registration.

For this task, we utilized the Brain Extraction Tool (BET) provided within the FMRIB Software Library (FSL).

8.1.1 Parameter Tuning and Visual Inspection

BET utilizes a fractional intensity threshold (f) to estimate the boundaries of the brain. By default, FSL sets this threshold to 0.5. However, because anatomical geometries and scanner contrasts vary, relying solely on the default can lead to suboptimal extractions. The mathematical rule for BET is inverse: smaller threshold values generate larger brain outline estimates, while larger values result in smaller brain outlines.

To determine the optimal boundary for our dataset, we empirically tested several fractional intensity thresholds and visually inspected the results by overlaying the skull-stripped outputs onto the original T1-weighted images using FSLeyes. We tested the following parameters:

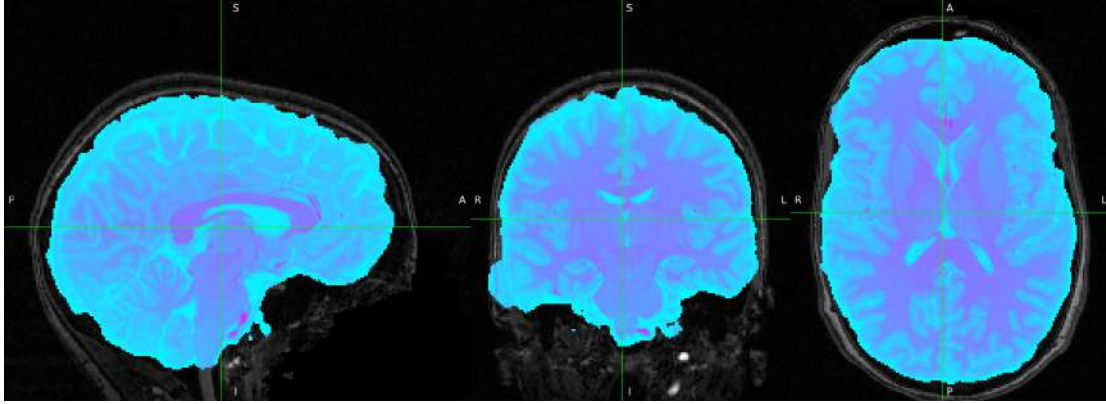


Figure 6: Brain extraction result with fractional intensity threshold $f = 0.1$. At this extremely low threshold, the estimated brain outline was too expansive. While all cortical tissue was successfully preserved, a significant amount of non-brain tissue (including the skull and dura mater) remained in the output image.

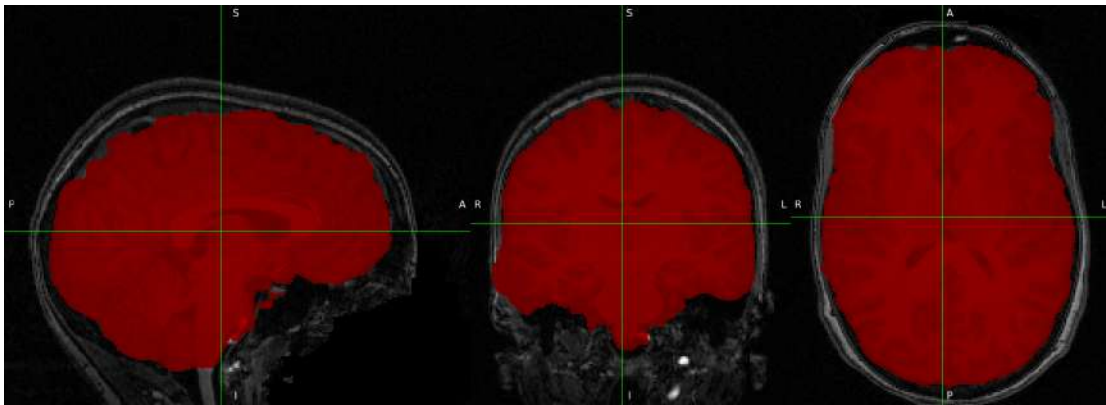


Figure 7: Brain extraction result with fractional intensity threshold $f = 0.2$. This value yielded the optimal extraction boundary. The resulting image accurately contoured the brain, perfectly isolating the cortex, subcortical structures, brainstem, and cerebellum, while successfully discarding the surrounding skull.

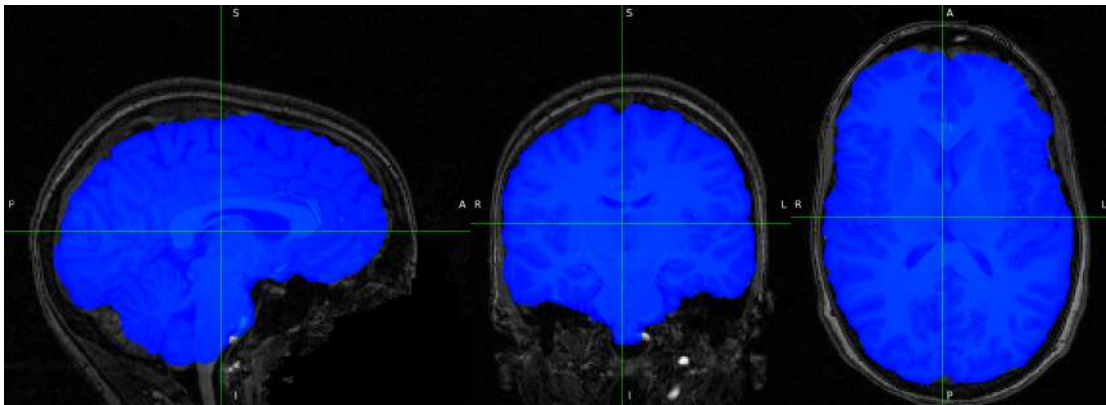


Figure 8: Brain extraction result with fractional intensity threshold $f = 0.3$. At this slightly higher threshold, the extraction boundary became too restrictive. Visual inspection revealed that small peripheral sections of the cortex (particularly in the frontal and parietal regions) were inadvertently stripped away.

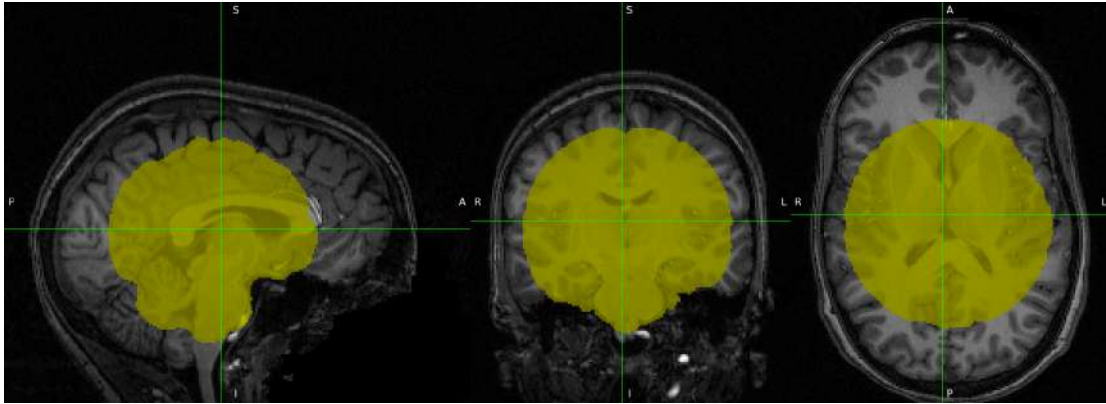


Figure 9: Brain extraction result with fractional intensity threshold $f = 0.9$. Applying an extremely high threshold resulted in a grossly underestimated brain outline. The extraction was far too small, aggressively cutting away massive portions of actual brain tissue and rendering the image entirely unusable for downstream analysis.

8.1.2 Conclusion of Brain Extraction

Based on this systematic visual quality assurance, a fractional intensity threshold ranges between **0.2 0.3** was selected for processing the anatomical scans. In neuroimaging preprocessing, it is a fundamental rule to err on the side of leaving a slight amount of skull rather than accidentally clipping the cortex. While minor residual non-brain tissue rarely impacts normalization, discarded cortical tissue is permanently lost and will compromise the integrity of the functional mapping.

8.2 Head Motion Correction (MCFLIRT)

One of the most significant challenges in fMRI data analysis is participant head movement during the scanning session. Because the BOLD signal is measured across hundreds of sequential volumes over several minutes, even minor head movements can severely degrade data quality. Head motion introduces two primary issues. First, it causes spatial blurring and misalignment; if a participant moves, a specific spatial coordinate (voxel) that initially contained gray matter may later contain white matter or cerebrospinal fluid, destroying the longitudinal integrity of the time-series. Second, motion introduces artificial signal variance. If head movement is systematically correlated with the experimental paradigm (e.g., the subject shifts slightly every time they press a button during an Incongruent Flanker trial), the analysis may falsely interpret the motion artifact as stimulus-evoked neural activity.

Motion correction was executed using FSL's MCFLIRT (Motion Correction FMRIB's Linear Image Registration Tool). The procedure works by selecting a single stationary volume from the fMRI time-series to act as a definitive "reference volume" (typically the middle volume of the run). MCFLIRT then algorithmically registers every other volume in the time-series to this reference. If the software detects that the subject's head shifted a millimeter to the right in volume 45, it applies an equal and opposite translation (a millimeter to the left) to realign volume 45 with the reference. This ensures that every voxel represents the exact same anatomical location from the first second of the Flanker task to the last.

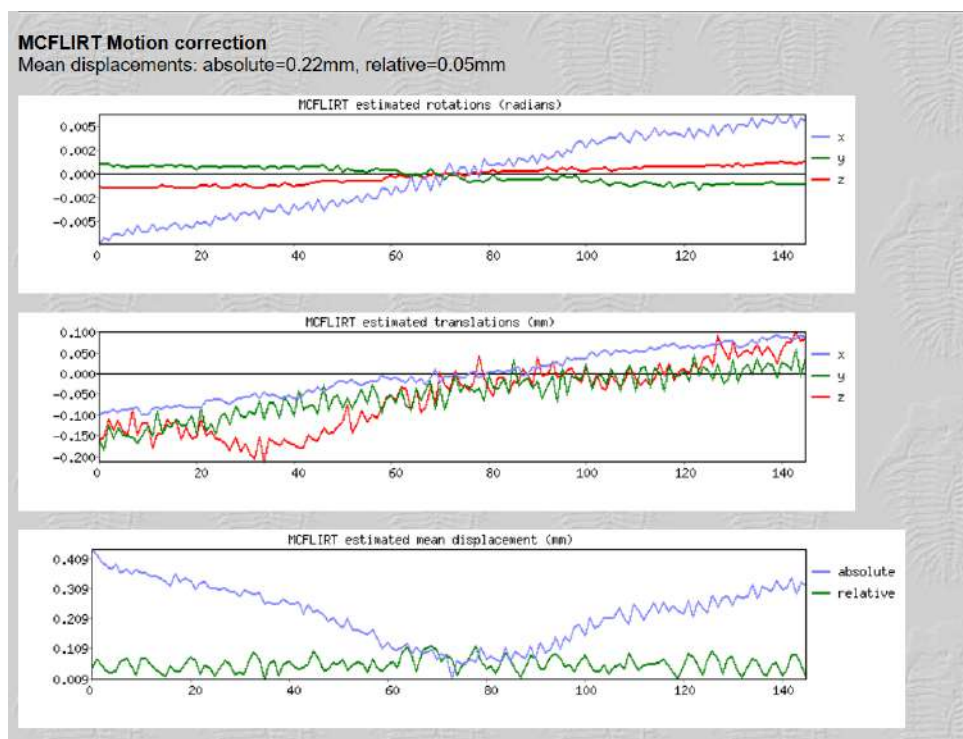


Figure 10: Motion Correction for sub-26 run-1

8.3 Slice-Timing Correction

Unlike a photograph, in which the entire picture is taken in a single moment, an fMRI volume is acquired in slices. Each of these slices takes time to acquire - from tens to hundreds of milliseconds. Depending on the scanner protocol, these slices are typically gathered either sequentially (e.g., consecutively from the bottom of the brain to the top) or in an interleaved fashion (e.g., acquiring all even-numbered slices first, followed by all odd-numbered slices). fMRI volumes are acquired slice-by-slice over the duration of the Repetition Time (TR). For this dataset (TR = 2,000 ms), there is a significant temporal delay between the acquisition of the first and final slice of each volume.

Because the General Linear Model (GLM) assumes all voxels within a single volume are acquired simultaneously, this delay must be addressed to maintain statistical power. Rather than altering the raw data through temporal interpolation, we followed FSL's default approach: slice-timing correction was bypassed during preprocessing. Instead, a temporal derivative was included later in the first-level GLM. This method mathematically absorbs the variance introduced by the 2-second acquisition delay without artificially smoothing the raw time-series data.

8.4 Spatial Smoothing

Spatial smoothing applies a Gaussian filter—measured in millimeters of Full Width at Half Maximum (FWHM)—to average the fMRI signal across neighboring voxels. While this process intentionally blurs the images, it is essential for reducing high-frequency noise, boosting the Signal-to-Noise Ratio (SNR), and helping to smooth over minor anatomical differences between subjects for group-level analysis.

8.4.1 The Trade-off of Kernel Size

Because smoothing inherently reduces spatial resolution, the size of the kernel must be carefully chosen based on the research goals:

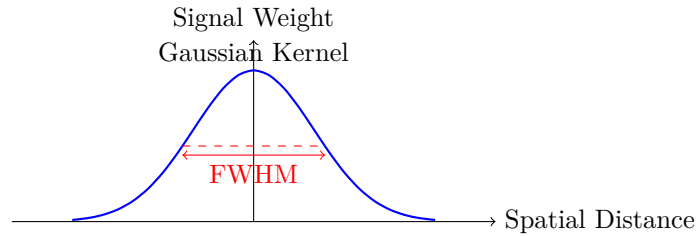


Figure 11: Illustration of a Gaussian smoothing kernel. The Full Width at Half Maximum (FWHM) determines the extent of spatial averaging. A larger FWHM results in broader smoothing and lower spatial resolution.

- **3 mm Kernel:** A conservative filter that preserves fine-grained spatial details. This is ideal for investigating small, highly localized regions of interest where precise anatomical localization is critical (Under-Smoothing)
- **5 mm Kernel:** The general rule is that smoothing works best when the blur size roughly matches the size of the brain activity we expect to find. Given the original voxel dimensions, a 5 mm FWHM applies a gentle, targeted blur. It successfully cleans up the noise while keeping the true “blobs” of cognitive control activation intact (e.g., within the Anterior Cingulate Cortex). (Perfect-Smoothing)
- **12 mm Kernel:** A broad filter that heavily averages the signal over a larger area. This is excellent for detecting large-scale network responses and increasing sensitivity to diffuse activations, but it will wash out fine anatomical details and potentially merge activations from adjacent but functionally distinct regions (Over-Smoothing)

Ultimately, the appropriate smoothing kernel must strike a balance: large enough to adequately reduce background noise and satisfy the assumptions of Gaussian Random Field theory for multiple comparison correction, yet small enough to preserve the spatial specificity required to map the targeted neural activity.

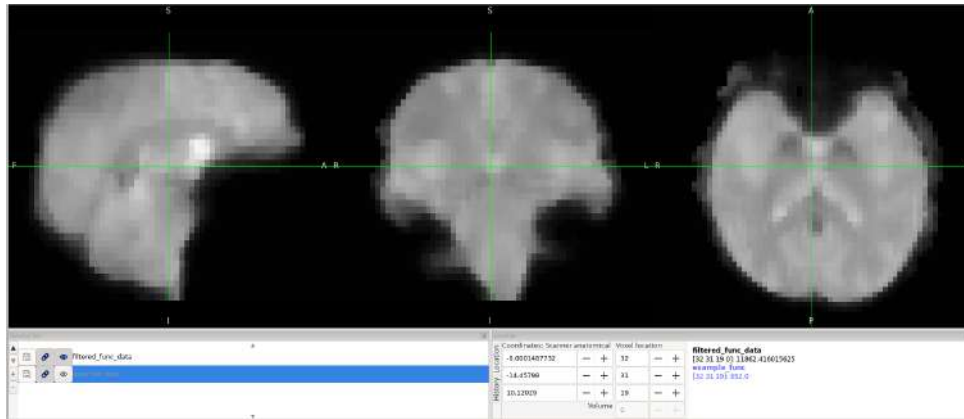


Figure 12: sub-26 run 1 with smoothing of 3 mm.

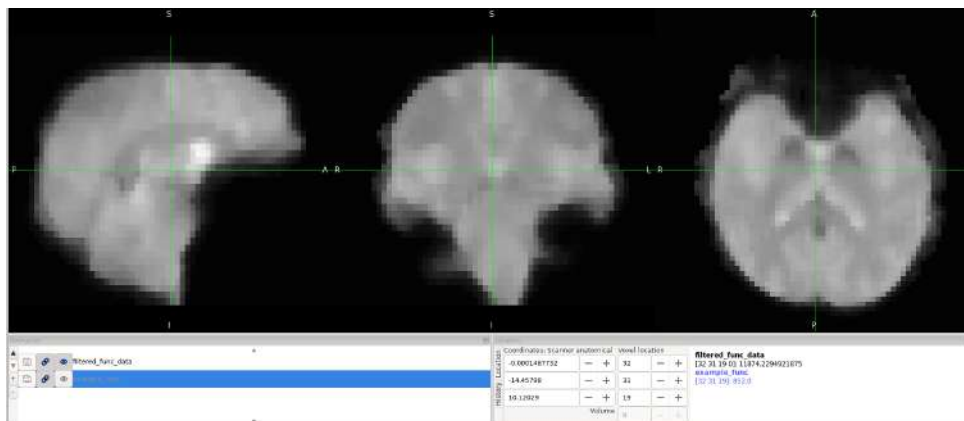


Figure 13: sub-26 run 1 with smoothing of 5 mm.

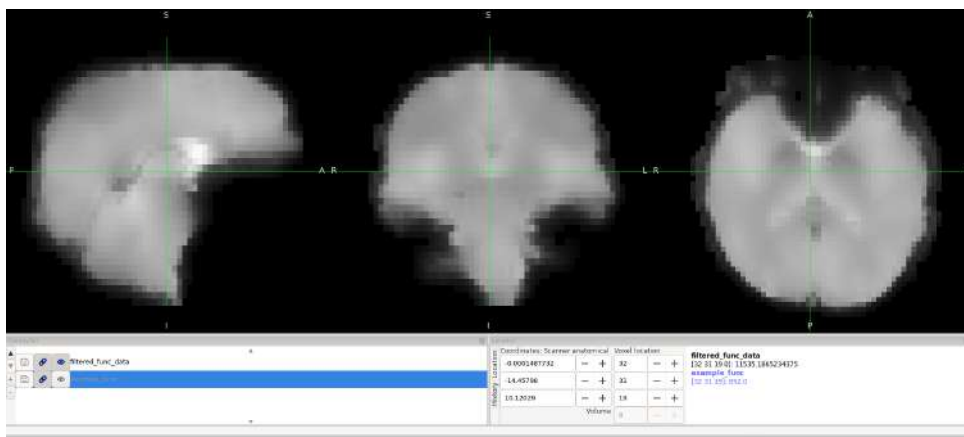


Figure 14: sub-26 run 1 with smoothing of 12 mm.

8.5 Registration and Normalization

To perform group-level statistical analyses, every participant’s fMRI data must be spatially aligned to a common anatomical framework. Because individual brains naturally vary in size and shape, we project the data into a standardized coordinate system (e.g., the MNI152 template) so that a specific spatial coordinate represents the exact same anatomical region across all subjects.

Because functional EPI images are relatively low-resolution, warping them directly to a high-resolution standard template often yields poor results. Therefore, we employed a standard two-step alignment pipeline using FSL’s FEAT tool:

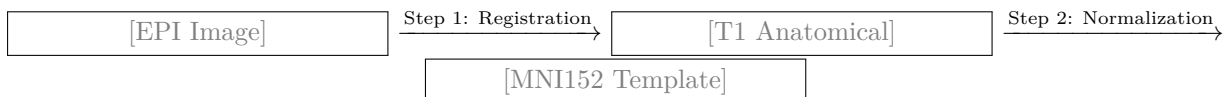


Figure 15: Two-step registration pipeline: functional EPI $\rightarrow$ anatomical T1 (affine, 12 DOF) $\rightarrow$ MNI152 standard space (nonlinear warp).

1. **Registration (Functional to Anatomical):** First, the low-resolution functional time-series is aligned to the subject’s own high-resolution, skull-stripped T1-weighted anatomical image. This alignment relies on maximizing

"mutual information" (matching inverse contrast weights, like bright CSF in functional images to dark CSF in anatomical ones). For optimal accuracy, FSL utilizes affine transformations—which include 12 degrees of freedom (translations, rotations, zooms, and shears) to stretch and scale the image appropriately.

2. **Normalization (Anatomical to Standard Space):** Next, the subject's high-resolution anatomical image is warped to match the standard MNI152 template. Once this optimal warp field is calculated, those exact same mathematical transformations are applied to the functional data.

8.5.1 Output Files in the `reg/` Folder

After running FEAT, the `reg/` subfolder contains several files that document the registration and normalization steps. Understanding these files is essential for quality assurance and for reproducing the analysis:

- **Transformation matrices (no extension):** Files such as `example_func2highres`, `highres2standard`, and `example_func2standard` store the affine matrices (and later nonlinear warp fields) that describe how each image was moved, rotated, stretched, or warped to align with the target space. These are the mathematical recipes used to transform the data.
- **Resampled images (.nii):** These are the actual image volumes after transformation. The most important ones are:
 - `highres.nii` – The original skull-stripped T1-weighted anatomical image.
 - `standard.nii` – The MNI152 template image used as the target for normalization.
 - `example_func2highres.nii` – Functional data resampled into anatomical space (after functional-to-anatomical registration).
 - `highres2standard.nii` – The anatomical image warped into MNI152 template space.
 - `example_func2standard.nii` – Functional data transformed directly into MNI space (using the combined transformations).
- **Visual check images (.png):** FEAT automatically generates PNG snapshots of the alignment at each stage. These figures, shown below, are crucial for quickly verifying that registration succeeded.

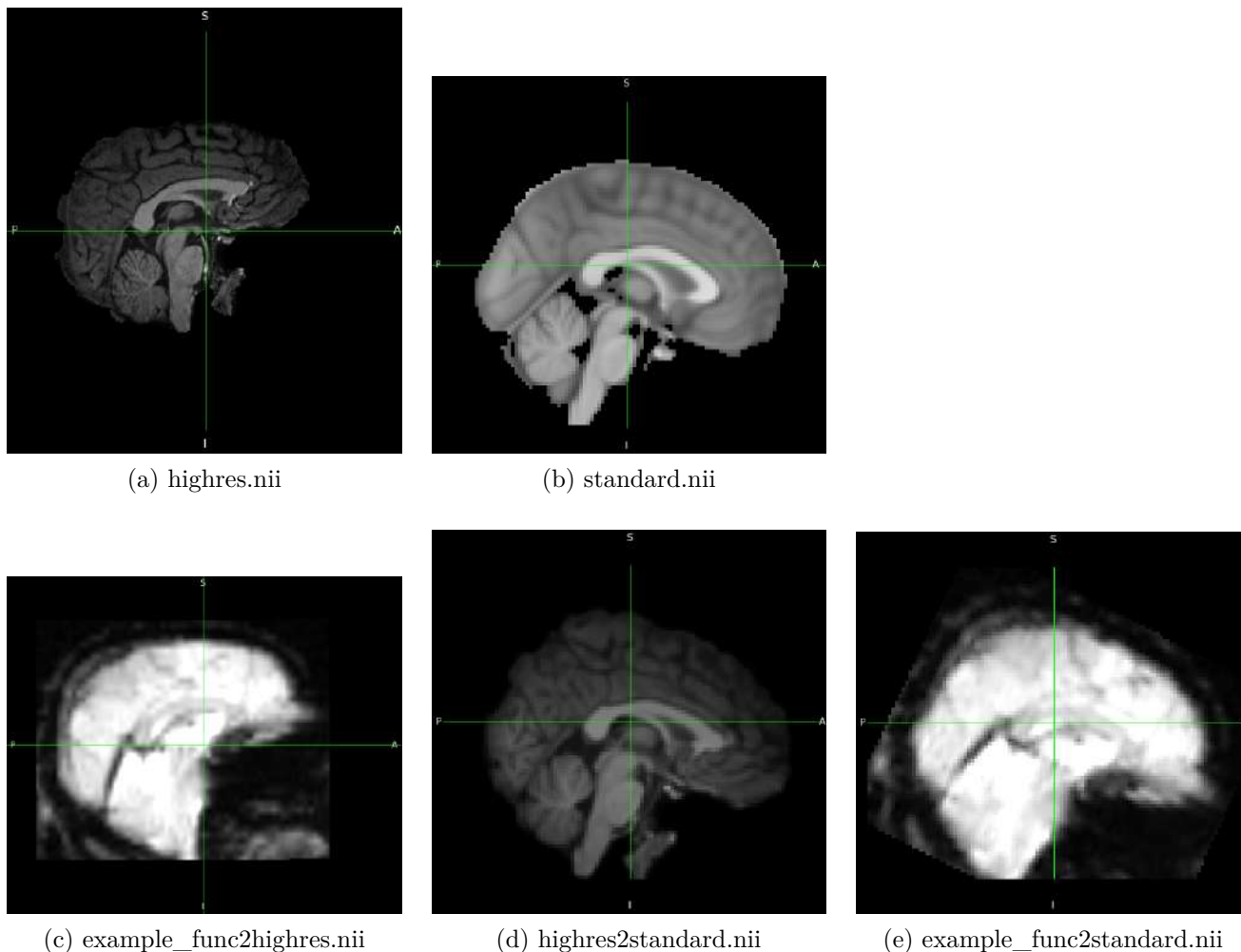


Figure 16: Resampled images from the reg/ folder.

The transformation matrices and resampled NIFTI files are the actual data products used in subsequent steps (e.g., applying the same warp to statistical maps), while the PNG files serve as quality control reports.

8.6 Implementation Parameters

For this dataset, both registration and normalization were configured to perform a "Full Search" to ensure the most thorough initial alignment. The normalization to standard space utilized a 12 Degree-of-Freedom (DOF) affine transformation, effectively bringing the functional Flanker task data into a standardized space ready for cross-subject statistical comparison.

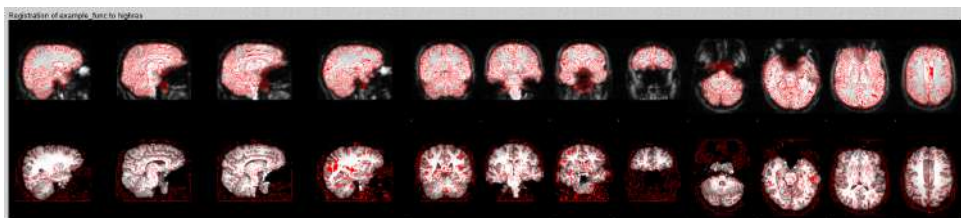


Figure 17: sub-26 run 1 12DOF Registration of 12 DOF

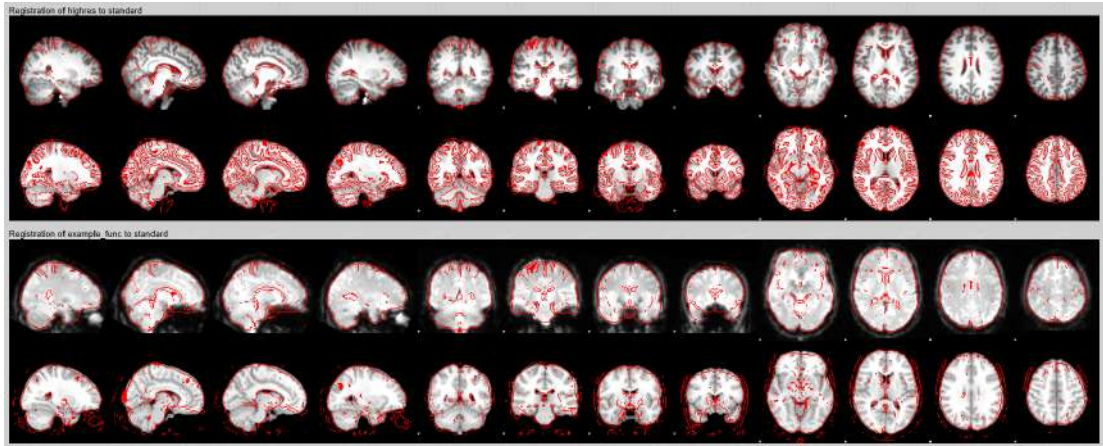


Figure 18: sub-26 run 1 12DOF Normalization of 12 DOF

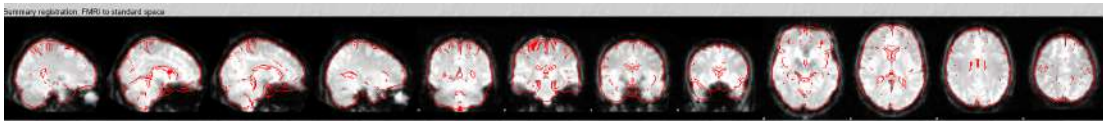


Figure 19: sub-26 run 1 12DOF Final Result from fMRI to Standard Space

Table 5: Comparison of Registration Degrees of Freedom (DOF) and Algorithms

Method	DOF	Transformations	Application
3 DOF	3	Translations only (X, Y, Z)	Very rarely used; cannot account for head rotation
6 DOF (Rigid-Body)	6	3 Translations + 3 Rotations (Pitch, Roll, Yaw)	Motion Correction (MCFLIRT); functional-to-anatomical alignment (no distortion)
12 DOF (Affine)	12	3 Translations + 3 Rotations + 3 Zooms + 3 Shears	Spatial normalization; warping subject brain to MNI152 template
BBR (Brain-Boundary Registration)	N/A	Advanced algorithm using white matter/gray matter boundary	Functional-to-anatomical registration; accounts for EPI susceptibility artifacts

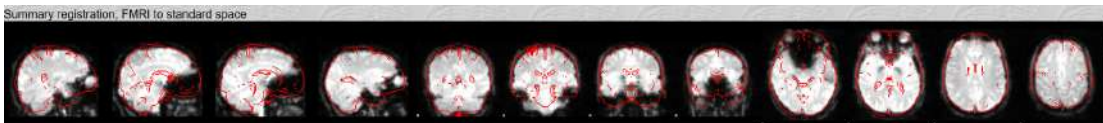


Figure 20: sub-26 run 1 3DOF Final Result from fMRI to Standard Space

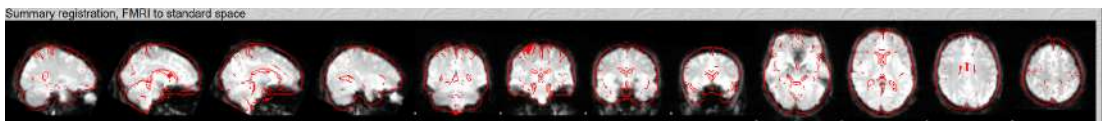


Figure 21: sub-26 run 1 6DOF Final Result from fMRI to Standard Space

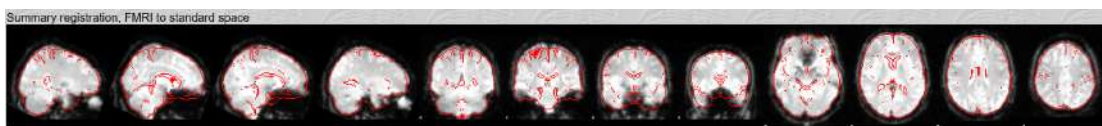


Figure 22: sub-26 run 1 BBR Final Result from fMRI to Standard Space

9 First-Level Statistical Analysis and the General Linear Model (GLM)

9.1 Introduction to the GLM in fMRI

With the preprocessing pipeline complete, the functional data is now formatted as a fully prepared 4D time-series, comprising 146 continuous volumes per run. The primary objective at this stage is to determine whether the BOLD signal within any given voxel exhibits fluctuations that systematically align with our experimental paradigm—specifically, the presentation of congruent and incongruent Flanker trials. Because this statistical evaluation is conducted independently across hundreds of thousands of individual spatial units, the procedure is formally termed a mass univariate analysis. To execute this, we employ the General Linear Model (GLM), the foundational statistical engine in fMRI research used to model expected neural responses and test hypotheses about task-evoked brain activation. [7].

9.2 The Core Equation

At its core, the GLM asserts that the observed BOLD data can be expressed as a linear combination of explanatory variables (the model) plus unexplainable error (the noise). Mathematically, this is expressed in matrix notation as:

$$Y = X\beta + \epsilon$$

- Y (Data): A column vector containing the observed BOLD time-series for a single voxel. If the fMRI run has 146 volumes, Y has 146 data points.
- X (Model / Design Matrix): A matrix containing our explanatory variables (regressors). For the Flanker task, this includes the exact onset times of the Congruent and Incongruent trials, convolved with a canonical Hemodynamic Response Function (HRF). It also includes "nuisance regressors," such as the 6 degrees of freedom of head motion calculated during preprocessing.
- β (Beta / Parameter Estimate): A vector of unknown weights that we must calculate. The beta value represents the *amplitude* or *effect size* of the neural response. It tells us how much of the voxel's variance is explained by that specific column in the design matrix.
- ϵ (Noise / Error): The residual variance in the BOLD signal that cannot be explained by our model (e.g., scanner drift, breathing, random thermal noise).

9.3 Hypothesis Testing: The Coin Toss Analogy

To understand how we determine if a calculated β weight is "significant," it is helpful to start with a fundamental statistical analogy: a coin toss.

9.3.1 How to Calculate a Null Distribution in a Coin Toss

Suppose we want to test if a coin is rigged. Our Null Hypothesis (H_0) states that the coin is perfectly fair, meaning the probability of landing on heads is 0.5. If we flip the coin 20 times, what are the expected outcomes? Using the binomial distribution equation, we can calculate the exact probability of getting k heads out of N flips:

$$P(X = k) = \binom{N}{k} p^k (1 - p)^{N-k}$$

Plotting these probabilities yields our **Null Distribution**—a bell curve peaking at 10 heads. If we conduct our experiment and the coin lands on heads 18 times, we look at our null distribution and ask: "*Assuming the coin is fair, what is the probability of getting 18 or more heads purely by chance?*" This calculation yields a **p-value** of approximately 0.0002. Because this p-value is extremely low (typically < 0.05), we reject the null hypothesis and conclude the coin is rigged.

9.4 Solving the Math: Estimating Betas and fMRI Null Distributions

We apply the exact same logical framework to our fMRI data.

9.4.1 Estimating the Betas

To solve the GLM equation $Y = X\beta + \epsilon$ and find the actual β weights that best fit our data, we use the Ordinary Least Squares (OLS) estimator. The math minimizes the squared errors (ϵ^2), yielding the following equation for the Beta estimates ($\hat{\beta}$):

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

This matrix algebra calculates how much the voxel "cared" about the Congruent and Incongruent Flanker trials.

9.4.2 Calculating the fMRI Null Distribution

Just like the coin, we need a Null Hypothesis (H_0). In fMRI, H_0 states that the experimental task had absolutely no effect on the voxel, meaning the true $\beta = 0$.

Because fMRI data contains continuous noise, the null distribution of a voxel is modeled as a **Student's t-distribution**. The degrees of freedom for this distribution are determined by the number of time points (scans) minus the number of regressors in our model ($df = N - p$).

9.4.3 The Relationship Between Beta (β) and the p-value

A large Beta value means the BOLD signal responded strongly to the task. However, a large Beta is meaningless if the data is incredibly noisy (large ϵ). Therefore, we must calculate a **t-statistic**, which is the ratio of the Beta estimate to its standard error (the noise):

$$t = \frac{\hat{\beta}}{SE(\hat{\beta})}$$

We then compare this t -statistic against our theoretical t -distribution (the null distribution).

The **p-value** is the probability of observing a t -statistic this extreme if the voxel was actually doing nothing ($\beta = 0$).

- **Beta** (β) is the *magnitude* of the brain's response (Effect Size).
- **p-value** is the *statistical confidence* that this response wasn't just a random fluke of scanner noise.

9.5 The Multiple Comparisons Problem in fMRI

When evaluating statistical significance at the group level, we encounter a massive statistical hurdle: the **Multiple Comparisons Problem** [8].

In our coin toss example, we ran one test. In fMRI, a standard normalized brain volume contains roughly 100,000 to 200,000 distinct voxels. If we run a GLM on every single voxel using a standard alpha threshold of $p < 0.05$ (meaning we accept a 5% false-positive rate), simple math dictates that we will find

$$100,000 \times 0.05 = 5,000$$

"active" voxels purely by random chance.

If left uncorrected, scanner noise will falsely masquerade as cognitive control networks. To solve this, strict mathematical corrections must be applied to the group-level p-values:

- **Family-Wise Error Rate (FWER):** The probability of making even one false discovery across the entire brain. The most famous FWER correction is the Bonferroni Correction, where the alpha level is divided by the number of tests ($0.05/100,000 = 0.0000005$). While mathematically safe, Bonferroni is often too conservative for fMRI, causing true neural signals to be ignored (Type II errors).
- **Cluster-Based Thresholding (Gaussian Random Field Theory):** The modern standard for fMRI. Instead of looking at individual voxels, this method relies on the spatial layout of the brain. True neural activity tends to cluster across adjacent voxels, whereas random noise is scattered. The software calculates the probability of a cluster of a certain size appearing by chance, offering a powerful balance between sensitivity and false-positive control.

9.6 Visualization Strategies for fMRI Group-Level Analysis

Description: After processing functional Magnetic Resonance Imaging (fMRI) data through a General Linear Model (GLM) across a cohort of subjects (e.g., using FSL's FLAME tool), biomedical engineers and neuroscientists are left with massive arrays of statistical data. These data are not raw images, but rather Statistical Parametric Maps (SPMs)—volumes where each voxel contains a statistical value (like a Z-score or t-statistic) representing the significance of brain activation or connectivity. Effectively displaying these 4D datasets (3D space + statistics) is critical for interpreting neurological function, communicating pathological differences between clinical groups, and planning neurosurgical interventions.

9.6.1 Core Visualization Methods

- **2D Orthogonal Multi-Planar Views:**
 - This is the most common clinical and academic standard. The statistical map is thresholded (e.g., showing only voxels where $p < 0.05$) and overlaid onto a high-resolution, standard anatomical template (like the MNI152 brain discussed previously).
 - The data is displayed across three intersecting 2D planes: Axial (top-down), Coronal (front-back), and Sagittal (side profile).
 - **Color Mapping:** Engineers typically use a "heatmap" color palette. Warm colors (red/yellow) indicate significant positive activation or increased connectivity, while cool colors (blue/light blue) indicate negative activation (deactivation) or decreased connectivity compared to the baseline.
- **3D Cortical Surface Rendering (Surface-Based Mapping):**
 - Because the cerebral cortex is highly folded, 2D slices can obscure continuous patterns of activation along a sulcus or gyrus. Engineers use tools like FreeSurfer or Connectome Workbench to construct a 3D mesh of the cortex.

- The brain is virtually "inflated" (smoothing out the folds) so that the deep sulci and superficial gyri are both visible simultaneously. The fMRI statistical maps are then projected onto this 3D surface, providing a striking, continuous map of cortical networks.

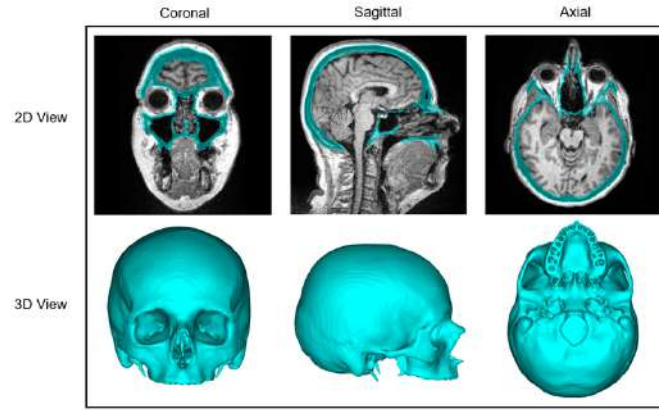


Figure 23: Multiplanar 2D views (coronal, sagittal, axial) and 3D reconstructions illustrating computed tomography (CT)-based skull segmentations registered onto corresponding magnetic resonance imaging (MRI) data. Source: Redefining MRI-Based Skull Segmentation Through AI-Driven Multimodal Integration

- **Maximum Intensity Projections (MIP) / "Glass Brain":**

- In this method, the brain is rendered as a transparent, 3D wireframe outline.
- The algorithm projects the highest statistical values from the deep brain structures to the surface of the image. This allows researchers to view all clusters of activation throughout the entire volume of the brain at a single glance, without having to scroll through individual 2D slices.

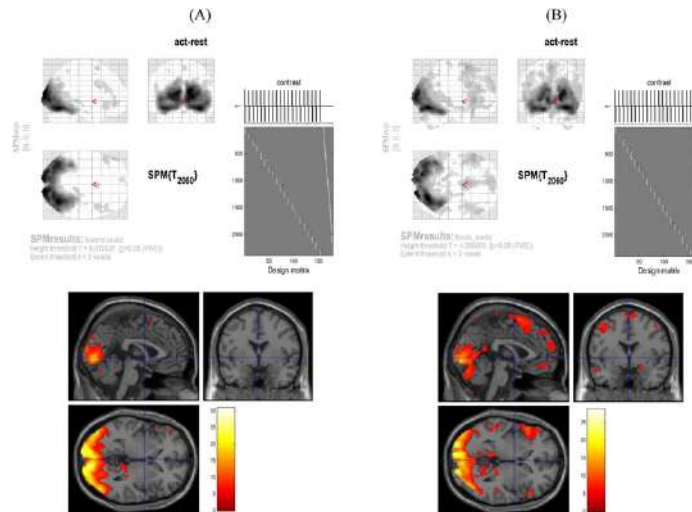


Figure 24: Maximum intensity projection (MIP) on a glass brain in three orthogonal planes, Design matrix, Brain activation map changing according to T value (a) 20 healthy group (b) 20 depressed group Source: Machine learning based detection of depression from task-based fMRI using weighted-3D-DWT denoising method

- **Quantitative ROI Bar Charts and Boxplots:**

- While heatmaps show *where* activation occurs, they are poor at conveying the exact magnitude (effect size) of the activation.
- Engineers use parcellation atlases (like the Harvard-Oxford atlas) to extract the average BOLD signal change (beta values) from specific Regions of Interest (ROIs). These are then plotted as bar charts, boxplots, or violin plots to clearly illustrate the quantitative statistical differences between distinct cohorts (e.g., Healthy Controls vs. Alzheimer’s Patients).

• Connectograms and Adjacency Matrices (For Resting-State Networks):

- When analyzing group-level functional connectivity (how different brain regions communicate over time), standard anatomical overlays become cluttered.
- **Adjacency Matrices:** Display a grid where both the X and Y axes represent specific brain regions, and the color of the intersecting square dictates the strength of their functional correlation.
- **Connectograms:** A circular graph where the perimeter is lined with brain regions, and thick, color-coded lines are drawn across the circle to represent the strength and directionality of the functional networks connecting them.

9.7 Building the Design Matrix: Timing Files and the Ideal Time-Series

Let us examine the time series data of **sub-04**, run 1, voxel of location $[32\ 25\ 19\ 32]$ and coordinates $(-2.32, -21.54, -2.30)$:

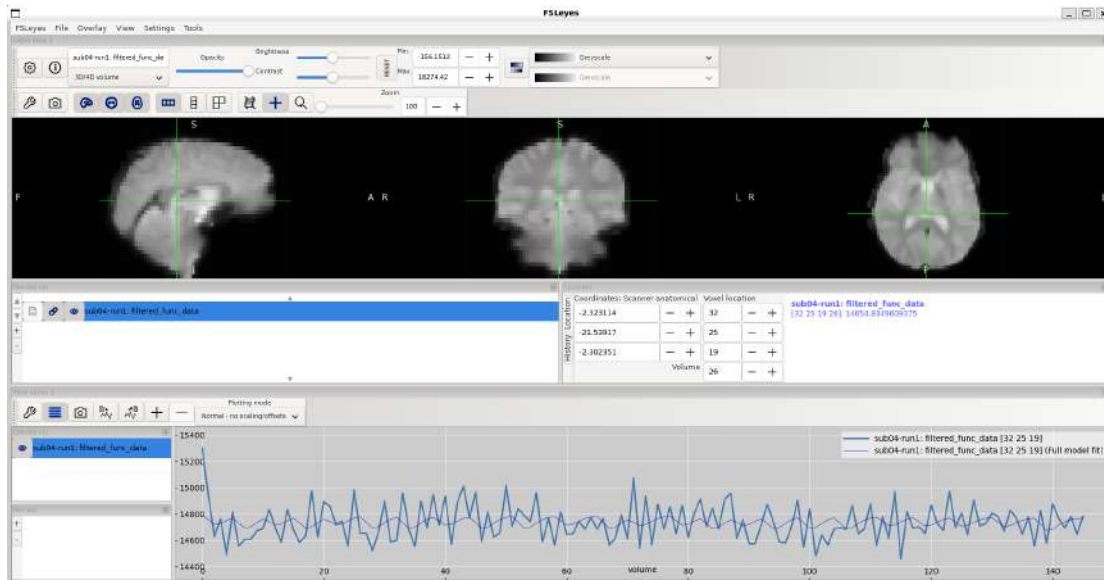


Figure 25: Time series of a voxel in sub-04, run 1

Because the analysis software is completely blind to what the participant actually experienced inside the scanner, the experimental paradigm must be manually translated into a readable format. This involves extracting behavioral data from the raw **events.tsv** logs to create text-based timing files (or onset files). These files strictly contain three columns: the exact onset time of the trial relative to the start of the scan, the duration of the trial, and a parametric modulation weight (typically set to 1).

onset	duration	trial_type	response_time	correctness	Stimvar	Response	Stimulus
0.0	2.0	incongruent_correct	1.342	correct 2	1	incongruent	cond003
10.0	2.0	congruent_correct	1.043	correct 1	1	congruent	cond001
20.0	2.0	incongruent_correct	1.141	correct 2	1	incongruent	cond003
30.0	2.0	congruent_correct	0.961	correct 1	1	congruent	cond001
40.0	2.0	incongruent_correct	0.997	correct 2	1	incongruent	cond003
52.0	2.0	congruent_correct	0.783	correct 1	1	congruent	cond001
64.0	2.0	incongruent_correct	0.787	correct 1	1	incongruent	cond003
76.0	2.0	incongruent_correct	1.046	correct 2	1	incongruent	cond003
88.0	2.0	congruent_correct	0.8	correct 1	1	congruent	cond001
102.0	2.0	congruent_correct	0.77	correct 1	1	congruent	cond001
116.0	2.0	incongruent_correct	0.883	correct 2	1	incongruent	cond003
130.0	2.0	congruent_correct	0.765	correct 1	1	congruent	cond001
140.0	2.0	incongruent_correct	0.858	correct 2	1	incongruent	cond003
154.0	2.0	incongruent_correct	0.851	correct 2	1	incongruent	cond003
164.0	2.0	congruent_correct	0.67	correct 1	1	congruent	cond001
174.0	2.0	incongruent_correct	0.77	correct 2	1	incongruent	cond003
184.0	2.0	congruent_correct	0.613	correct 1	1	congruent	cond001
196.0	2.0	congruent_correct	0.777	correct 1	1	congruent	cond001
208.0	2.0	congruent_correct	0.62	correct 1	1	congruent	cond001
222.0	2.0	incongruent_correct	0.821	correct 2	1	incongruent	cond003
234.0	2.0	incongruent_correct	0.825	correct 2	1	incongruent	cond003
246.0	2.0	incongruent_correct	0.73	correct 2	1	incongruent	cond003
260.0	2.0	incongruent_correct	0.781	correct 2	1	incongruent	cond003
274.0	2.0	congruent_correct	0.726	correct 1	1	congruent	cond001

(a) Paradigm sub-04-task-flanker-run-1-events.tsv

10.0	2.0	1
30.0	2.0	1
52.0	2.0	1
64.0	2.0	1
88.0	2.0	1
102.0	2.0	1
130.0	2.0	1
164.0	2.0	1
184.0	2.0	1
196.0	2.0	1
208.0	2.0	1
274.0	2.0	1

(b) Congruent-run1 stimuli data separated

Figure 26: Event-related design visualization for sub-04 run 1. (a) shows the complete experimental paradigm with trial timing extracted from the events.tsv file, illustrating the sequence of congruent and incongruent trials across the run. (b) displays the separated stimuli data specifically for the congruent condition, highlighting the trial onsets and durations used for model construction.

By inputting these precise timing files as Explanatory Variables (EVs), the software convolves the exact trial onsets with the theoretical HRF to construct the Design Matrix. The design matrix visually and mathematically generates the "ideal time-series"—a theoretical curve representing exactly what a voxel's signal should look like if it was perfectly responsive to that specific experimental condition (e.g., an incongruent Flanker trial).

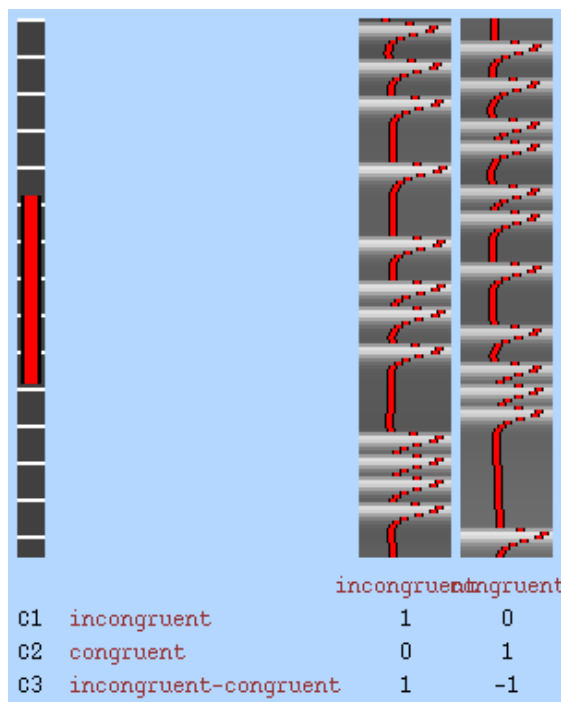
9.8 Configuring the First-Level GLM in FSL FEAT

Following the preparation of our onset files, the single-subject statistical modeling was carried out using FSL's FMRI Expert Analysis Tool (FEAT). The parameters configured within the software's interface directly correspond to the mathematical components of our General Linear Model (GLM):

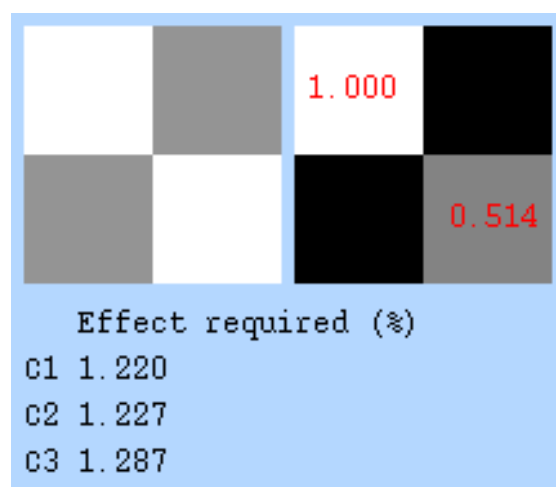
- **Explanatory Variables (EVs):** The EVs serve as the independent regressors that make up the columns of our design matrix (X). For this analysis, two primary EVs were established to represent the congruent and incongruent task conditions. To generate the theoretical "ideal" BOLD time-series for each condition, FEAT mathematically convolved our three-column timing logs with a canonical double-gamma Hemodynamic Response Function (HRF). To preserve the simplicity of the model, temporal derivatives were excluded.
- **High-Pass Filtering:** To account for low-frequency scanner drift and physiological noise, a high-pass filter with a 100-second cutoff was applied. FEAT mathematically achieves this filtration by integrating a series of cosine basis functions directly into the GLM's design matrix (X).
- **Statistical Contrasts and COPEs:** To directly evaluate our experimental hypotheses, we specified three contrast vectors (c). FSL formally refers to the statistical maps generated by these tests as Contrasts of Parameter

Estimates (COPEs).

- **Contrast 1 (Incongruent vs. Baseline):** Defined by the vector $[1, 0]$, this evaluates whether the BOLD response to incongruent trials is significantly positive ($\beta_{inc} > 0$).
- **Contrast 2 (Congruent vs. Baseline):** Defined by the vector $[0, 1]$, this tests if the brain's response to congruent trials is significantly greater than zero ($\beta_{cong} > 0$).
- **Contrast 3 (Incongruent > Congruent):** Defined by the vector $[1, -1]$, this isolates the neural cost of conflict resolution by testing whether the activation during incongruent trials significantly exceeds that of congruent trials ($\beta_{inc} - \beta_{cong} > 0$).



(a) Design matrix (X) for sub-04 run 1.



(b) Design covariance matrix for sub-04 run 1.

Figure 27: Visual representations of the General Linear Model generated by FSL FEAT. (a) Time flows top-to-bottom. Columns left-to-right show: high-pass filter basis functions, convolved regressor for incongruent trials (EV1), and congruent trials (EV2). (b) This evaluates experimental efficiency. Darker off-diagonal elements indicate low correlation (collinearity) between EVs, allowing the GLM to cleanly separate the beta weights.

9.9 Understanding .feat Output Files

Comprehending the architecture of the FSL output directory is essential, as these files represent the direct, spatial realization of our GLM equations converted into three-dimensional neuroimaging maps. The primary .feat root directory contains several foundational files:

- **mask.nii.gz:** A binary 3D image defining the boundaries of the brain (encoded as 1s for brain tissue and 0s for external space). FSL utilizes this mask to restrict statistical computations strictly to intracranial voxels, thereby optimizing processing time by ignoring the empty space surrounding the head.

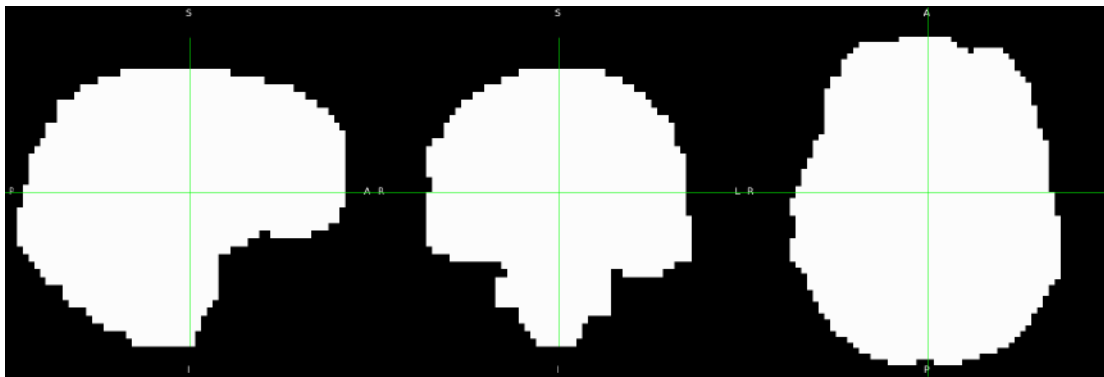


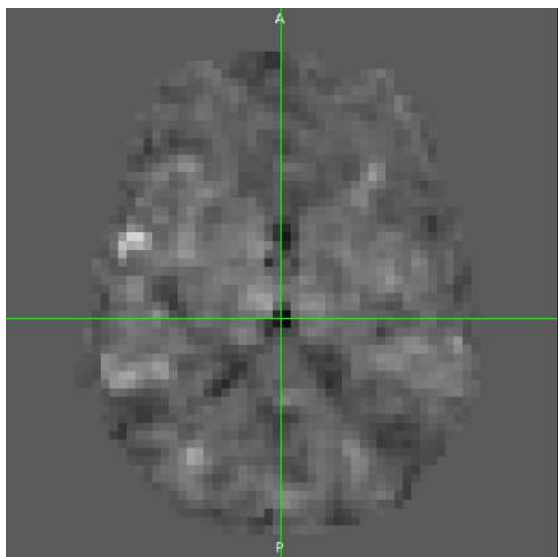
Figure 28: mask.nii.gz of sub-04 run1

- `design.mat` & `design.con`: Text-based representations of our exact design matrix (X) and the specified contrast vectors (c).

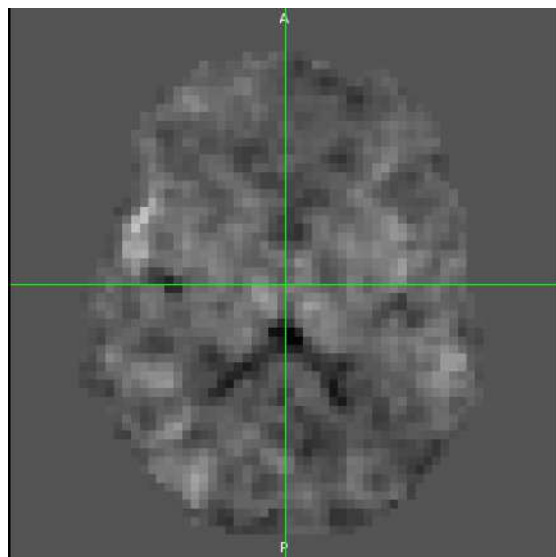
9.9.1 The stats/: The Mathematical Engine

The most critical outputs are housed within the `stats/` subdirectory. Exploring this folder provides a transparent, step-by-step view of FSL's underlying statistical computations. The software effectively deconstructs the core GLM framework ($Y = X\beta + \epsilon$) into individual 3D NIfTI files, with each file representing a specific variable from the equation:

1. `pe*.nii.gz` (**Parameter Estimates**): These images store the raw $\hat{\beta}$ weights computed for every Explanatory Variable (EV) defined in the design matrix. For instance, `pe1` captures the raw modeled signal amplitude for the incongruent condition ($\hat{\beta}_{inc}$), while `pe2` captures the same for the congruent condition ($\hat{\beta}_{cong}$).



(a) PE 1 (Incongruent)



(b) PE 2 (Congruent)

Figure 29: Parameter Estimates (pe). These images represent the raw $\hat{\beta}$ weights for sub-04 run 1

2. `cope*.nii.gz` (**Contrast of Parameter Estimates**): These files embody the calculated effect sizes ($\hat{c}$) for our specific experimental hypotheses. For Contrasts 1 and 2, the COPE is functionally identical to the individual parameter estimate (pe) of that condition. For Contrast 3, it reflects the mathematical subtraction of the estimates ($\beta_{inc} - \beta_{cong}$). Essentially, the COPE maps the raw amplitude of the targeted neural response.

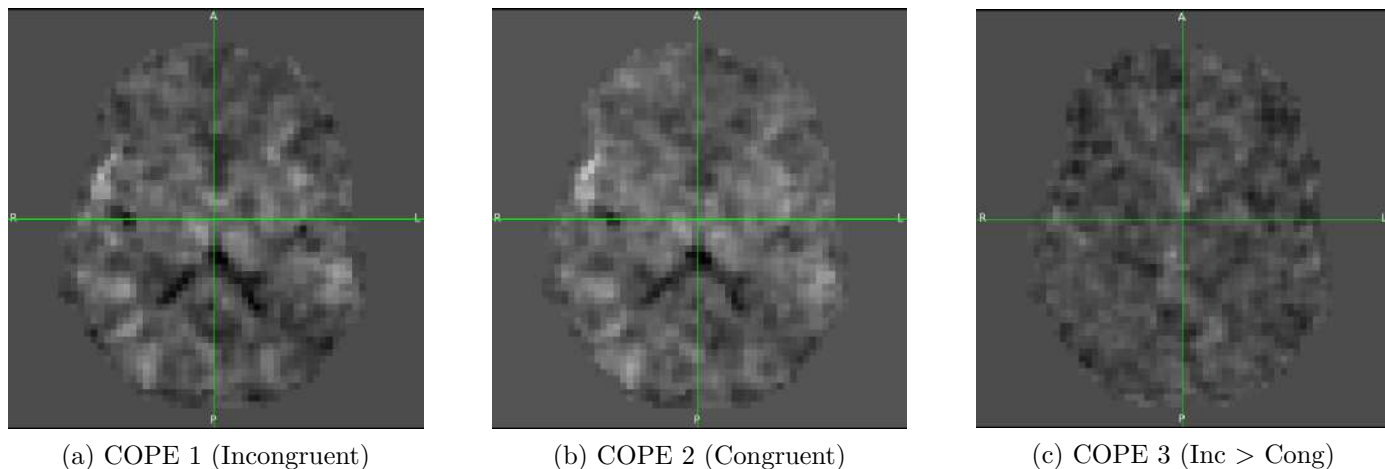


Figure 30: Raw Effect Size Maps (COPEs) for sub-04 run 1

3. **tstat*.nii.gz (t-statistic)**: This map represents the signal-to-noise ratio at every voxel. FSL generates this by dividing the isolated effect size by the square root of its variance (the standard error):

$$t = \hat{c} / \sqrt{\text{Var}(\hat{c})}$$

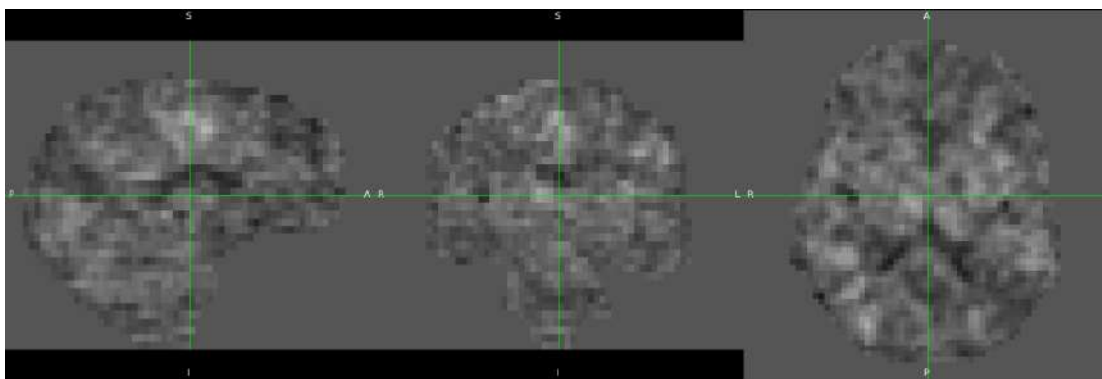


Figure 31: T-Statistic Map (tstat3) for sub-04 run1

4. **zstat*.nii.gz (Z-statistic)**: Because the shape of a t-distribution relies heavily on the scan's degrees of freedom, FSL transforms the **tstat** maps into standardized Z-scores (conforming to a standard normal distribution). This standardization ensures statistical uniformity; a Z-score of 3.1 will reliably correspond to a p-value of approximately 0.001, independent of the number of functional volumes acquired during the scan.

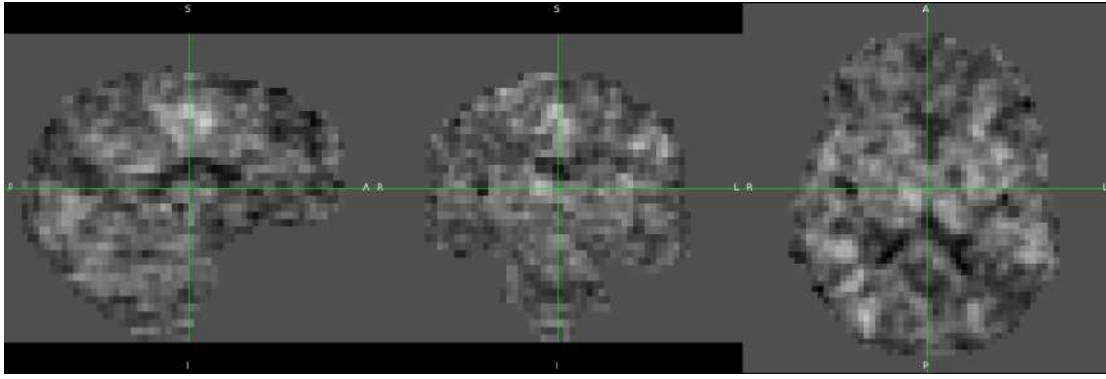


Figure 32: Z-Statistic Map (zstat3) for sub-04 run1

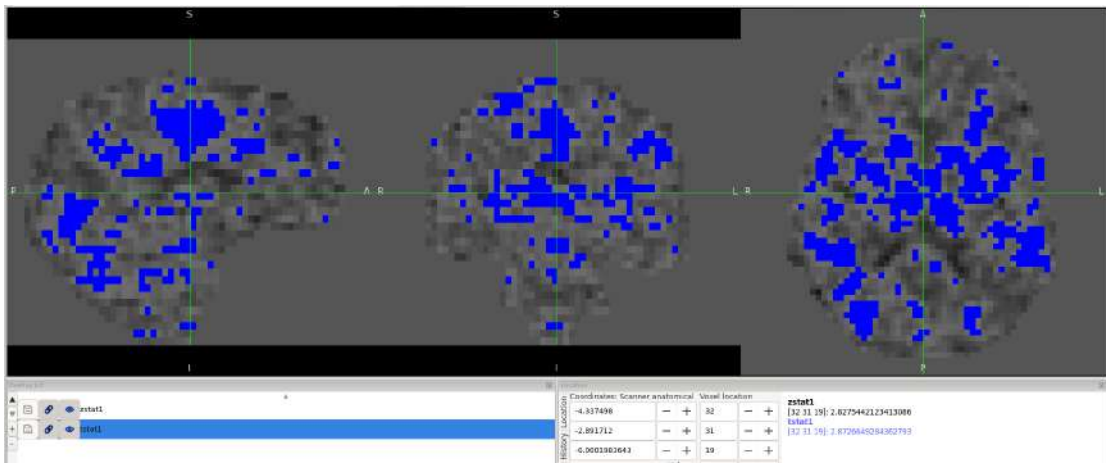


Figure 33: Over layering Z-Statistic Map on T-Statistic Map for sub-04 run1

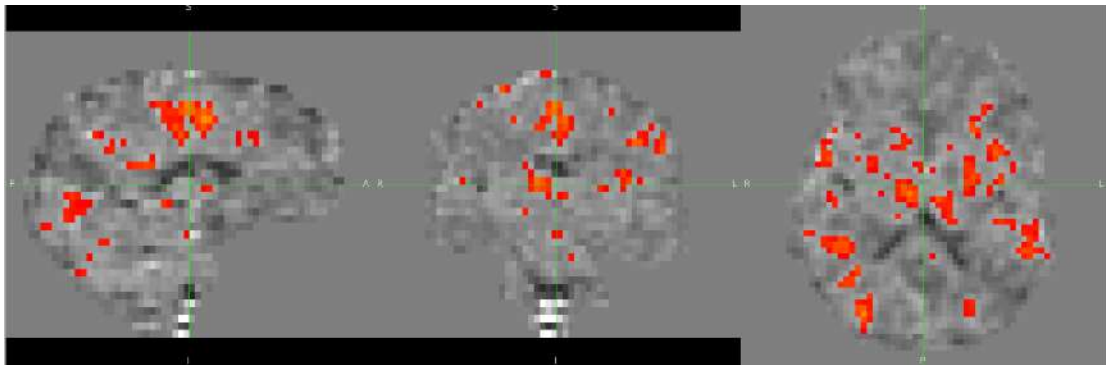


Figure 34: Over layering Z-Statistic Map on PE 1 (Incongruent) for sub-04 run1

5. `res4d.nii.gz` (**Residuals**): Representing the noise or error term (ϵ) of the GLM, this 4D dataset contains the unexplained BOLD time-series variance that remains after the model's predicted responses have been subtracted from the raw data.
6. `sigmasquareds.nii.gz` ($\hat{\sigma}^2$): This file maps the estimated variance of the model's residuals. It provides a spatial

summary of the noise contained within the `res4d` file, which is a mathematically strictly required component for deriving the standard error in subsequent steps.

7. `varcope*.nii.gz` (**Variance of the COPE**): This image quantifies the statistical uncertainty surrounding our effect size estimate, mathematically expressed as $Var(\hat{c})$. It is derived using both the design matrix and the residual variance (`sigmasquareds`).
8. `smoothness` & `logfile`: These are supporting background text documents. The `logfile` tracks the underlying mathematical computations executed by the software, whereas the `smoothness` file records the data's estimated spatial smoothness—a metric FSL relies upon later to perform cluster-based corrections for multiple comparisons.

9.9.2 Finalizing the Output: Clusters and Thresholded Maps

Once the Z-statistic maps are generated within the `stats/` directory, the software returns to the primary `.feat` root folder. Here, it applies necessary corrections for the multiple comparisons problem and produces the finalized files intended for visual interpretation:

- `cluster_mask_zstat*.nii.gz`: Generated alongside the thresholded maps, this utility image assigns a distinct integer identifier (1, 2, 3, etc.) to each surviving significant cluster. This numerical indexing simplifies the subsequent extraction of quantitative data from specific Regions of Interest (ROIs).

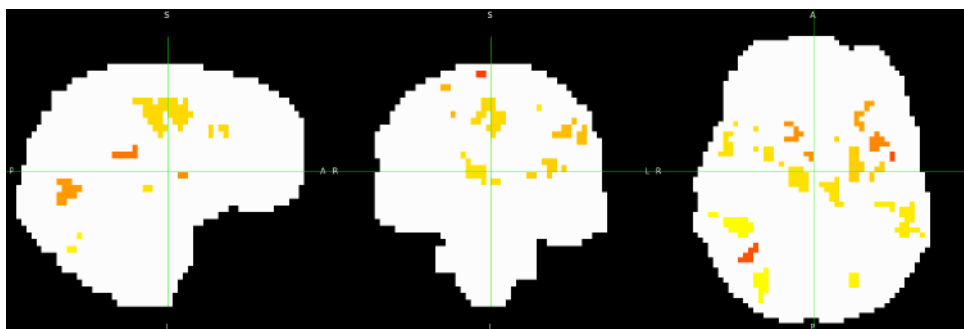


Figure 35: Cluster Mask Map for sub-04 run1

- `thresh_zstat*.nii.gz`: These represent the final, statistically thresholded brain maps. FSL subjects the raw `zstat` images to a stringent two-step cluster correction (e.g., an initial voxel-level threshold of $Z > 3.1$, followed by a cluster-level significance of $p < 0.05$). Any voxel failing to meet these strict statistical criteria is forced to zero, retaining only the significant clusters (or "blobs") of true neural activation. Notably, FSL relies entirely on these thresholded Z-statistic images to convey significance, bypassing the need to output maps of raw p-values.

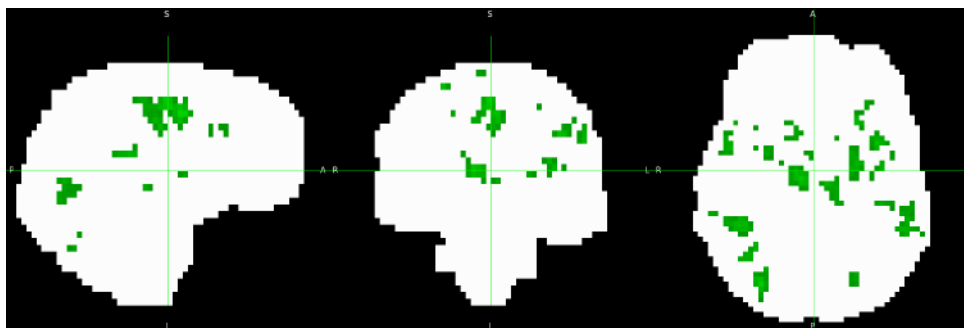


Figure 36: Over layering Z-Statistic Map on T-Statistic Map for sub-04 run1

9.9.3 Pre-Rendered Statistical Overlays (rendered_thresh_zstat*.nii.gz)

To eliminate the need for manually co-registering and overlaying thresholded statistical maps onto anatomical underlays, FSL's FEAT pipeline automatically generates pre-rendered image files. These **rendered_thresh_zstat** outputs feature the significant activation clusters permanently "baked" onto a background structural image, such as a high-resolution T1 scan or the standard MNI template, allowing for immediate visual inspection.

The color intensity within these superimposed clusters maps directly to the underlying statistical Z-score of each voxel. Utilizing a standard Red-Yellow gradient, deep red signifies voxels that barely clear the minimum statistical threshold (e.g., $Z \approx 3.1$), whereas bright yellow denotes regions exhibiting highly robust, intensely significant neural activation.

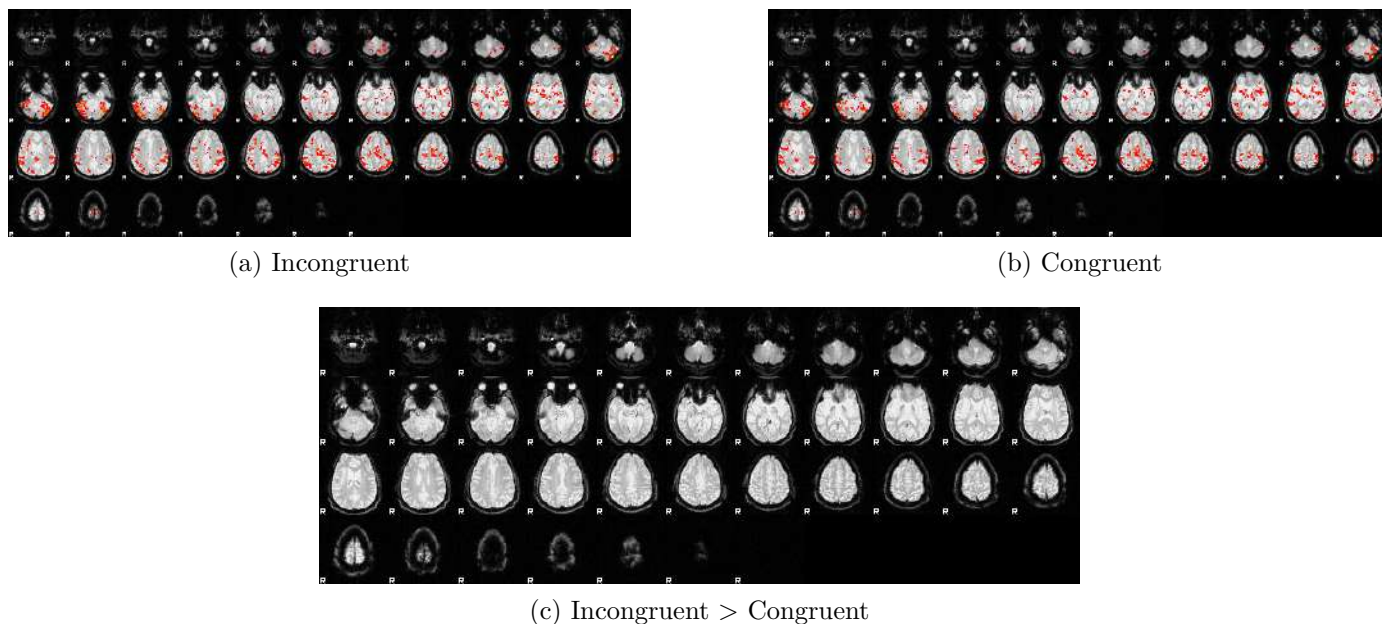


Figure 37: Pre-Rendered Activation Maps. These figures display the statistically significant voxel clusters (utilizing a voxel-level threshold of $Z > 3.1$ and a cluster-forming threshold of $p < 0.05$) automatically superimposed onto a structural brain image to facilitate clear spatial localization.

9.10 Temporal Signal Visualization (tsplot/)

While the rendered anatomical maps illustrate the spatial distribution of brain activity (the **where**), the outputs housed within the **tsplot/** directory depict the temporal dynamics of that signal (the **how**). This folder generates detailed graphical plots that map the theoretical GLM prediction against the actual, recorded BOLD signal, specifically extracted from the peak active voxel within the significant clusters.

Examining these plots serves as a vital quality-assurance mechanism. It allows researchers to visually verify that the GLM equations accurately modeled a genuine physiological hemodynamic response, ensuring the software did not erroneously fit the data to a sudden motion artifact or random noise spike.

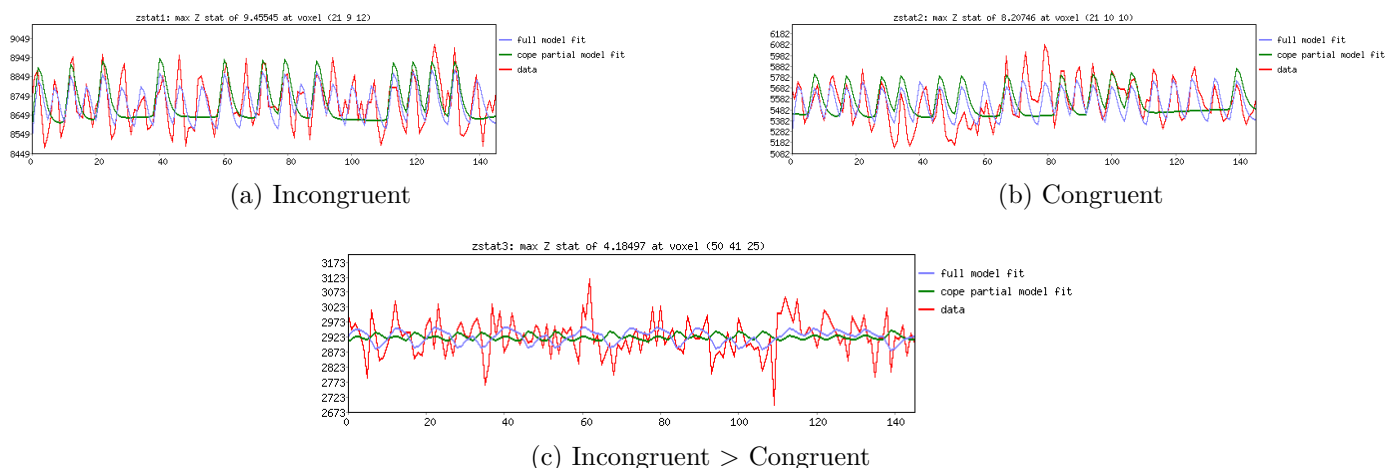


Figure 38: Time Series Plots. Derived from the most statistically significant voxel within each respective contrast, these graphs illustrate the model’s goodness-of-fit. They offer a direct visual comparison between the theoretical BOLD trajectory predicted by the GLM (typically represented by a solid red line) and the raw, observed data points (typically represented in blue) across the entire duration of the functional run.

10 Scripting

10.1 Configuration and Reproducibility: The design.fsf File

In FSL, every parameter selected within the FEAT Graphical User Interface (GUI) is permanently saved into a plain-text configuration script known as the FEAT Setup File (design.fsf). Understanding this file is critical for methodological transparency, as it contains the exact mathematical and computational instructions used to execute the single-subject analysis for sub-04 (Run 1). By reviewing and saving this script, researchers can rapidly replicate the analysis or apply the exact same GLM parameters across hundreds of subjects using batch processing. Below is a detailed deconstruction of the parameters defined in our model.

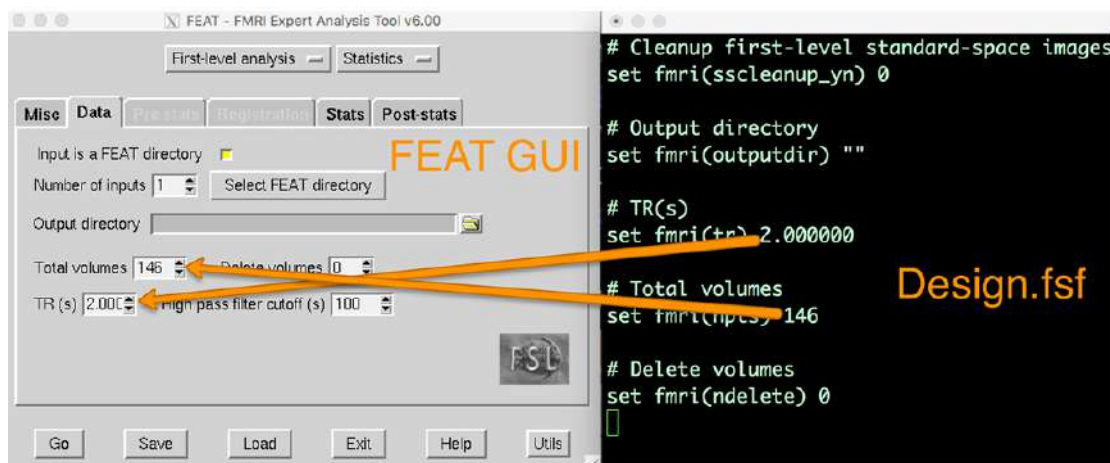


Figure 39: Design.fsf file for sub-04 run 1. This text-based configuration script contains the exact parameters and settings used to execute the first-level GLM analysis in FSL FEAT. Each line corresponds to a specific aspect of the preprocessing and statistical modeling pipeline, ensuring complete reproducibility and transparency of the analysis.

10.1.1 Global System and Analysis Scope Parameters

The initial block of the script defines the computational environment and the scope of the analysis.

- **Version and Environment:** `fmri(version) 6.00` and `fmri(inmelodic) 0` log the specific version of FSL being used and confirm that the software is running a standard FEAT analysis rather than an Independent Component Analysis (ICA/MELODIC).
 - `fmri(help_yn) 1` and `fmri(featwatcher_yn) 1` are GUI utilities that enable balloon help and the Featwatcher progress monitor during processing.
- **Analysis Level:** `fmri(level) 1` explicitly dictates that this is a first-level (single-subject) analysis rather than a higher-level group analysis.
- **Execution Stages:** Crucially, `fmri(analysis) 2` instructs the software to only run the statistical modeling stage. FSL defines '0' as registration only, '1' as pre-processing only, '2' as statistics only, and '7' as a full analysis. Setting this to '2' implies that the input data (defined later as `feat_files(1) /.../sub04-run1.feat`) has already undergone preprocessing.

10.1.2 Functional Data Acquisition Parameters

This section directly maps the scanner's physical acquisition settings to the software so it can accurately calculate the time-series math.

- **Timing and Length:** `fmri(tr) 2.000000` defines the Repetition Time (TR) as exactly 2 seconds, which is the sampling rate of the scanner. `fmri(npts) 146` tells the software to expect exactly 146 3D volumes in this run.
- **Volume Deletion:** `fmri(ndelete) 0` indicates that no initial volumes are being discarded. Often, the first few "dummy" scans are deleted to allow the magnetic field to reach a steady state, but here it is set to zero, meaning all 146 volumes are utilized.
- **Total Voxels:** `fmri(totalVoxels) 23920640` is a system-calculated variable representing the absolute total number of data points (spatial voxels $\times$ timepoints) the model must process.

10.1.3 Pre-Statistics (Preprocessing) Configuration

Even though the pipeline is set to execute only the statistics (analysis 2), the `design.fsf` meticulously logs the preprocessing parameters to maintain a complete record of the data's history.

- **Motion and Slice Timing:** `fmri(mc) 1` confirms that MCFLIRT motion correction was applied to realign the subject's head over time. `fmri(st) 1` indicates regular sequential slice-timing correction (acquiring slices from bottom to top, 0, 1, 2...).
- **Spatial and Temporal Filtering:** `fmri(smooth) 5` dictates a spatial smoothing kernel of 5mm Full-Width at Half-Maximum (FWHM), which blurs the data slightly to increase the signal-to-noise ratio. `fmri(tempph_py) 1` and `fmri(paradigm_hp) 100` enable a High-Pass temporal filter with a 100-second cutoff, which is critical for mathematically removing slow, low-frequency scanner drift from the baseline signal.
- **Brain Extraction:** `fmri(bet_yn) 1` with a `fmri(brain_thresh) 10` confirms that the Brain Extraction Tool (BET) was used to strip the skull and non-brain tissue from the image, utilizing a conservative 10% threshold to ensure no peripheral cortex was accidentally removed.

10.1.4 The General Linear Model (GLM) and Explanatory Variables (EVs)

This section is the mathematical engine of the hypothesis test, defining how the task design is transformed into the design matrix (X).

- **Statistical Toggles:** `fmri(stats_yn) 1` confirms the GLM will be executed. `fmri(prewhiten_yn) 1` is a vital command that runs FSL's FILM algorithm to correct for temporal autocorrelation (the fact that brain activity at TR 2 is inherently correlated with activity at TR 1), ensuring our p-values are not artificially inflated.
- **Defining the Regressors:** `fmri(eps_orig) 2` instructs the GLM to construct two primary Explanatory Variables (EVs).
 - **EV 1 (Incongruent):** `fmri(evttitle1) "incongruent"` names the condition. `fmri(shape1) 3` dictates that the software should read a custom 3-column text file containing the exact behavioral trial onsets (`custom1`). To model the physiological delay of blood flow, `fmri(convolve1) 2` instructs the software to convolve these exact trial times with a standard Gamma Hemodynamic Response Function (HRF), utilizing a spatial sigma of 3 (`gamma_sigma1 3`) and a peak delay of 6 seconds (`gamma_delay1 6`). `fmri(tempfilt_yn1) 1` ensures this theoretical model is high-pass filtered exactly like the raw data, maintaining mathematical symmetry. `fmri(deriv_yn1) 0` turns off temporal derivatives to keep the model strictly basic.
 - **EV 2 (Congruent):** The exact same parameters are mirrored for EV 2, referencing the `congruent_run1.txt` timing file. Orthogonalization variables (e.g., `fmri(ortho1.0)`) are set to 0, ensuring the GLM treats the congruent and incongruent trials as entirely independent variables without forcing them to mathematically compete for variance.

10.1.5 Statistical Contrasts and Hypothesis Testing

Once the beta weights for the EVs are estimated, the script defines how to mathematically compare them using Contrasts of Parameter Estimates (COPEs).

- **Contrast Setup:** `fmri(ncon_orig) 3` tells the model to run three distinct statistical tests.
- **Contrast 1 (Incongruent vs. Baseline):** Defined mathematically by the vectors `fmri(con_real1.1) 1` and `fmri(con_real1.2) 0`. This translates to the vector $[1, 0]$, testing where brain activity during incongruent trials is significantly greater than zero.
- **Contrast 2 (Congruent vs. Baseline):** Defined by vectors 0 and 1.0 ($[0, 1]$), isolating positive activation during congruent trials.
- **Contrast 3 (Incongruent > Congruent):** The primary hypothesis test. It is defined by `con_real3.1) 1.0` and `con_real3.2) -1.0` ($[1, -1]$). This subtraction effectively subtracts the neural baseline of the congruent task out of the incongruent task, mapping the isolated cognitive cost of conflict resolution.

10.1.6 Thresholding and Post-Statistics

This block dictates how the software separates true neural signals from random statistical noise using Multiple Comparisons Correction.

- **Thresholding Methodology:** `fmri(thresh) 3` instructs FSL to use Cluster-based thresholding (based on Gaussian Random Field theory), which is the standard for fMRI.
- **Statistical Limits:** `fmri(z_thresh) 3.1` sets the initial voxel-forming threshold (meaning a voxel must have a Z-score > 3.1 to even be considered). `fmri(prob_thresh) 0.05` dictates that the resulting spatial clusters of activation must have a family-wise error probability of $p < 0.05$ to survive and be rendered.

- **Note on Execution:** Interestingly, the script reads `fmri(poststats_yn)` 0. This indicates that while the thresholding parameters are strictly defined and saved, the automated post-stats execution was toggled off for this specific computational run, likely to be executed manually in a subsequent step.

10.1.7 Spatial Registration to Standard Space

The final relevant block configures how the individual subject's functional data will be warped to fit standard human anatomy.

- **Standard Space Target:** `fmri(regstandard_yn)` 1 enables standard registration, setting the target template to the MNI152 2mm brain (`fmri(regstandard)` `"../MNI152_T1_2mm_brain"`).
- **Transformation Parameters:** `fmri(regstandard_search)` 180 tells the alignment algorithm to perform a full 180-degree mathematical search space to find the optimal anatomical fit. `fmri(regstandard_dof)` 12 confirms that a 12 Degrees of Freedom (affine) transformation will be used, allowing the software to translate, rotate, scale, and shear the subject's brain to perfectly match the MNI template for future group-level analysis.

10.2 Automating the Pipeline: Custom Batch Processing Across Subjects and Runs

While FSL's FEAT Graphical User Interface (GUI) is highly effective for establishing the initial mathematical model, configuring the parameters manually for two separate runs across 26 participants (a total of 52 individual analyses) is computationally inefficient and highly susceptible to human error.

To overcome this, a prerequisite manual analysis was first established. The pipeline was rigorously configured and successfully executed for a single session (sub-04, Run 1) via the GUI. FSL automatically saved this confirmed configuration into a plain-text file (`design_sub04_run1.fsf`). This file was then utilized as the "master template" within a custom Bash shell script to automate the entire first-level dataset processing.

Below is a detailed deconstruction of the custom script engineered for this study:

10.2.1 Environment Setup and Nested Looping

```

1 #!/bin/bash
2
3 # Navigate to your dataset folder
4 cd /mnt/d/Third-Year/Second-Term/Flanker-task-dataset/ds000102-download/
5
6 # Loop through all subjects 01 to 26
7 for id in {01..26}; do
8     subj="sub- $\{id\}$ "
9
10    # Loop through run 1 and run 2
11    for run in 1 2; do
12
13        echo "===== "
14        echo " Setting up FEAT for  $\{subj\}$  Run  $\{run\}$ ..."
15        echo "===== "
```

Listing 1: Environment setup and nested looping structure

- **Initialization and Navigation:** The script initializes the Bash environment and systematically navigates to the root directory containing the BIDS-formatted Flanker dataset.

- **The Outer Subject Loop:** The command `for id in {01..26}` utilizes Bash brace expansion to generate a zero-padded sequential array (01, 02... 26). This assigns the current participant ID to the variable `${subj}` (e.g., `sub-05`).
- **The Inner Run Loop:** Because the experimental design features two separate functional runs per subject, a nested loop (`for run in 1 2`) is introduced. This ensures that for every individual subject, the script will independently iterate through and configure both Run 1 and Run 2 before moving on to the next participant. Status echoes are printed to the terminal to allow for real-time monitoring of the script's progression.

10.2.2 Dynamic Parameter Substitution (sed)

```

1 # This command creates a new .fsf file for the specific subject and run.
2 sed -e "s/sub-04/${subj}/g" \
3     -e "s/run-1/run-${run}/g" \
4     -e "s/run1/run${run}/g" \
5     design_sub04_run1.fsf > design_${subj}_run${run}.fsf

```

Listing 2: Dynamic parameter substitution with sed

- **The Stream Editor Engine:** This block represents the core automation mechanism. The `sed` command takes the manually validated master template (`design_sub04_run1.fsf`) and executes a series of "search and replace" operations to dynamically customize the pathways for the active loop iteration.
- **Targeted String Replacement (-e flags):**
 - `"s/sub-04/${subj}/g"`: Locates all instances of the template subject (`sub-04`) and replaces them with the current loop's subject ID (e.g., `sub-05`).
 - `"s/run-1/run-${run}/g"`: Replaces the BOLD NIfTI file naming convention. If the script is on Run 2, it updates the functional data pathway from `run-1` to `run-2`.
 - `"s/run1/run${run}/g"`: Replaces the timing file naming convention. Because the onset `.txt` files lack the hyphen found in the BOLD data (e.g., `incongruent_run1.txt`), this specific expression ensures the behavioral pathways are accurately updated to `run2`.
- **Output Redirection (>):** Rather than overwriting the master template, the `>` operator pipes the modified text into a brand-new configuration file uniquely named for the current iteration (e.g., `design_sub-05_run2.fsf`).

10.2.3 Execution and Pipeline Completion

```

1 # Run FSL FEAT using the newly generated file
2 feat design_${subj}_run${run}.fsf
3
4     done
5 done
6
7 echo "ALL SUBJECTS AND RUNS ARE COMPLETELY FINISHED!"

```

Listing 3: FEAT execution and loop completion

- **Headless FEAT Execution:** With the newly generated, subject-specific `.fsf` file correctly pointing to the appropriate structural, functional, and behavioral data, the `feat` command is called. This triggers FSL's mathematical modeling engine to run the GLM pipeline entirely in the background (headless), bypassing the GUI.

- **Loop Termination:** The dual `done` commands sequentially close the inner run loop and the outer subject loop. Once the script systematically parses through all 52 runs, it prints a final confirmation message to the terminal, indicating that the dataset is fully prepared for Second-Level (Group) analysis.

10.2.4 Complete Batch Processing Script (Nested Loops)

Below is the complete, consolidated bash script with nested loops for processing all subjects and runs:

```

1  #!/bin/bash
2
3  # =====
4  # Batch processing script for NYU Slow Flanker dataset
5  # Processes all subjects (01-26) and both runs (1-2)
6  # =====
7
8  # Navigate to your dataset folder
9  cd /mnt/d/Third-Year/Second-Term/Flanker-task-dataset/ds000102-download/
10
11 # Loop through all subjects 01 to 26
12 for id in {01..26}; do
13     subj="sub-01"
14
15     # Loop through run 1 and run 2
16     for run in 1 2; do
17
18         echo "===== "
19         echo " Setting up FEAT for ${subj} Run ${run}..."
20         echo "===== "
21
22         # Create subject-specific design file from master template
23         sed -e "s/sub-04/${subj}/g" \
24             -e "s/run-1/run-${run}/g" \
25             -e "s/run1/run${run}/g" \
26             design_sub04_run1.fsf > design_${subj}_run${run}.fsf
27
28         # Run FSL FEAT using the newly generated file
29         echo "===> Running FEAT for ${subj} Run ${run}"
30         feat design_${subj}_run${run}.fsf
31
32         echo "===> Completed FEAT for ${subj} Run ${run}"
33         echo ""
34     done
35 done
36
37
38 echo "===== "
39 echo "ALL SUBJECTS AND RUNS ARE COMPLETELY FINISHED!"
40 echo "Total analyses completed: 52 (26 subjects \times 2 runs)"
41 echo "===== "
```

Listing 4: Complete batch processing script with nested loops for all subjects and runs

10.2.5 Required Software and Dependencies

To successfully execute this batch processing script, the following software and libraries must be installed on the system:

- **FSL (FMRIB Software Library):** Version 6.0 or higher. The script specifically uses:
 - **feat** - FMRI Expert Analysis Tool for first-level GLM
- **Bash Shell:** Unix/Linux environment (macOS terminal, Linux terminal, or Windows Subsystem for Linux)
- **Standard Unix Utilities:**
 - **sed** - Stream editor for text replacement
 - **echo** - For terminal output
 - **cd** - For directory navigation

10.2.6 Setup Instructions

Before running the script, ensure the following directory structure and files are in place:

```
ds000102-download/
|-- design_sub04_run1.fsf          # Master template (based on sub-04, Run 1)
|-- sub-01/
|   |-- anat/
|   |   '-- sub-01_T1w.nii.gz
|   '-- func/
|       |-- sub-01_task-flanker_run-1_bold.nii.gz
|       |-- sub-01_task-flanker_run-2_bold.nii.gz
|       |-- incongruent_run1.txt
|       |-- incongruent_run2.txt
|       |-- congruent_run1.txt
|       '-- congruent_run2.txt
|-- sub-02/
|   '-- ...
'-- (sub-03 through sub-26)
```

10.2.7 Executing the Script

Save the script as `run_batch_nested.sh` and execute it from the terminal:

```
1 # Make the script executable
2 chmod +x run_batch_nested.sh
3
4 # Run the script
5 ./run_batch_nested.sh
```

11 Second Level fMRI Analysis: Within-Subject Aggregation

Following the preprocessing and 1st-level analysis of raw fMRI time-series data, neuroimaging pipelines must aggregate the data to draw meaningful conclusions. In the FMRIB Software Library (FSL), a 2nd-level analysis is specifically defined as the process of averaging together the parameter estimates and contrast estimates from the 1st-level analyses

within each individual subject. Because subjects frequently undergo multiple scanning sessions or "runs" to increase statistical power, this intermediate step is required to condense each subject's multiple runs into a single, comprehensive statistical map before proceeding to group-level (population) inferences.

11.1 Computational Implementation and Data Management

- **Initialization:** The 2nd-level workflow is initiated via FSL's FEAT graphical user interface (GUI) by selecting "Higher-Level Analysis" from the primary dropdown menu. Engineers can choose to input either lower-level FEAT directories directly or specific 3D cope (Contrast of Parameter Estimates) images. Directly selecting 3D cope images provides greater architectural flexibility if the initial data was not processed using FSL's default directory structure.
- **Automated Data Routing:** In large neuroimaging cohorts, manual data selection is highly inefficient and prone to human error. For example, a study with 26 subjects completing 2 runs each yields 52 distinct FEAT directories. Instead of manual selection, engineers utilize terminal command-line scripting with wildcards (e.g., `ls -d $PWD/sub-??/run*`) to automatically generate absolute file paths. This ensures exact, error-free routing of the 52 directory inputs into the FEAT pipeline.

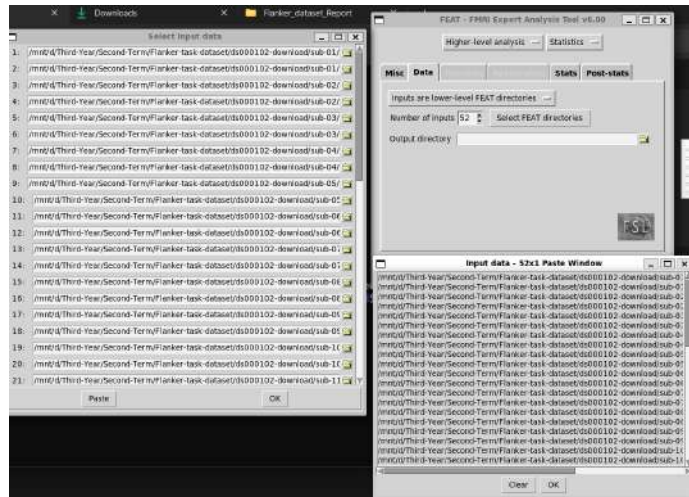


Figure 40: Example of terminal command-line scripting to automate data routing for 2nd-level analysis.

11.2 Lower-Level Copes and Contrast Estimates

During this stage, the engineer selects which specific 1st-level statistical outcomes (copes) to propagate forward. For example, in a standard cognitive task, the pipeline can independently analyze:

1. The isolated contrast estimate for "Condition A" (e.g., an Incongruent stimulus).
2. The isolated contrast estimate for "Condition B" (e.g., a Congruent stimulus).
3. The mathematical subtraction of the two parameter estimates (e.g., Incongruent minus Congruent) to isolate specific functional networks.

11.3 Statistical Models for 2nd-Level Analysis

FSL offers several statistical models for 2nd-level analysis, each with distinct assumptions and applications:

- **Fixed Effects:**

- This model does not attempt to generalize findings from your sample to the wider population; it simply calculates the average.
- It is the appropriate choice when you simply want to take the average of the parameter estimates across the different runs within each individual subject.

- **Mixed Effects: Simple OLS (Ordinary Least Squares):**

- This approach performs a t-test on the average parameter estimates calculated for each subject.
- Critically, it does not take into account the variability that occurs between the runs for each individual subject.

- **Mixed Effects: FLAME 1:**

- This sophisticated model weights each subject's parameter estimate by the variance of that specific contrast estimate.
- In practice, this means a subject with relatively low variance will be weighted more heavily, while a subject with relatively high variance (noisier data) will be weighted less.

- **Mixed Effects: FLAME 1+2:**

- This is a more rigorous and computationally demanding version of the FLAME 1 model.
- Because it takes much longer to process, it is typically only helpful for analyzing small samples, such as those consisting of 10 subjects or fewer.

- **Randomise:**

- This option utilizes a non-parametric statistical test.
- Its specific applications and mechanics are generally reserved for later, more advanced stages of neuroimaging lectures.

11.4 General Linear Model (GLM) Configuration for Within-Subject Averaging

To execute the Fixed Effects averaging across runs, the engineer must manually configure the General Linear Model (GLM) matrix:

- **The EV Matrix (Explanatory Variables):** In the Full Model Setup, the total number of rows corresponds to the total number of individual parameter estimates (e.g., 52 runs), while the "Number of main EVs" is set to equal the total number of subjects (e.g., 26). The matrix is populated by placing a "1" in the specific column corresponding to the exact runs that belong to a single subject, instructing the algorithm to average them together.

EVs

Contrasts & F-tests

Number of main EVs 26

Number of additional, voxel-dependent EVs 0

Paste

Group	EV1	EV2	EV3	EV4	EV5	EV6	EV7	EV8
Input 1	1	0	0	0	0	0	0	0
Input 2	1	1.0	0	0	0	0	0	0
Input 3	1	0	1.0	0	0	0	0	0
Input 4	1	0	1.0	0	0	0	0	0
Input 5	1	0	0	1.0	0	0	0	0
Input 6	1	0	0	1.0	0	0	0	0
Input 7	1	0	0	0	1.0	0	0	0
Input 8	1	0	0	0	1.0	0	0	0
Input 9	1	0	0	0	0	1.0	0	0
Input 10	1	0	0	0	0	1.0	0	0
Input 11	1	0	0	0	0	0	1.0	0
Input 12	1	0	0	0	0	0	1.0	0
Input 13	1	0	0	0	0	0	0	1.0
Input 14	1	0	0	0	0	0	0	1.0
Input 15	1	0	0	0	0	0	0	0
Input 16	1	0	0	0	0	0	0	1.0

Figure 41: Example of GLM EV matrix configuration for within-subject averaging.

- **Contrasts Generation:** In the Contrasts & F-tests configuration, the number of contrasts is matched to the number of subjects (26), and a diagonal line of "1"s is entered across the matrix. This architecture outputs a single, clean contrast estimate for every individual subject that represents their averaged functional data.

EVs		Contrasts & F-tests									
		Contrasts		26		F-tests		0			
Paste		Title	EV1	EV2	EV3	EV4	EV5	EV6	EV7	EV8	
C1			1	0	0	0	0	0	0	0	
C2			0	1.0	0	0	0	0	0	0	
C3			0	0	1.0	0	0	0	0	0	
C4			0	0	0	1.0	0	0	0	0	
C5			0	0	0	0	1.0	0	0	0	
C6			0	0	0	0	0	1.0	0	0	
C7			0	0	0	0	0	0	1.0	0	
C8			0	0	0	0	0	0	0	1.0	

Figure 42: Example of GLM contrast matrix configuration for within-subject averaging.

- **Post-Stats:** Because this specific 2nd-level pipeline is purely an intra-subject averaging mechanism and not a population inference, post-statistical thresholding is ignored at this stage.

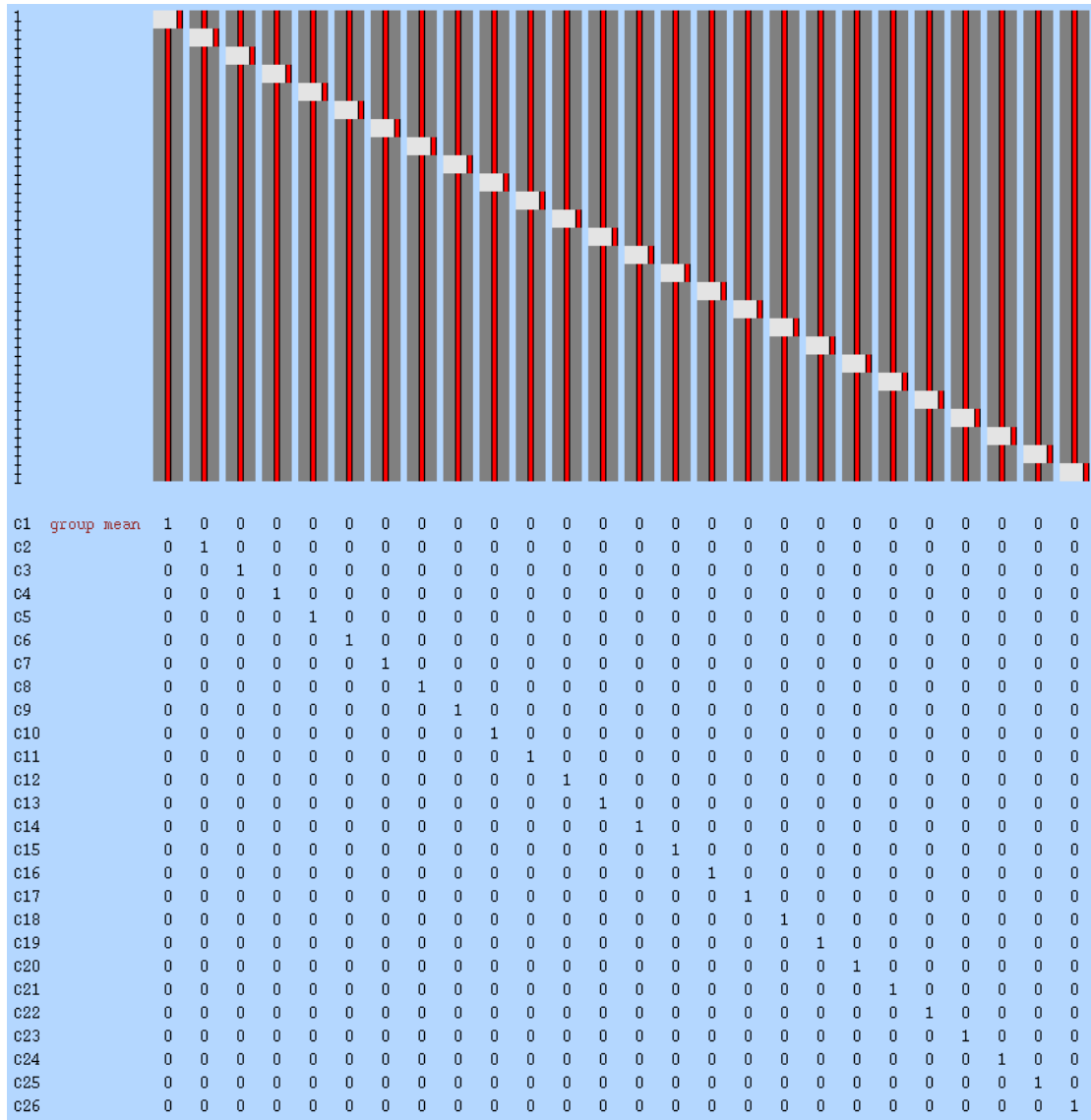
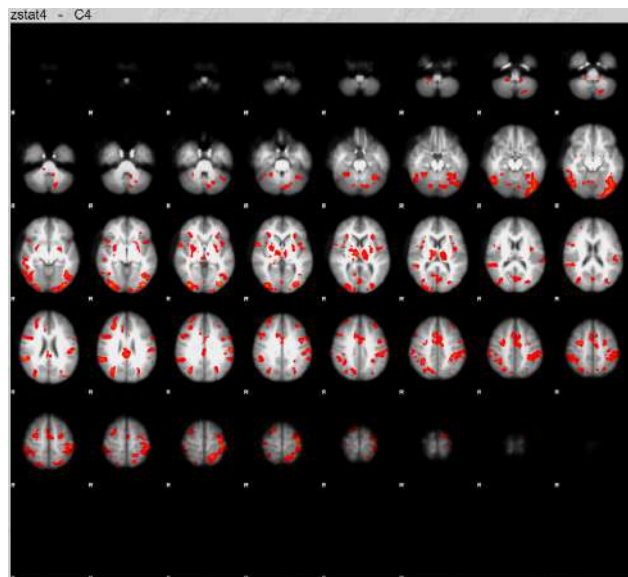
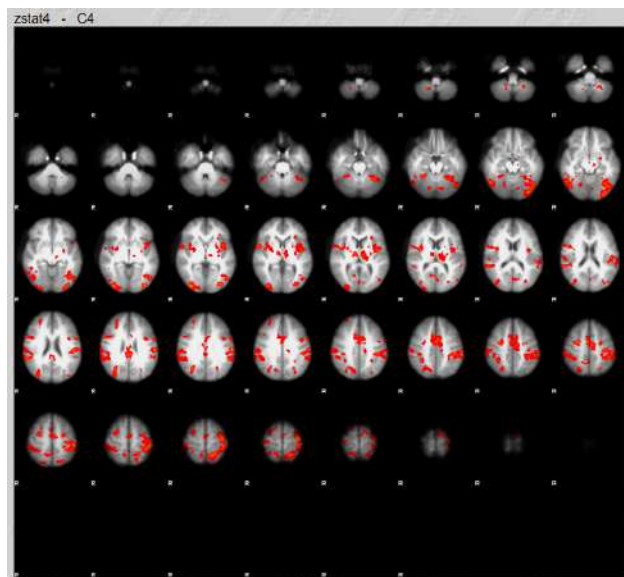


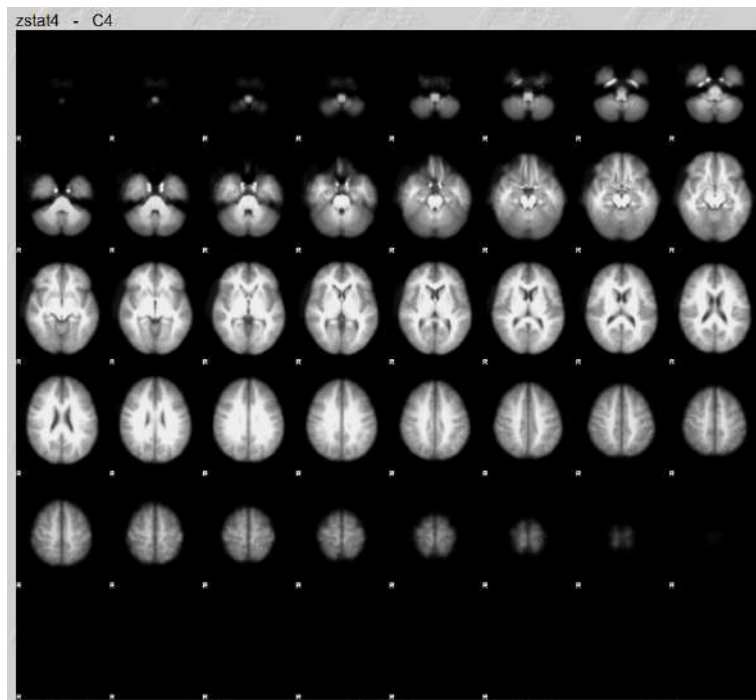
Figure 43: GLM design matrix for 2nd-level within-subject averaging.



(a) Incongruent (COPE1) for Subject 4



(b) Congruent (COPE2) for Subject 4



(c) Incongruent - Congruent (COPE3) for Subject 4

Figure 44: Example of 2nd-level output contrast estimates for Subject 4, showing the averaged functional activation patterns for Incongruent, Congruent, and their subtraction.

12 Third-Level fMRI Analysis: Population Inference and Generalization

While the first- and second-level fMRI analyses focus on intra-subject parameter estimation and multi-run aggregation, the primary objective of a third-level analysis is to generalize the findings from the specific study sample (e.g., $N = 26$ subjects) to the broader population from which the sample was drawn. In the FMRIB Software Library (FSL) framework,

the third-level analysis functions as a group-level evaluation that calculates the mean and standard error for specific contrast estimates across all participants, testing whether this average brain activation is statistically significant.

12.1 Data Inputs and GLM Setup

The direct inputs for a third-level FEAT analysis are the 3D Contrast of Parameter Estimates (COPE) images generated during the second-level analysis. Each subject contributes a single, averaged COPE image representing their functional neural response to a specific condition or mathematical subtraction (e.g., Incongruent minus Congruent stimuli). Because the contrast for each subject has already been isolated, the General Linear Model (GLM) for this stage is established using a "single group average" setup. The design matrix consists of a single Explanatory Variable (EV) representing the group mean, populated entirely by 1s for every subject included in the cohort.

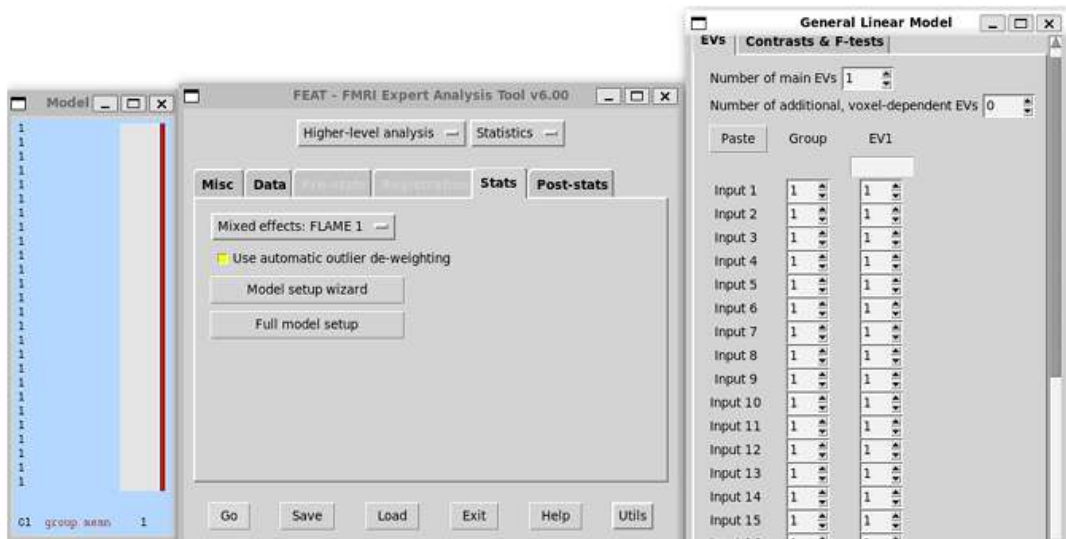


Figure 45: GLM design matrix for 3rd-level group analysis.

12.2 Statistical Inference Models (The Stats Tab)

To accurately generalize results to a population, the statistical model must appropriately handle both the variance within a single subject's scanning session and the variance between different subjects. FSL offers several distinct computational methods for modeling this variance:

- **Fixed Effects:** This model averages the data across subjects but does not factor in inter-subject variance. As a result, inferences drawn from Fixed Effects are restricted specifically to the participants in the study and cannot be generalized to a broader population.
- **Mixed Effects (Simple OLS):** Ordinary Least Squares performs a standard t-test on the average parameter estimates but is considered a less robust Mixed Effects model because of how it handles variance.
- **Mixed Effects (FLAME 1):** FSL's Local Analysis of Mixed Effects (FLAME 1) is the standard for group-level inference. It provides highly accurate parameter estimates by incorporating information regarding both within-subject and between-subject variability. It appropriately weights subjects based on the noisiness of their data.
- **Mixed Effects (FLAME 1+2):** This represents a more rigorous and computationally intensive extension of FLAME 1. While theoretically more accurate, the additional mathematical benefit is typically minimal, and it requires significantly longer processing times, making it best suited for very small clinical samples.

- **Randomise:** Unlike the previous models that rely on standard Gaussian assumptions, Randomise utilizes a non-parametric permutation-based approach. This method is statistically valid and highly effective when the standard assumptions regarding data normality are violated.



Figure 46: FSL’s statistical inference options for 3rd-level group analysis.

12.3 Post-Statistical Thresholding Strategies

Following the generation of group-level Z-statistic maps, rigorous post-statistical thresholding must be applied to determine which voxels or clusters represent true biological signal versus random mathematical noise. The topological approaches evaluated include:

- **None:** No thresholding is applied. This raw output displays the parameter estimate at every single voxel in the brain, regardless of its statistical significance.
- **Uncorrected:** This method utilizes a strict Z-threshold (e.g., $Z > 3.1$) but fails to correct for the multiple comparisons problem inherent in evaluating hundreds of thousands of voxels. It allows any individual voxel passing the threshold to be displayed.
- **Voxel-Based Correction:** This approach employs maximum height thresholding based on Gaussian Random Field (GRF) theory. It corrects for multiple comparisons at the level of individual voxels and is considered mathematically less conservative than standard Bonferroni correction.
- **Cluster-Based Correction:** The standard and most robust method for fMRI inference. It utilizes a two-step process: a Cluster-Defining Threshold (CDT, e.g., $Z = 3.1$) to identify contiguous groupings of significant voxels, followed by a cluster probability threshold (e.g., $p = 0.05$). This method acknowledges that neighboring brain voxels are not independent. FSL runs computational simulations based on the CDT to create a null distribution of expected cluster sizes; any neuroanatomical clusters in the real data that occur less than 5% of the time in these simulations are deemed statistically significant.

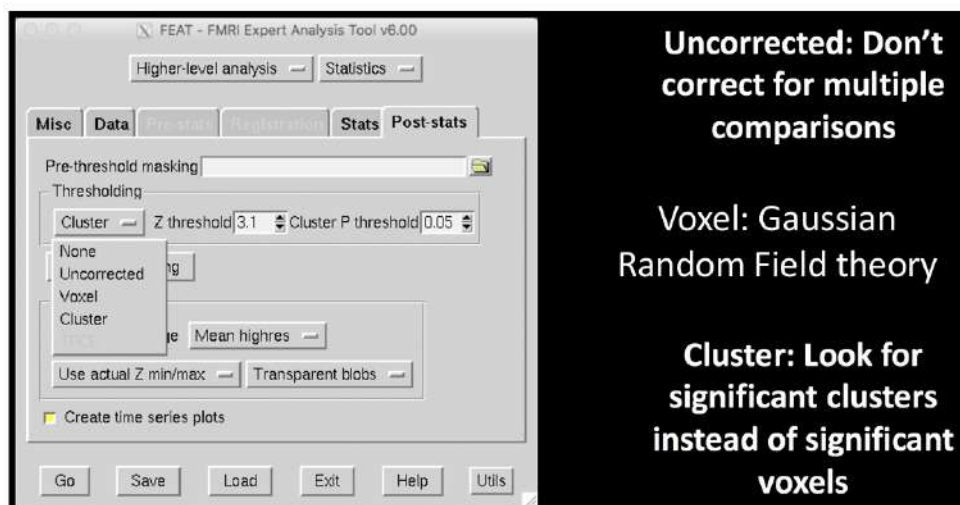


Figure 47: FSL's post-statistical thresholding options for 3rd-level group analysis.

12.4 Computational Implementation: Bash Scripting Automation

To systematically eliminate human error and ensure computational reproducibility, the generation of the diverse statistical and thresholding models was automated. Relying on FSL's GUI for repetitive pipeline adjustments is inefficient for large-scale methodological comparisons; instead, the underlying .fsf (FEAT setup) files were manipulated programmatically.

Automating Statistical Models:

A master design file (`design_level3_cope3_master.fsf`) was established. A Bash script was then engineered to iterate through the five FSL statistical inference codes: 0 (Fixed), 1 (OLS), 2 (FLAME1), 3 (FLAME1+2), and 4 (Randomise). Using the stream editor (`sed`), the script dynamically targeted and replaced the statistical variable `fmri(mixed_yn)` and updated the output directory name to reflect the applied model.

The core operational command within the loop was structured as follows:

```
1 sed -e "s/set fmri(mixed_yn) ./set fmri(mixed_yn) ${stat_code}/g" \
2     -e "s/Flanker_Level3_cope3_master/Flanker_Level3_cope3_${stat_name}/g" \
3     design_level3_cope3_master.fsf > design_level3_cope3_${stat_name}.fsf
```

Listing 5: Bash script for automating statistical model selection

Following the generation of the unique .fsf files, the script automatically launched the FEAT analysis for each model via the terminal command:

```
1 feat design_level3_cope3_${stat_name}.fsf
```

Automating Thresholding Methods:

A similar architectural approach was utilized to compare post-statistical thresholding techniques. Holding the statistical model constant at FLAME 1 (`mixed_yn = 2`), the script looped through the thresholding codes: 0 (None), 1 (Uncorrected), and 2 (Voxel).

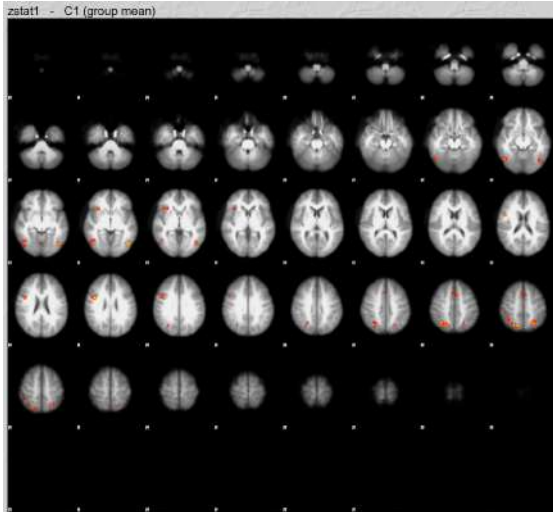
The targeted inline replacement command was:

```
1 sed -e "s/set fmri(thresh) ./set fmri(thresh) ${thresh_code}/g" \
2     -e "s/set fmri(mixed_yn) ./set fmri(mixed_yn) 2/g" \
3     -e "s/Flanker_Level3_cope3_master/Flanker_Level3_cope3_FLAME1_${thresh_name}/g" \
4     design_level3_cope3_master.fsf > design_level3_cope3_FLAME1_${thresh_name}.fsf
```

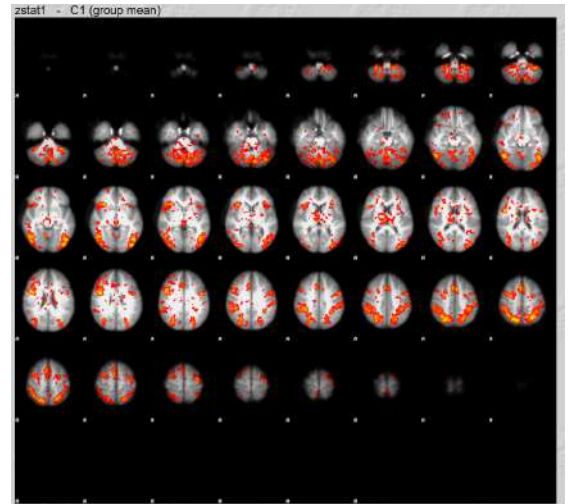
Listing 6: Bash script for automating thresholding methods

12.5 3rd-Level Analysis (COPE 3)– Group Activation Map

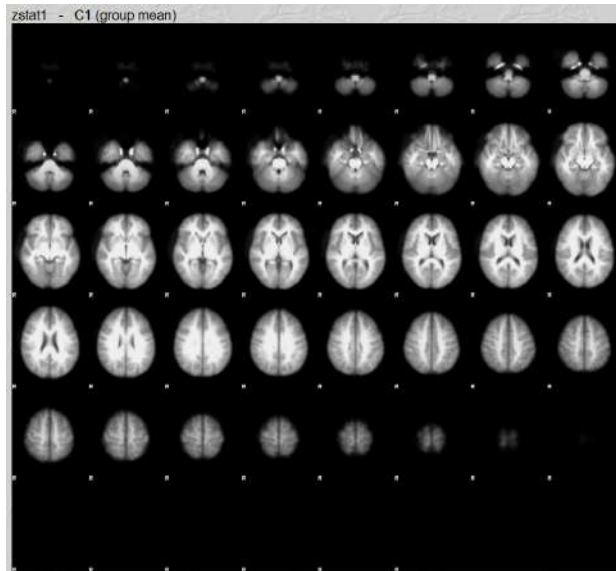
The final output of the 3rd-level analysis is a group-level Z-statistic map for the specific contrast of interest (e.g., Incongruent minus Congruent). This map represents the average functional activation pattern across all subjects in the cohort, with each voxel's intensity reflecting the statistical significance of that activation relative to the null hypothesis. The application of rigorous post-statistical thresholding ensures that only clusters of activation that are unlikely to have occurred by chance are visualized, providing a robust foundation for subsequent neuroanatomical interpretation and scientific reporting.



(a) Expected activation map with sparse, localized clusters.



(b) Incorrect activation map showing widespread, dense activation.



(c) Voxel-level thresholding map.

Figure 48: Comparison of 3rd-level COPE 3 activation maps in the Flanker task.

12.6 Visualization and Interpretation of Group-Level Maps

Upon the successful completion of the third-level FEAT directories (e.g., `Flanker_3rdLevel_inc-con.gfeat`), the resulting generalized population data must be visualized to interpret the anatomical loci of functional activation. The output directory contains a `.feat` folder (e.g., `cope1.feat`) which houses the fully processed statistical maps, most notably the `thresh_zstat1.nii.gz` file. This volume contains only the voxel clusters that survived the specified thresholding criteria (e.g., Cluster-based correction with a $Z > 3.1$ and $p < 0.05$).

FSLeves Visualization Protocol:

To visualize these results for academic reporting and clinical interpretation, the `thresh_zstat1` image is overlaid onto a high-resolution standard structural template (MNI152) within FSLeves. To optimize the visualization of these statistical parametric maps:

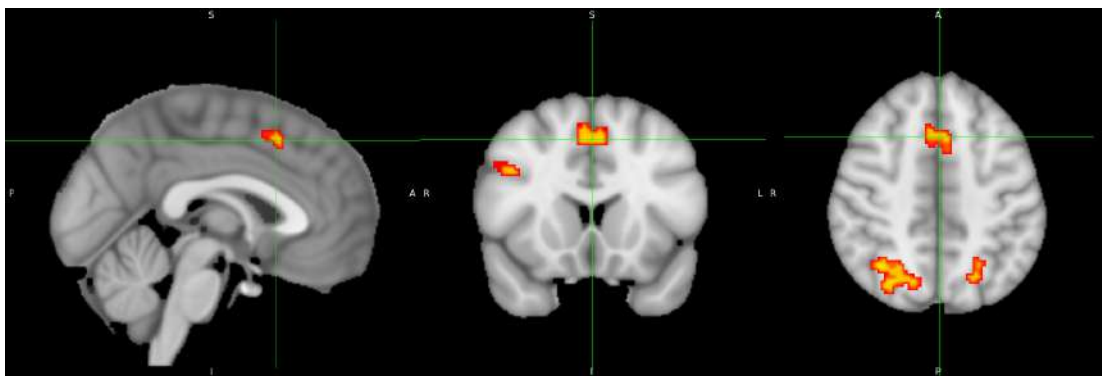


Figure 49: Visualization of 3rd-level group activation maps in FSLeves.

- **Color Mapping:** The functional overlay is assigned a "Red-Yellow" color palette. This standardizes the display so that warmer colors represent higher Z-statistics (greater statistical significance).
- **Threshold Trimming:** The "Min." display value is manually aligned with the mathematical Cluster-Defining Threshold utilized during the analysis (e.g., a minimum value of 3.1). This ensures that any residual noise below the analytical threshold is not visually rendered on the cortex.
- **Smoothing and Rendering:** Interpolation settings are often applied to the functional overlay to smooth the jagged edges of raw voxels, creating a more continuous and biologically plausible map of cortical networks. Finally, crosshairs are disabled to capture clean, unobstructed multi-planar snapshots (axial, coronal, sagittal) for inclusion in the final manuscript.

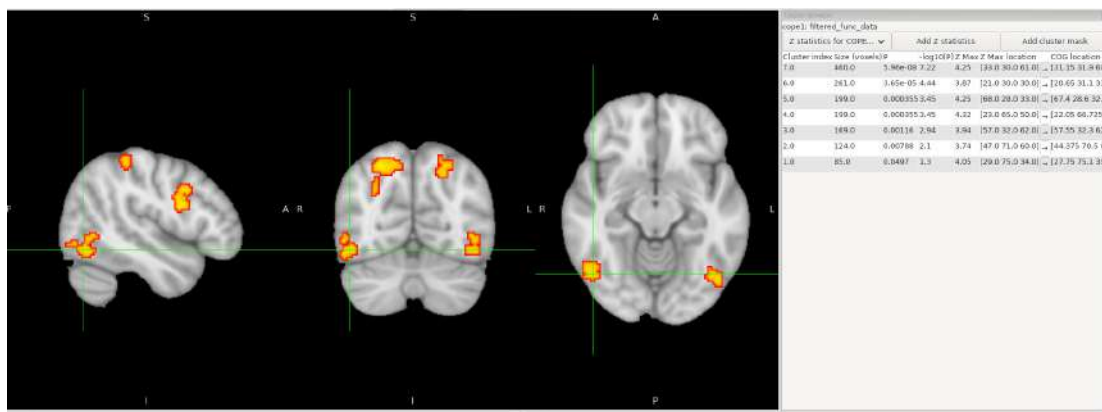


Figure 50: Final visualization of 3rd-level group activation maps with optimized color mapping and thresholding.

13 Region of Interest (ROI) Analysis: Confirmatory Functional Mapping

While third-level whole-brain analyses are highly effective for exploratory mapping of widespread functional networks, they impose severe statistical penalties due to the necessity of correcting for hundreds of thousands of multiple comparisons (voxel-wise). To overcome this, and to test specific *a priori* neuroanatomical hypotheses, a Region of Interest (ROI) analysis is conducted. Often termed a "confirmatory analysis," this method restricts the evaluation of Blood-Oxygen-Level-Dependent (BOLD) signal changes to a highly specific subset of voxels. By isolating specific cortical or subcortical structures, biomedical engineers can directly quantify and compare the magnitude of activation between experimental conditions (e.g., Congruent vs. Incongruent stimuli) within a targeted network.

13.1 Mask Generation Strategies

The foundational step of an ROI analysis is the creation of a spatial boundary, or "mask," that dictates exactly which voxels will be sampled. Based on the functional clusters identified during the exploratory third-level analysis (e.g., the Lateral Occipital Cortex, Paracingulate Gyrus, and Insular Cortex), two primary masking strategies were employed:

1. **Anatomical Masks:** This method relies on standardized probabilistic atlases, such as the Harvard-Oxford Cortical and Subcortical atlases. It extracts the boundaries of entire structural regions. For example, to investigate the brain's "salience network"—which identifies relevant internal and external stimuli (such as the "oddball" incongruent arrow in the Flanker task)—an anatomical mask of the Insular Cortex can be explicitly generated.
2. **Spherical Masks:** Rather than mapping an entire anatomical region, this method isolates the precise functional epicenter of a task. Using the peak MNI coordinates (x, y, z) identified from the group-level cluster analysis, a geometric sphere with a defined spatial radius (e.g., $r = 7$ mm) is computationally drawn around the specific activation peak.

13.2 Computational Extraction Pipeline

Prior to data extraction, the individual statistical maps must be temporally and spatially aligned. The lower-level parameter estimates (e.g., Z-statistic files) for all subjects across all relevant COPEs (e.g., COPE 1, 2, and 3) are concatenated, or "stacked," into a single merged 4D file.

The data extraction is then executed utilizing FSL's command-line tool, `fslmeants`. This algorithm superimposes the generated ROI mask over the stacked subject data and extracts the average parameter estimate from the confined voxels.

When executing this extraction, two distinct mathematical approaches for weighting the voxels within the mask are evaluated:

- **Binarized Extraction:** The mask is strictly binary. Any voxel within the mask's boundary containing a numerical value greater than zero is converted to a "1". The algorithm extracts and averages the data from all "1" voxels equally, regardless of whether a voxel sits at the core of the region or on its biological periphery.
- **Probabilistic Weighting (`-w` flag):** Because anatomical boundaries derived from atlases (like Harvard-Oxford) are inherently probabilistic, engineers can apply the `-w` option within the `fslmeants` command. Instead of a binary cutoff, this mathematically weights the extracted contrast estimates based on the underlying probability distribution of the atlas (e.g., data from a voxel with a 90% probability of being in the amygdala contributes more heavily to the final average than a voxel with a 20% probability).

13.3 Statistical Evaluation

Following the extraction of average parameter estimates from each ROI, statistical evaluation is conducted to test the significance of differences between experimental conditions (e.g., Incongruent vs. Congruent). Typically, paired t-tests or repeated-measures ANOVA are applied across subjects to determine whether the mean BOLD signal change within

the ROI differs significantly from zero or between conditions, providing confirmatory evidence for the functional role of the targeted brain region.

13.4 Automation of ROI Analysis and Data Extraction

To efficiently extract mean Z-statistics from multiple spherical ROIs across all subjects, a two-stage automated pipeline was implemented. First, a Bash script (`run_roi_analysis.sh`) generates the spherical masks and uses `fslmeants` to extract the data. Second, a Python script (`plot_roi_barcharts.py`) reads the extracted text files, performs statistical tests, and generates publication-ready bar charts with significance indicators.

Bash Script: Mask Generation and Data Extraction

```

1  #!/bin/bash
2
3  # =====
4  # ROI Analysis Automation Script
5  # =====
6
7  STATS_DIR="Flanker_2ndLevel.gfeat/cope3.feats/stats"
8  STANDARD_BRAIN="$FSLDIR/data/standard/MNI152_T1_2mm.nii.gz"
9
10 echo "=== Starting ROI Analysis Automation ==="
11
12 # 1. Check for the merged Z-stats file
13 if [ ! -f "allZstats.nii.gz" ]; then
14     echo "-> allZstats.nii.gz not found. Merging from $STATS_DIR..."
15     if [ -d "$STATS_DIR" ]; then
16         fslmerge -t allZstats.nii.gz $(ls ${STATS_DIR}/zstat*.nii.gz | sort -V)
17         echo "-> allZstats.nii.gz created successfully."
18     else
19         echo "ERROR: $STATS_DIR does not exist. Check your path."
20         exit 1
21     fi
22 else
23     echo "-> allZstats.nii.gz already exists. Skipping merge."
24 fi
25
26 # 2. Define the regions from your image
27 # Format: "RegionName Voxel_X Voxel_Y Voxel_Z Radius_in_mm"
28 # Note: These MNI coordinates have been converted to 2mm Voxel coordinates for FSL
29 declare -a regions=(
30     "Fusiform_Gyrus_Right 28 37 28 5"          # MNI: 34 -52 -16
31     "LOC_sup_Right 29 36 49 5"                  # MNI: 32 -54 26
32     "PCG 27 62 60 5"                            # MNI: 36 -2 48
33     "LOC_inf_Right1 34 39 27 5"                 # MNI: 22 -48 -18
34     "Fusiform_Gyrus_Left 63 59 28 5"            # MNI: -36 -8 -16
35     "LOC_sup_Left 61 33 47 5"                   # MNI: -32 -60 22
36     "Insular_Cortex_Left 59 64 30 5"            # MNI: -28 2 -12
37     "LOC_inf_Right2 19 32 34 5"                 # MNI: 52 -62 -4
38 )
39
40 # 3. Loop through each region to create masks and extract stats
41 echo "=== Generating Masks and Extracting Stats ==="
42

```

```

43 for region_info in "${regions[@]"; do
44     # Read the variables from the array string
45     read -r name x y z radius <<< "$region_info"
46
47     echo "Processing $name at voxel coords ($x, $y, $z)..."
48
49     # Step A: Create a single voxel point at the coordinates
50     fslmaths "$STANDARD_BRAIN" -mul 0 -add 1 -roi $x 1 $y 1 $z 1 0 1 "ROI_${name}.nii.gz"
51     -odt float
52
53     # Step B: Expand the point into a sphere
54     fslmaths "ROI_${name}.nii.gz" -kernel sphere $radius -fmean "Sphere_${name}.nii.gz"
55     -odt float
56
57     # Step C: Binarize the spherical mask
58     fslmaths "Sphere_${name}.nii.gz" -bin "Sphere_bin_${name}.nii.gz"
59
60     # Step D: Extract the mean time series (z-stats) from the mask
61     fslmeants -i allZstats.nii.gz -m "Sphere_bin_${name}.nii.gz" > "Stats_${name}.txt"
62
63     echo "-> Saved extracted data to Stats_${name}.txt"
64     echo "-----"
65 done
66
67 echo "=== All ROI analyses completed successfully! ==="

```

Listing 7: Automated ROI analysis script (run_roi_analysis.sh)

Contrast Estimates (COPE 3) for the dmPFC ROI The following values are the extracted parameter estimates for the **dorsomedial Prefrontal Cortex (dmPFC)** region of interest, corresponding to the `Stats_dmPFC_cope3.txt` file.

Table 6: Individual subject contrast estimates (COPE 3) for dmPFC.

Subject	Estimate
1	1.607354
2	0.791012
3	0.671231
4	0.636057
5	0.715009
6	1.532448
7	1.868126
8	2.186121
9	-0.566627
10	-0.333565
11	0.075891
12	1.144021
13	-0.468592
14	-0.001120
15	0.178788
16	0.184022
17	1.218302
18	-0.785554
19	-0.486537
20	0.264204
21	0.646710
22	0.798099
23	1.309559
24	-0.155901
25	-0.862956
26	0.956068

Contrast Estimates (COPE 3) for the PCG ROI The following values are the extracted parameter estimates for the **Paracingulate Gyrus (PCG)** region of interest, corresponding to the `Stats_PCG_cope3.txt` file.

Table 7: Individual subject contrast estimates (COPE 3) for PCG.

Subject	Estimate
1	0.851869
2	-0.345220
3	1.131169
4	0.478695
5	-0.237098
6	0.948197
7	1.003871
8	1.859882
9	0.366511
10	0.115892
11	0.468893
12	0.091392
13	-0.219626
14	0.318064
15	0.362344
16	-0.892906
17	0.115691
18	0.924945
19	-0.795685
20	0.840411
21	0.345504
22	-0.022915
23	2.320574
24	0.221759
25	0.058844
26	0.789840

Python Script: Bar Chart Generation and Statistical Evaluation

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy import stats
4 import os
5
6 # List of all the regions we generated masks for
7 regions = [
8     "Fusiform_Gyrus_Right",
9     "LOC_sup_Right",
10    "PCG",
11    "LOC_inf_Right1",
12    "Fusiform_Gyrus_Left",
13    "LOC_sup_Left",
14    "Insular_Cortex_Left",
15    "LOC_inf_Right2"
16 ]
17
18 # Labels for the X-axis
19 conditions = ['incong', 'cong', 'incong - cong']
20 copes = ['cope1', 'cope2', 'cope3']

```

```
21
22 print("Generating bar charts...")
23
24 for region in regions:
25     data_means = []
26     data_ci = []
27     data_pvals = []
28
29     # Read data for all 3 copes for the current region
30     for cope in copes:
31         filename = f"Stats_{region}_{cope}.txt"
32
33         if not os.path.exists(filename):
34             print(f"Warning: {filename} not found. Skipping {region}.")
35             continue
36
37         # Load the 26 subjects' data
38         data = np.loadtxt(filename)
39
40         # Calculate Mean
41         mean_val = np.mean(data)
42
43         # Calculate 95% Confidence Interval (1.96 * Standard Error)
44         se = stats.sem(data)
45         ci = 1.96 * se
46
47         # Calculate 1-sample t-test against 0 for significance
48         t_stat, p_val = stats.ttest_1samp(data, 0)
49
50         data_means.append(mean_val)
51         data_ci.append(ci)
52         data_pvals.append(p_val)
53
54     # Skip plotting if data is missing
55     if len(data_means) < 3:
56         continue
57
58     # Create the figure and axis
59     fig, ax = plt.subplots(figsize=(8, 6))
60     positions = [1, 2, 3]
61
62     # Draw bars and error lines
63     for i in range(3):
64         pos = positions[i]
65         mean = data_means[i]
66         ci = data_ci[i]
67         p_val = data_pvals[i]
68
69         # Absolute mean for the bar height (as done in the study's script)
70         abs_mean = abs(mean)
71         y_min = abs_mean - ci
72         y_max = abs_mean + ci
73
```

```

74     # Draw the bar
75     ax.bar(pos, abs_mean, color='gray', width=0.4, alpha=0.7)
76
77     # Draw the error line (95% CI)
78     ax.plot([pos, pos], [y_min, y_max], color='black', linewidth=1.5)
79
80     # Determine the number of stars based on p-value
81     if p_val <= 0.001:
82         stars = '**'
83     elif p_val <= 0.05:
84         stars = '*'
85     else:
86         stars = ''
87
88     # Draw the stars slightly above the highest point of the error bar
89     if stars:
90         ax.text(pos, y_max + 0.05, stars, ha='center', va='bottom', color='green',
91                 fontsize=20)
92
93     # Formatting the plot
94     ax.set_title(f"{region.replace('_', ' ')} using Spherical Mask")
95     ax.set_xticks(positions)
96     ax.set_xticklabels(conditions)
97     ax.set_ylabel('Mean Z-Statistic')
98
99     # Set y-axis lower limit to 0 for better visualization
100    ax.set_ylim(bottom=0)
101
102    # Save the plot
103    output_filename = f"BarChart_{region}.png"
104    plt.savefig(output_filename, bbox_inches='tight')
105    plt.close() # Close the figure to free memory
106
107    print(f"Saved {output_filename}")
108 print("All charts generated successfully!")

```

Listing 8: Automated bar chart generation (plot_roi_barcharts.py)

Workflow Description

1. The Bash script first checks for the existence of a merged 4D file containing all subjects' Z statistic maps ('allZstats.nii.gz'). If missing, it creates it using 'fslmerge'.
2. For each predefined region (with voxel coordinates and a radius of 5 mm), the script generates a spherical mask, binarizes it, and extracts the mean Z statistic across all subjects using 'fslmeants'. The extracted values are saved as plain text files (Stats_RegionName.txt).
3. The Python script then processes each text file, calculating the mean, the 95% confidence interval (CI), and the p -value from a one-sample t-test against zero.
4. Bar charts are generated for each region, showing the mean absolute Z statistic for the three contrasts (incongruent, congruent, incongruent-congruent). Error bars represent the 95% CI, and green stars indicate statistical significance (* for $p \leq 0.05$, ** for $p \leq 0.001$).

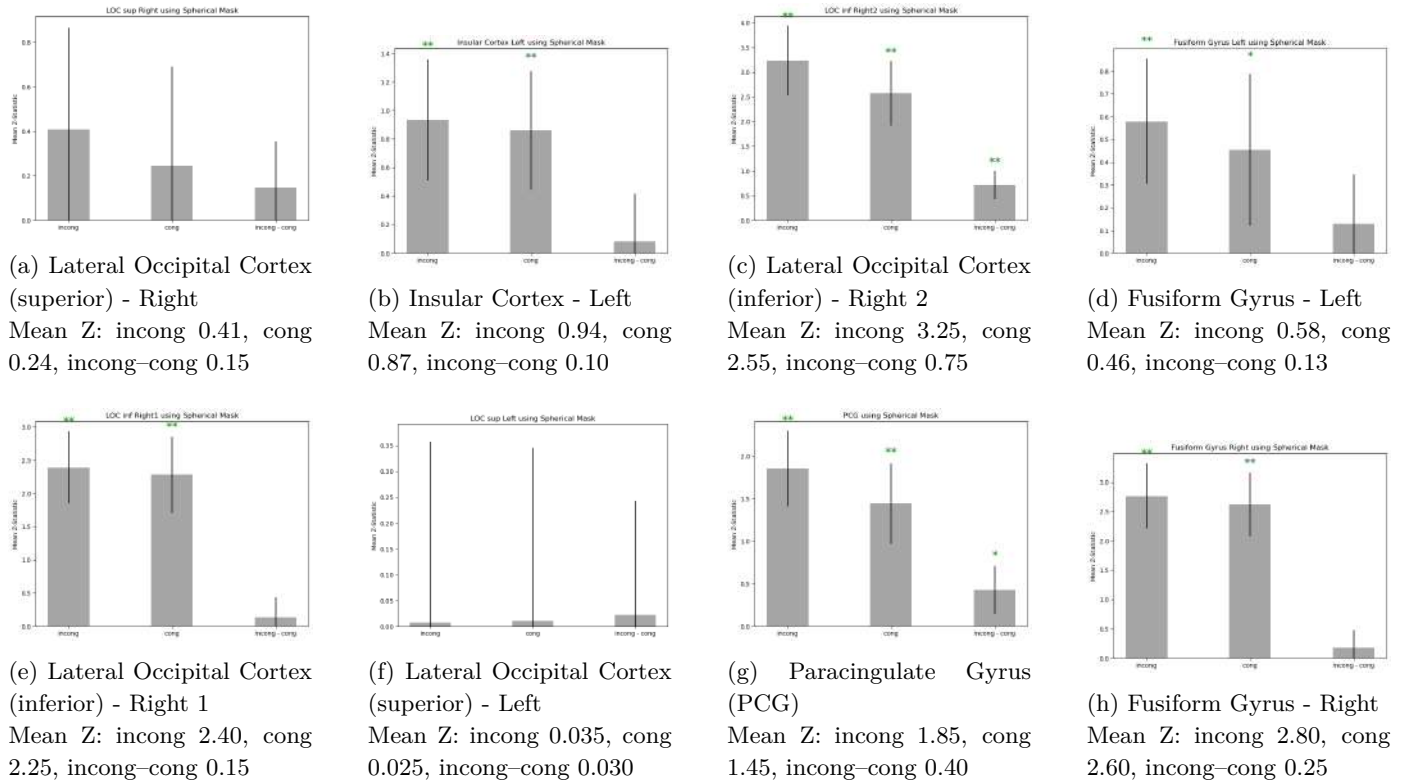


Figure 51: Region of Interest (ROI) bar charts showing mean Z-statistics (with 95% confidence interval error bars) for the three contrasts (incongruent, congruent, and incongruent-congruent). Each ROI was defined as a 5 mm radius sphere around the peak voxel coordinates from the third level analysis. Stars indicate statistical significance: * $p \leq 0.05$, ** $p \leq 0.001$.

14 Introduction to Neuroanatomy

Neuroanatomy is the study of the structural organization, connectivity, and complex architecture of the nervous system. While traditionally a domain of neuroscience and medicine, neuroanatomy is a fundamental prerequisite for Biomedical Engineering (BME). For biomedical engineers, the brain is not just a biological organ, but an intricate, highly optimized electrochemical network.

Understanding neuroanatomy is critical for engineers who design medical imaging technologies (such as the fMRI pipelines previously discussed), develop neural prosthetics, engineer Brain-Computer Interfaces (BCIs), or create neuro-modulation devices like Deep Brain Stimulation (DBS) implants. Without precise spatial mapping and an understanding of regional brain functions, it is impossible to target neural tissues safely, interpret physiological signals accurately, or design diagnostic tools that interface with the human nervous system.

14.1 The Brain

Description: The brain is the central command organ of the human nervous system. Composed of approximately 86 billion neurons and an equivalent number of supportive glial cells, it is a highly folded, metabolically demanding structure responsible for driving all human behavior, physiological regulation, and consciousness.

Location: Housed securely within the cranial cavity of the skull, suspended in protective cerebrospinal fluid (CSF).

Core Functions:

- Processing and integration of all external sensory data and internal physiological signals.
- Execution of complex cognitive processes (thought, memory, language, and emotion).
- Maintenance of physiological homeostasis.
- Generation and coordination of voluntary and involuntary motor commands.

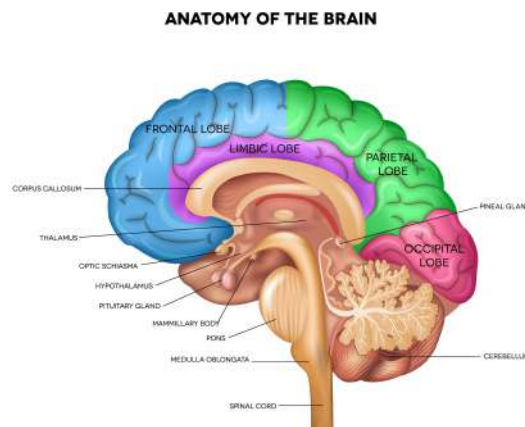
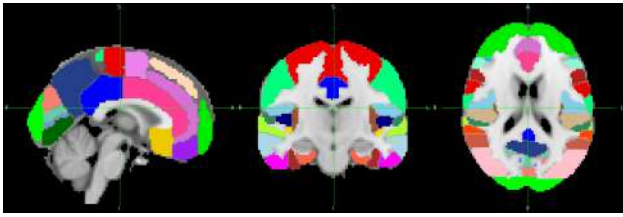


Figure 52: Anatomical illustration of the human brain. *Source: Shutterstock*

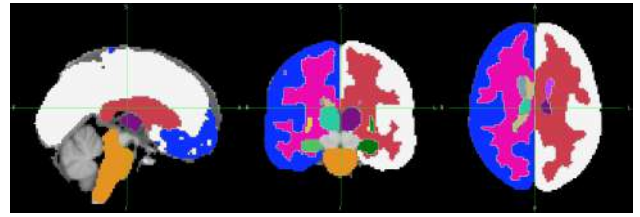
14.2 Anatomical Localization in FSL: The MNI152 Standard and Harvard-Oxford Atlas

In neuroimaging and biomedical engineering, standardizing brain anatomy across diverse subject populations is absolutely critical for accurate group-level statistical analysis and functional brain mapping. Within the FMRIB Software Library (FSL) environment, this anatomical standardization is achieved by mathematically registering (warping) an individual patient's native-space MRI scan to the MNI152 standard space. This is a universal, three-dimensional stereotaxic coordinate system (x, y, z) created by the Montreal Neurological Institute by averaging 152 high-resolution structural

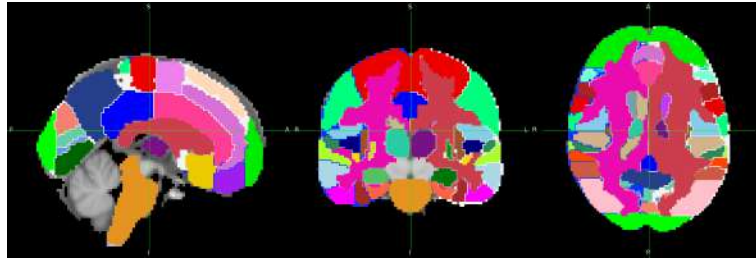
MRI scans, serving as the neuroimaging field’s standard biological template. Once a brain volume is successfully normalized to this MNI152 template, engineers and neuroscientists utilize the Harvard-Oxford Atlas—a robust, built-in tool within FSL—to automatically and quantitatively identify specific structural regions.



(a) Harvard-Oxford Cortical Atlas highlighted.



(b) Harvard-Oxford Subcortical Atlas highlighted.



(c) Harvard-Oxford Atlas highlighted.

Figure 53: The Harvard-Oxford Atlas within the MNI152 standard space. (a) Cortical regions, (b) Subcortical regions, (c) Both combined.

14.3 The Brainstem

Description: The brainstem is a vital, stalk-like structural and functional conduit that connects the higher cognitive centers of the cerebrum with the spinal cord. It is evolutionarily one of the oldest parts of the brain, strictly dedicated to survival and automated physiological regulation rather than conscious thought.

Location: Situated at the base of the brain, lying continuous with the spinal cord inferiorly and the diencephalon (thalamus/hypothalamus) superiorly. It sits just anterior to the cerebellum.

Core Functions:

- Autonomic regulation of essential life-sustaining functions (e.g., cardiac and respiratory rhythms).
- Acting as the primary relay hub for all ascending (sensory) and descending (motor) nerve tracts passing to and from the body.
- Serving as the origin point for the majority of the cranial nerves (Nerves III through XII), which innervate the face and neck.

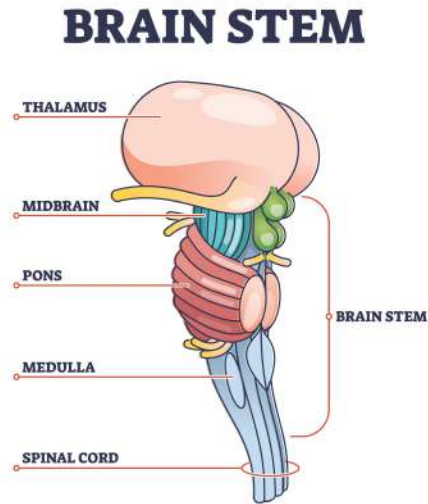


Figure 54: Anatomical illustration of the brainstem. *Source: Shutterstock*

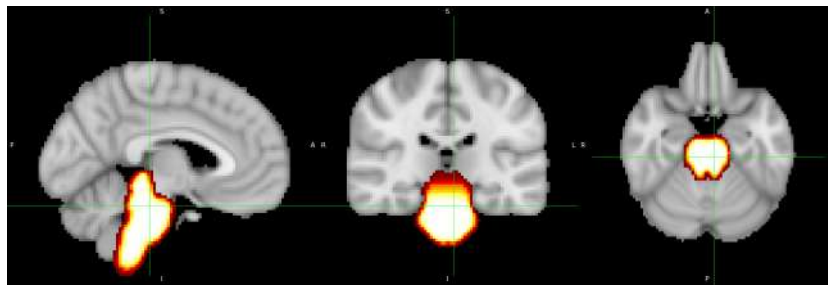


Figure 55: MNI152 2mm brain template with the brainstem highlighted.

14.4 The Medulla Oblongata

Description: The medulla oblongata (often simply called the medulla) is the terminal, most inferior segment of the brainstem. It serves as the vital autonomic control center for the body; severe injury to this region is almost always instantly fatal.

Location: Positioned between the pons (above) and the spinal cord (below), transitioning into the spinal cord at the foramen magnum (the large opening at the base of the skull).

Core Functions:

- **Cardiovascular Regulation:** Controls baseline heart rate, cardiac contractility, and blood vessel dilation/constriction to manage blood pressure.
- **Respiratory Rhythm:** Generates the basal rate of breathing and coordinates the mechanics of inhalation and exhalation.
- **Reflex Coordination:** Manages critical involuntary survival reflexes, including swallowing, coughing, sneezing, and vomiting.
- **Motor Decussation:** Houses the "pyramids," the structural site where primary motor nerve tracts cross over (decussate) to the opposite side of the body, explaining why the left brain controls the right side of the body.

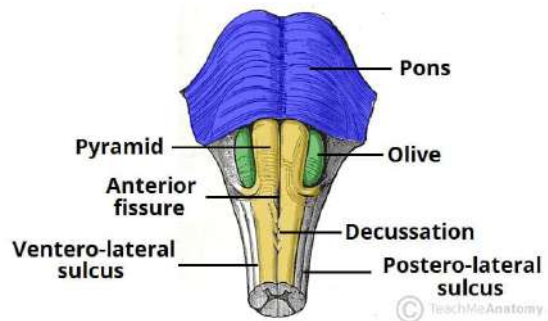


Figure 56: Anatomical illustration of the medulla. *Source: TeachMEAnatomy*

14.5 The Pons

Description: The pons (Latin for "bridge") is a distinct, bulbous structure on the anterior surface of the brainstem. It earns its name by acting as the primary neuroanatomical bridge connecting various parts of the brain, most notably routing signals between the cerebral cortex and the cerebellum.

Location: Positioned centrally within the brainstem—superior to the medulla, inferior to the midbrain, and directly anterior to the cerebellum.

Core Functions:

- **Cerebellar Communication:** Relays vast amounts of information regarding motor planning and execution from the forebrain to the cerebellum, allowing for smooth, coordinated movement.
- **Respiratory Modulation:** Contains the pneumotaxic center, which fine-tunes the basic breathing rhythm generated by the medulla, adjusting breathing rate and depth.
- **Sleep Regulation:** Plays a critical role in the regulation of sleep cycles, particularly in initiating Rapid Eye Movement (REM) sleep and causing sleep paralysis to prevent acting out dreams.
- **Facial Sensomotor Control:** Houses the nuclei for several cranial nerves (V, VI, VII, VIII), driving functions like facial expressions, chewing, and eye movement.

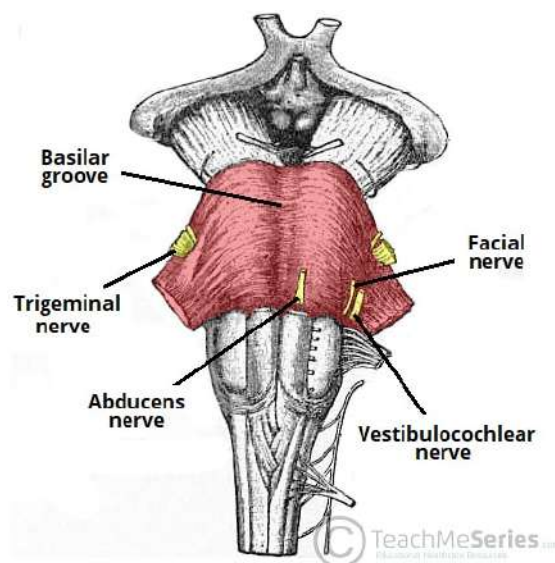


Figure 57: Anatomical illustration of the pons. *Source: TeachMEAnatomy*

14.6 The Midbrain (Mesencephalon)

Description: The midbrain is the shortest and most superior segment of the brainstem. It is a highly specialized region heavily involved in basic sensory processing, motor tract routing, and dopamine production.

Location: Situated directly below the diencephalon (the thalamus and hypothalamus) and directly above the pons.

Core Functions:

- **Sensory Reflexes:** Controls automatic visual and auditory reflexes via structures called the superior and inferior colliculi (e.g., automatically turning the head toward a loud sound or tracking a moving object).
- **Motor Control & Dopamine:** Contains the substantia nigra, a cluster of neurons that produce the neurotransmitter dopamine, which is crucial for initiating voluntary movement (the degradation of this area causes Parkinson's disease).
- **Arousal and Alertness:** Houses parts of the reticular formation, which controls overall brain arousal and transitions between sleep and wakefulness.
- **Pain Modulation:** Contains the periaqueductal gray matter, which plays a primary role in the descending modulation and suppression of pain signals.

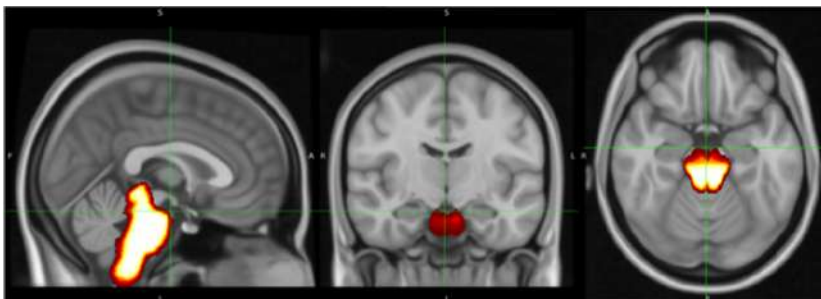


Figure 58: MNI152 2mm brain template with the midbrain highlighted.

14.7 The Cerebral Peduncles

Description: The cerebral peduncles are two massive, stalk-like bundles of nerve fibers (white matter tracts) that form the pillars of the anterior midbrain. They are the physical highways through which the higher brain communicates with the lower brain and body.

Location: Comprising the anterior (front) portion of the midbrain, extending downwards from the cerebral cortex to attach to the pons.

Core Functions:

- **Motor Signal Transport:** Acts as the primary conduit for the major descending motor pathways (the corticospinal and corticobulbar tracts).
- **Proprioceptive Relay:** Assists in transmitting sensory information regarding body position back up to the thalamus and cortex.
- **Motor Refinement:** Works in conjunction with the cerebellum and basal ganglia to help refine raw motor commands into smooth, purposeful actions before they reach the spinal cord.

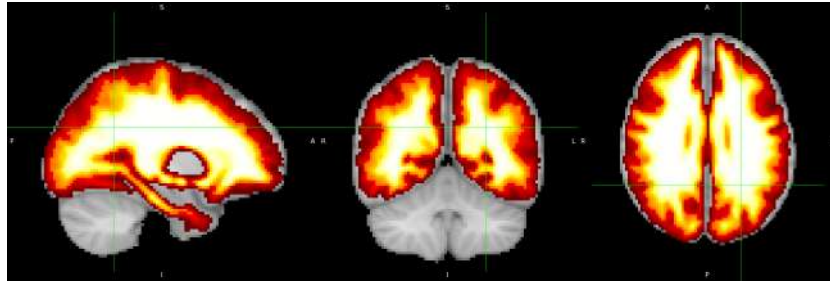


Figure 59: MNI152 2mm brain template with the cerebral white matter highlighted.

14.8 The Cerebellum

Description: Translating to "little brain" in Latin, the cerebellum is a highly folded, densely packed structure that houses more than half of the total neurons in the human brain. For biomedical engineers, the cerebellum is essentially a biological microprocessor; it does not initiate movement itself, but rather receives massive amounts of sensory and motor data, computing error-corrections in real-time to optimize human biomechanics and motor learning.

Location: Tucked beneath the occipital lobes of the cerebrum and situated directly posterior to the brainstem (specifically the pons and medulla).

Core Functions:

- **Motor Coordination:** Synchronizes the timing and force of different muscle groups to produce smooth, fluid voluntary movements.
- **Balance and Posture:** Integrates vestibular (inner ear) and proprioceptive (body position) signals to maintain upright posture and equilibrium.
- **Motor Learning:** Acts as the primary neural substrate for procedural memory, allowing for the adaptation and perfection of skilled motor tasks (e.g., riding a bicycle or playing an instrument).
- **Cognitive Timing:** Plays an emerging role in the timing of non-motor cognitive processes, including language processing and shifting attention.

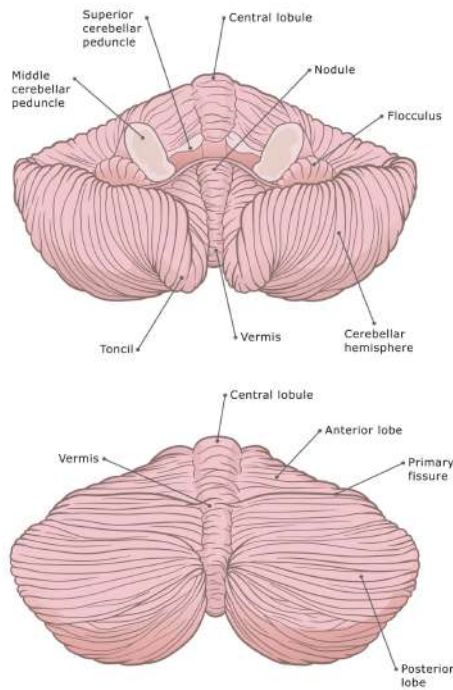


Figure 60: Anatomical illustration of the cerebellum. *Source: Getty Images*

14.9 The Vermis

Description: The vermis is the narrow, worm-like midline structure that physically and functionally bridges the two large cerebellar hemispheres. It is phylogenetically one of the oldest parts of the cerebellum, primarily dedicated to the stability of the body's central axis.

Location: Running longitudinally down the exact center (midline) of the cerebellum.

Core Functions:

- **Axial Motor Control:** Regulates the tone and movement of the proximal and axial muscles (the trunk, shoulders, and hips).
- **Locomotion Stability:** Coordinates the rhythmic, automated physical mechanics of walking and running.
- **Cardiovascular Modulation:** Modulates autonomic cardiovascular and respiratory responses during physical exertion or postural changes.
- **Speech Coordination:** Assists in the precise motor coordination of the vocal cords and mouth required for fluent speech production.

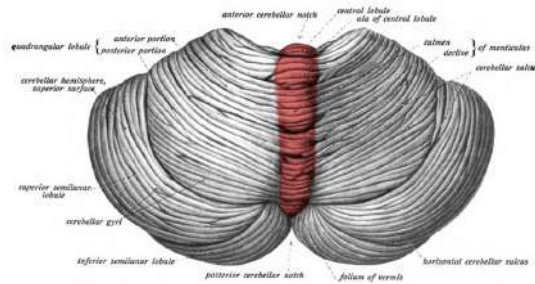


Figure 61: Anatomical illustration of the vermis. *Source: Wikipedia*

14.10 The Left and Right Hemispheres

Description: Flanking the central vermis, the expansive left and right cerebellar hemispheres make up the vast majority of the cerebellum's volume. Unlike the cerebral cortex (where the right brain controls the left body), the cerebellar hemispheres control motor function on the ipsilateral (same) side of the body.

Location: Expanding laterally outward from the vermis, filling the lower posterior cranial fossa of the skull.

Core Functions:

- **Distal Motor Control:** Manages the fine, precise movements of the distal extremities, particularly the hands, fingers, and feet.
- **Motor Planning:** Works in a continuous feedback loop with the cerebral cortex to plan the trajectory, speed, and force of an intended movement before it occurs.
- **Error Correction:** Compares the brain's intended motor command with the actual physical movement taking place, instantly transmitting corrective signals if a deviation is detected.
- **Complex Skill Storage:** Stores the complex neural engrams required for highly practiced, specialized physical skills.

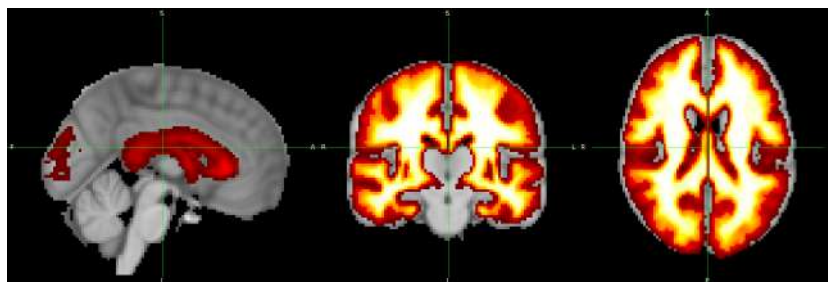


Figure 62: MNI152 2mm brain template with the Left and Right Cerebellar Hemispheres highlighted.

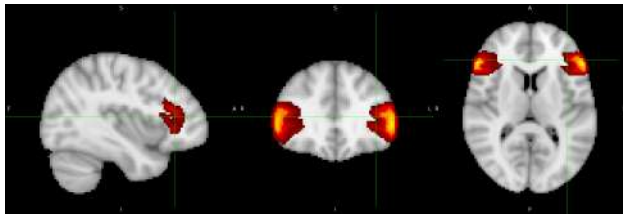
14.11 Anterior and Posterior Lobes

Description: Anatomically, the cerebellum is divided horizontally into distinct lobes separated by deep fissures (most notably the primary fissure). These lobes represent different evolutionary stages of the brain and handle different tiers of motor processing.

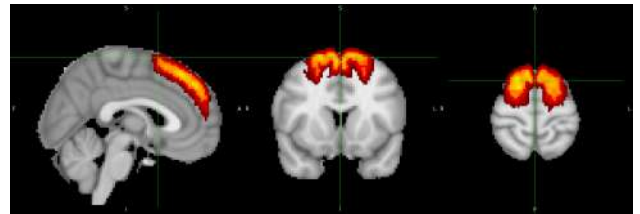
Location: The anterior lobe sits superiorly and forward (closest to the brainstem), while the posterior lobe is much larger, situated inferiorly and backwards.

Core Functions:

- **Proprioceptive Processing (Anterior):** The anterior lobe heavily processes continuous somatosensory feedback from the spinal cord regarding muscle stretch and joint position.
- **Unconscious Muscle Tone (Anterior):** Maintains baseline resting tension in the muscles to keep the body upright against gravity.
- **Fine Motor Execution (Posterior):** The posterior lobe is heavily involved in the execution of highly dexterous, voluntary actions.
- **Cognitive Processing (Posterior):** The posterior lobe connects densely with the cerebral cortex, supporting higher-order cognitive tasks, working memory, and spatial navigation.



(a) Inferior Frontal Gyrus (Pars Triangularis)



(b) Superior Frontal Gyrus

Figure 63: Anatomical illustration of the anterior and posterior lobes of the cerebellum.

14.12 The Cerebellar Tonsils

Description: The cerebellar tonsils are two small, rounded, peg-like lobules situated at the lowest extremity of the cerebellum. While they participate in general motor coordination, their primary importance in biomedicine and clinical engineering is pathological due to their proximity to the skull base.

Location: On the inferior (bottom) surface of the cerebellar hemispheres, resting immediately above the foramen magnum (the large opening where the brainstem exits the skull).

Core Functions (and Clinical Significance):

- **General Motor Support:** Functions as a lower extension of the posterior lobe, assisting in fine motor control.
- **Pathological Herniation Risk:** If intracranial pressure rises (due to tumors, hemorrhage, or severe traumatic brain injury), the tonsils are the structures most likely to be forced downward through the foramen magnum.
- **Brainstem Compression:** Known as "tonsillar herniation," this downward displacement physically crushes the medulla oblongata, rapidly causing respiratory arrest and death—making it a critical parameter to monitor in neuroimaging and intracranial pressure (ICP) engineering.

14.13 The Cerebral Hemispheres

Description: The cerebrum is the largest and most highly developed portion of the human brain, divided into two massive halves known as the left and right cerebral hemispheres. They are characterized by a highly folded outer layer of gray matter called the cerebral cortex (comprising ridges called gyri and grooves called sulci), which maximizes the surface area for billions of neuronal connections. For biomedical engineers, the cerebral cortex is the primary source of the electrical potentials measured by Electroencephalography (EEG) and the main interface for both invasive and non-invasive Brain-Computer Interfaces (BCIs).

Location: Filling the vast majority of the cranial vault, sitting superior to the brainstem and cerebellum. The two hemispheres are separated by a deep groove called the longitudinal fissure and are physically connected by a massive bundle of white matter nerve fibers called the corpus callosum.

Core Functions:

- **Contralateral Control:** The motor and sensory networks are crossed; the left hemisphere generally controls and receives input from the right side of the body, and vice versa.
- **Lateralization of Function:** Certain complex cognitive tasks are localized to specific sides (e.g., analytical processing and language are predominantly left-lateralized, while spatial reasoning and facial recognition are right-lateralized).
- **Higher-Order Consciousness:** Generates conscious awareness, subjective experience, and complex self-reflection.
- **Interhemispheric Communication:** Synthesizes isolated sensory data from both sides of the body into a single, unified perceptual experience via the corpus callosum.

14.14 The Lobes of the Cerebral Hemispheres

Anatomically, each cerebral hemisphere is subdivided into four primary lobes. These distinct geographical regions are heavily specialized in their functional processing. Understanding these subdivisions is critical in clinical neuroengineering for targeting neuromodulation devices, isolating the sources of epileptic seizures, and designing sensory or motor neuroprosthetics.

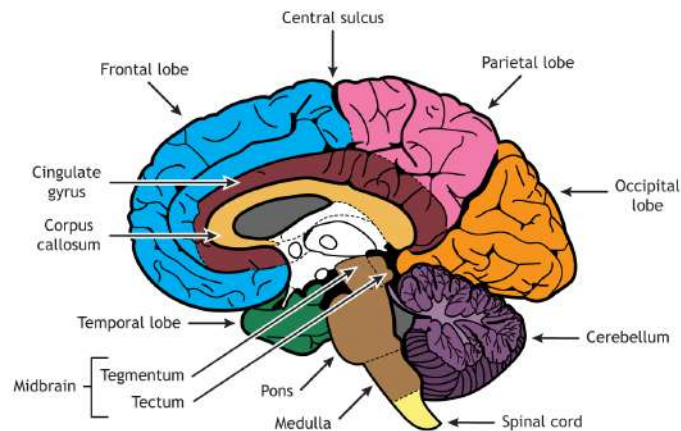


Figure 64: Anatomical illustration of the Midsagittal. *Source: Michigan State University*

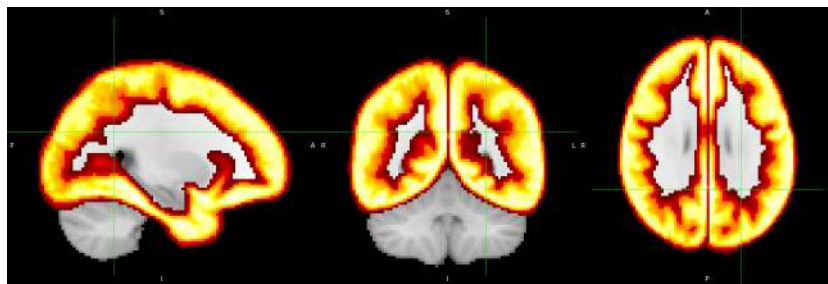


Figure 65: MNI152 2mm brain template with the cerebral cortex highlighted.

14.15 The Frontal Lobe

Description: The frontal lobe is the largest of the four lobes and acts as the brain's executive command center. It distinguishes humans from other species due to its vast size and complexity. Because it regulates everything from voluntary movement to mood, it is heavily targeted in engineering therapies, such as Transcranial Magnetic Stimulation (TMS) for depression and motor cortex microelectrode implants for paralyzed patients controlling robotic limbs.

Location: Located at the anterior (front) of the brain, extending from the forehead backward to the central sulcus, and resting above the lateral sulcus (Sylvian fissure).

Core Functions:

- **Executive Functioning (Prefrontal Cortex):** Drives high-level cognition, including decision-making, long-term planning, problem-solving, and impulse control.
- **Voluntary Motor Control (Primary Motor Cortex):** Located on the precentral gyrus, it generates the precise neural commands necessary to initiate deliberate physical movements.
- **Speech Production (Broca's Area):** Formulates the complex motor plans required for the mouth, vocal cords, and diaphragm to produce articulate, fluent speech (typically located in the left hemisphere).
- **Working Memory:** Temporarily holds and manipulates immediate information required for active cognitive tasks and reasoning.
- **Personality and Social Behavior:** Regulates emotional expression, social appropriateness, and the outward manifestation of individual personality.

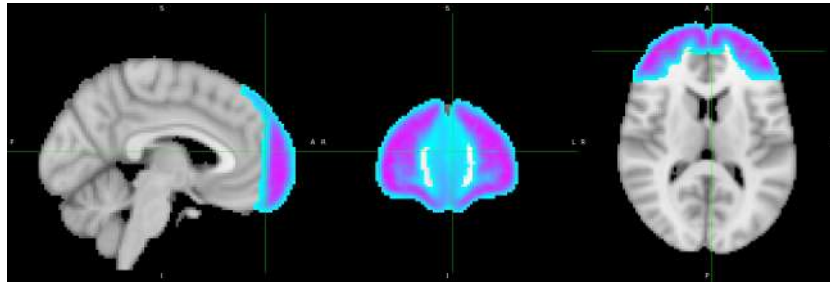


Figure 66: MNI152 2mm brain template with the frontal pole highlighted.

14.16 The Parietal Lobe

Description: The parietal lobe is the brain's primary sensory integration hub. It is responsible for mapping the physical body and synthesizing various sensory inputs into a cohesive understanding of the surrounding environment. In BME, understanding the parietal lobe is essential for developing closed-loop prosthetics that provide artificial tactile feedback (haptics) to amputees.

Location: Situated near the top and center of the brain—posterior to the frontal lobe (separated by the central sulcus), superior to the temporal lobe, and anterior to the occipital lobe.

Core Functions:

- **Somatosensory Processing (Primary Somatosensory Cortex):** Located on the postcentral gyrus, it processes tactile sensations originating from the skin and organs, including touch, pressure, pain, and temperature.
- **Proprioception:** Integrates continuous sensory feedback from the muscles and joints to maintain a real-time, 3D map of the body's physical position in space.
- **Spatial Awareness and Navigation:** Calculates the spatial relationships and distances between the body and external objects to accurately guide physical interactions.
- **Numeracy and Calculation:** Particularly within the inferior parietal lobule, it supports mathematical reasoning and the manipulation of numbers.
- **Visuomotor Coordination:** Synthesizes tactile mapping with visual input, enabling highly coordinated interactions with the physical world (e.g., reaching out to precisely grasp a cup).

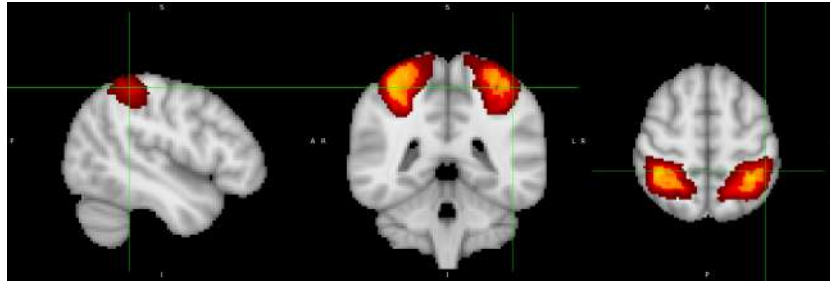


Figure 67: MNI152 2mm brain template with the parietal lobe highlighted.

14.17 The Occipital Lobe

Description: The occipital lobe is the smallest of the four main lobes and is almost exclusively dedicated to the complex processing of visual information. For engineers and researchers working on visual neuroprosthetics (such as cortical implants designed to bypass damaged retinas and restore sight to the blind), the neural architecture of the occipital lobe is the primary surgical and electrical target.

Location: Positioned at the extreme posterior (back) of the cerebral hemispheres, resting immediately above the cerebellum.

Core Functions:

- **Primary Visual Processing (V1):** Receives raw, unadulterated visual data straight from the retinas via the optic radiations.
- **Color Recognition:** Analyzes the wavelengths of incoming light to distinguish between different hues and colors.
- **Motion Perception:** Detects the speed, direction, and trajectory of dynamic, moving objects within the visual field.
- **Spatial and Depth Processing:** Assists in determining the depth, distance, and 3D layout of the visual environment (stereopsis).
- **Feature Extraction:** Breaks down visual scenes into fundamental lines, edges, and contrasts before routing this data forward to other lobes for complex object recognition.

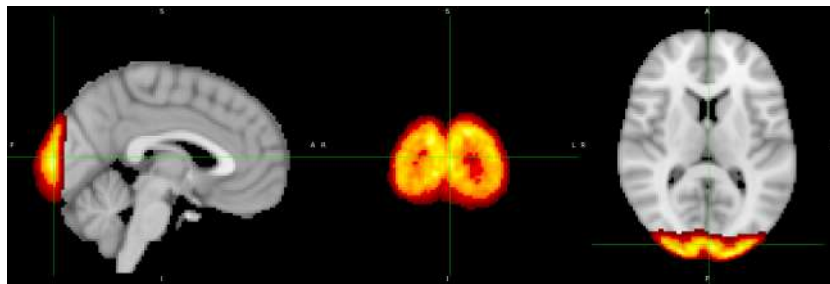


Figure 68: MNI152 2mm brain template with the occipital lobe highlighted.

14.18 The Temporal Lobe

Description: The temporal lobe serves as the brain's center for auditory processing, memory encoding, and emotional association. Because it houses deeply embedded, highly electrically active structures like the hippocampus, the temporal lobe is uniquely susceptible to seizure activity. Consequently, it is a primary focal point for engineers designing EEG-based seizure prediction algorithms and closed-loop responsive neurostimulation (RNS) devices for epilepsy.

Location: Situated on the lateral sides of the brain (near the temples), sitting inferior to the lateral sulcus and anterior to the occipital lobe.

Core Functions:

- **Auditory Processing (Primary Auditory Cortex):** Receives and interprets sound frequencies, pitch, and volume transmitted from the inner ear.
- **Language Comprehension (Wernicke's Area):** Translates raw auditory signals into meaningful, coherent language (typically lateralized to the left hemisphere).
- **Memory Formation (Hippocampus):** Encodes immediate short-term information into long-term declarative memories (facts) and episodic memories (events).
- **Complex Visual Recognition:** Acts as the end-point for the "ventral visual stream," allowing the brain to recognize highly complex visual stimuli, such as specific objects, scenes, and human faces.
- **Emotional Association (Amygdala):** Evaluates environmental stimuli and links memories to powerful emotional responses, particularly playing a critical role in fear processing, reward systems, and the "fight or flight" threat response.

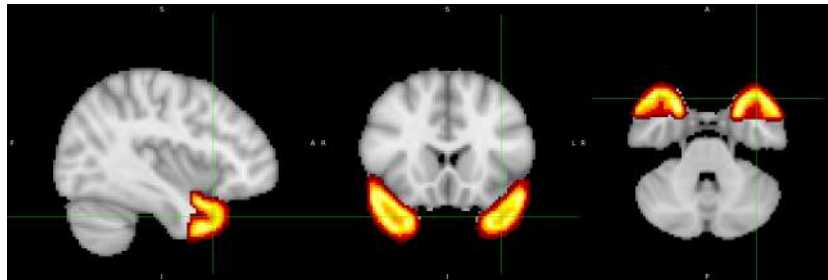


Figure 69: MNI152 2mm brain template with the temporal lobe highlighted.

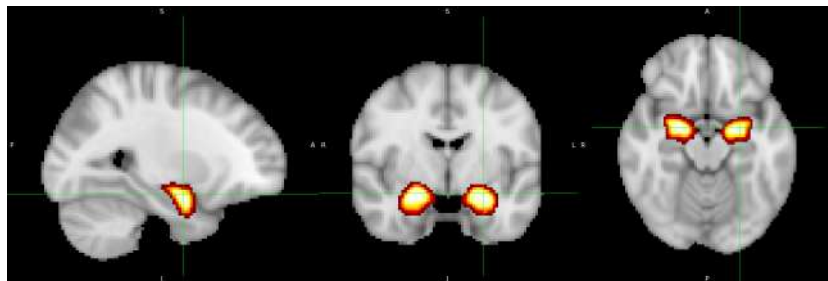


Figure 70: MNI152 2mm brain template with the amygdala highlighted.

14.19 The Topography of the Brain: Important Sulci and Gyri

Description: The human cerebral cortex is highly convoluted, folded into an intricate landscape of elevated ridges called gyri and deep cortical valleys called sulci (or fissures). This evolutionary adaptation dramatically increases the total cortical surface area—and thus computing power—within the confined spatial volume of the skull. For biomedical engineers, these topographical landmarks are indispensable. They provide the universal coordinate system used for neurosurgical navigation, functional brain mapping (like fMRI), and the precise placement of non-invasive EEG electrodes (the 10-20 system).

14.20 The Central Sulcus (Fissure of Rolando)

Description: The central sulcus is one of the most prominent and functionally significant landmarks in the human brain. It serves as the absolute anatomical border dividing the brain's motor execution networks from its sensory processing networks.

Location: Runs vertically down the middle of the lateral surface of each cerebral hemisphere, starting near the top-center (longitudinal fissure) and ending just above the Sylvian fissure.

Core Functions (and BME Relevance):

- **Functional Demarcation:** Physically separates the frontal lobe (anterior) from the parietal lobe (posterior).
- **Neurosurgical Landmark:** Acts as the critical reference line for neurosurgeons mapping safe entry zones to avoid causing paralysis during brain tumor resections.
- **Sensorimotor Segregation:** Ensures that descending motor commands and ascending sensory inputs are processed in distinct, albeit highly communicative, adjacent networks.
- **BCI Target Localization:** Serves as the primary stereotactic landmark for engineers surgically implanting Brain-Computer Interfaces (BCIs) to restore movement to paralyzed patients.

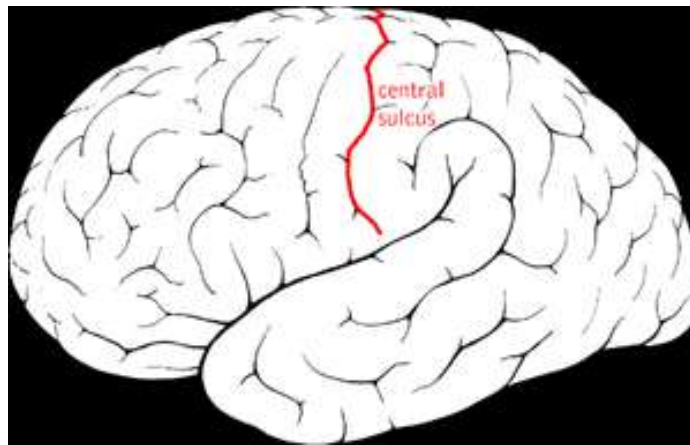


Figure 71: Anatomical illustration of the central sulcus. *Source: Wikipedia*

14.21 The Sylvian Fissure (Lateral Sulcus)

Description: The Sylvian fissure is the deepest and most prominent anatomical cleft in the cerebral hemisphere. Because it houses major blood vessels and covers deeply embedded cortical structures, it is a structure of immense importance in stroke pathology and neurovascular engineering.

Location: Runs horizontally along the lateral (side) surface of the brain, separating the temporal lobe below from the frontal and parietal lobes above.

Core Functions:

- **Structural Division:** Acts as the definitive anatomical ceiling for the temporal lobe.
- **Vascular Conduit:** Houses the Middle Cerebral Artery (MCA), making it the central focus for engineers modeling hemodynamics, ischemic strokes, and stent designs.
- **Language Network Hub:** The regions immediately surrounding this fissure (the perisylvian area) contain the brain's primary language comprehension and production centers (Wernicke's and Broca's areas).
- **Protection of the Insula:** Deeply conceals the insular cortex, a region critical for interoception (sensing the internal state of the body) and autonomic regulation.

14.22 The Parieto-occipital Sulcus

Description: This deep fissure serves as the definitive boundary separating the brain's somatosensory processing regions from its visual processing regions. It is highly prominent on the medial (inner) aspect of the brain but only slightly visible on the lateral surface.

Location: Situated near the posterior (back) of the brain, heavily cutting into the medial surface of the hemisphere.

Core Functions:

- **Lobe Segregation:** Defines the anatomical border between the parietal lobe and the occipital lobe.
- **Visual Pathway Routing:** Plays a structural role in the spatial organization of the optic radiations (white matter tracts) traveling to the primary visual cortex.
- **Medial Network Boundary:** Physically separates the precuneus (a core hub of the default mode network in the parietal lobe) from the cuneus (basic visual processing in the occipital lobe).

14.23 The Pre-central Gyrus

Description: The pre-central gyrus is the neuroanatomical home of the Primary Motor Cortex (M1). It houses massive upper motor neurons (Betz cells) that send their axons all the way down the spinal cord. In the field of neuroprosthetics, engineers extensively map this gyrus to decode the specific electrical firing patterns required to control robotic limbs.

Location: The prominent gyrus in the frontal lobe situated immediately anterior (in front of) the central sulcus.

Core Functions:

- **Voluntary Motor Execution:** Generates the raw, descending neural impulses that dictate conscious, voluntary muscle contractions.
- **Somatotopic Organization:** Arranged as a "motor homunculus," meaning specific anatomical segments of the gyrus map strictly to specific body parts (e.g., the medial section controls the legs, the lateral section controls the face).
- **Force and Trajectory Encoding:** Calculates and encodes the precise amplitude, direction, and force required for a specific physical action.
- **Motor Learning Consolidation:** Adapts structurally over time in response to learning new, complex physical skills.

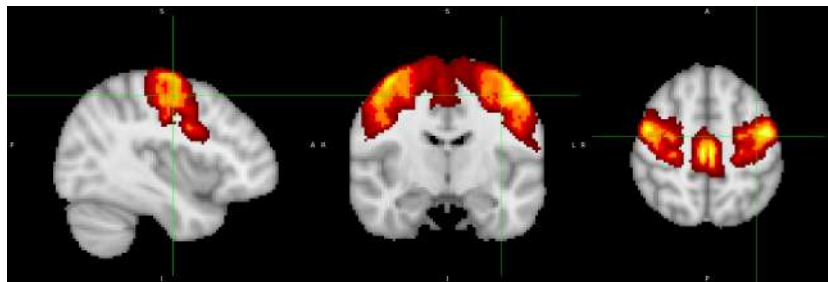


Figure 72: MNI152 2mm brain template with the pre-central gyrus highlighted.

14.24 The Post-central Gyrus

Description: The post-central gyrus is the seat of the Primary Somatosensory Cortex (S1). It is the brain's primary receptive area for the sense of touch. For BME researchers developing advanced closed-loop prosthetics, electrically

stimulating specific targets on this gyrus can provide an amputee with artificial tactile feedback, allowing them to "feel" what their robotic hand is grasping.

Location: The prominent gyrus in the parietal lobe situated immediately posterior (behind) the central sulcus.

Core Functions:

- **Tactile Perception:** Receives and processes raw tactile information, including fine touch, pressure, and vibration from the entire body.
- **Nociception and Thermoception:** Processes incoming signals regarding pain and temperature variations.
- **Sensory Homunculus Mapping:** Like the motor cortex, it maps the body spatially, dedicating massive cortical volume to highly sensitive areas like the fingertips and lips.
- **Sensorimotor Feedback Loop:** Instantly relays tactile and spatial data to the adjacent pre-central gyrus, allowing for the real-time adjustment of motor commands.

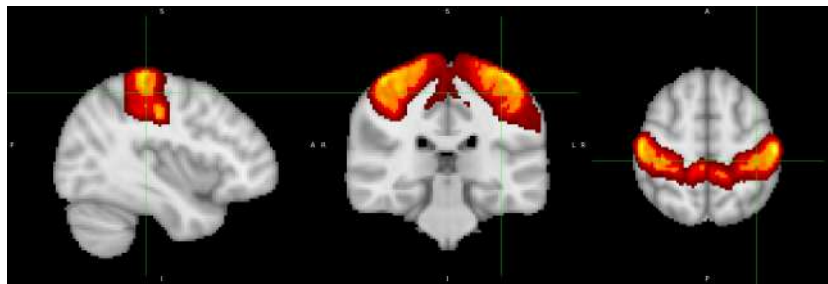


Figure 73: MNI152 2mm brain template with the post-central gyrus highlighted.

14.25 The Cingulate Gyrus

Description: The cingulate gyrus is a massive, C-shaped strip of cortex that forms a major component of the limbic system. It acts as a crucial biological bridge connecting the primitive, emotion-driven structures of the lower brain with the advanced, logic-driven neocortex. It is heavily evaluated in neuroimaging studies involving psychiatric disorders, pain management, and cognitive control.

Location: Located deep on the medial (inner) surface of the cerebral hemisphere, wrapping directly above and resting on the corpus callosum.

Core Functions:

- **Cognitive Conflict Resolution:** The anterior portion (the ACC) monitors for cognitive conflicts and errors. (Note: This is the exact brain region isolated in the fMRI Flanker task model detailed earlier in this report, as it works to suppress the incongruent distractor arrows).
- **Emotional Processing:** Links behavioral outcomes to motivation, reward, and emotional responses.
- **Pain Perception:** While the post-central gyrus maps the physical location of pain, the cingulate gyrus processes the emotional suffering and distress associated with pain.
- **Autonomic Regulation:** Interfaces with the brainstem to alter autonomic functions (like heart rate and blood pressure) in response to psychological stress or emotional stimuli.

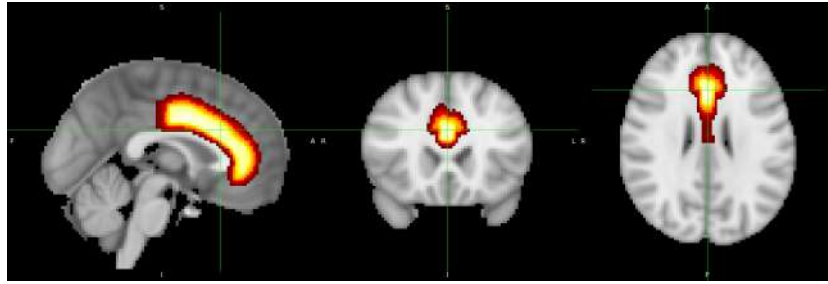


Figure 74: MNI152 2mm brain template with the cingulate gyrus highlighted.

14.26 The Thalamus

Description: The thalamus acts as the brain's grand central relay station. It is a dual-lobed mass of gray matter that filters, processes, and directs almost all sensory and motor information traveling to the cerebral cortex. In Biomedical Engineering, the thalamus—specifically the Ventral Intermediate (VIM) nucleus—is one of the most established surgical targets for Deep Brain Stimulation (DBS) to suppress severe essential tremors and Parkinsonian tremors.

Location: Deep within the core of the brain (the diencephalon), situated symmetrically on either side of the third ventricle and resting directly immediately above the brainstem.

Core Functions:

- **Sensory Relay:** Receives and routes all incoming sensory pathways (visual, auditory, tactile, and gustatory) to the appropriate cortical areas, with the sole exception of the olfactory (smell) system.
- **Motor Integration:** Receives highly processed motor signals from the basal ganglia and cerebellum, relaying them up to the motor cortex to execute movement.
- **Consciousness and Arousal:** Works continuously with the brainstem's reticular formation to maintain wakefulness, alertness, and conscious awareness.
- **Pain Processing:** Evaluates raw nociceptive (pain) signals before routing them to the somatosensory cortex and the cingulate gyrus for physical and emotional perception.
- **Sleep Regulation:** Modulates electrical rhythms during sleep, effectively acting as a "gate" that blocks sensory information from reaching the cortex so the brain can rest.

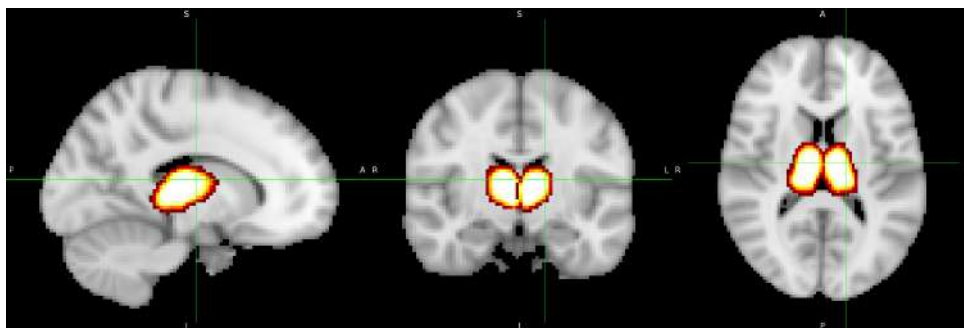


Figure 75: MNI152 2mm brain template with the thalamus highlighted.

14.27 The Hypothalamus

Description: The hypothalamus is the master regulator of the body's internal environment. Though relatively small (about the size of an almond), it acts as the primary interface between the central nervous system and the endocrine

(hormone) system. Understanding the hypothalamus is critical for engineers designing closed-loop neuromodulation devices for metabolic disorders, temperature regulation, and stress-related autonomic dysfunction.

Location: Located immediately inferior (below) the thalamus and directly superior (above) the pituitary gland, connected to the latter via a stalk called the infundibulum.

Core Functions:

- **Autonomic Control:** Acts as the highest-level command center for the autonomic nervous system, regulating baseline heart rate, blood pressure, and digestion.
- **Endocrine Interface:** Synthesizes releasing hormones that command the pituitary gland to secrete specific systemic hormones into the bloodstream.
- **Homeostasis Maintenance:** Monitors the blood continuously to regulate thirst (water balance), hunger (energy metabolism), and core body temperature.
- **Circadian Rhythms:** Houses the suprachiasmatic nucleus (the brain's biological clock), which regulates sleep-wake cycles based on light input from the eyes.
- **Stress Response:** Triggers the sympathetic nervous system and the HPA (Hypothalamic-Pituitary-Adrenal) axis during the "fight or flight" response.

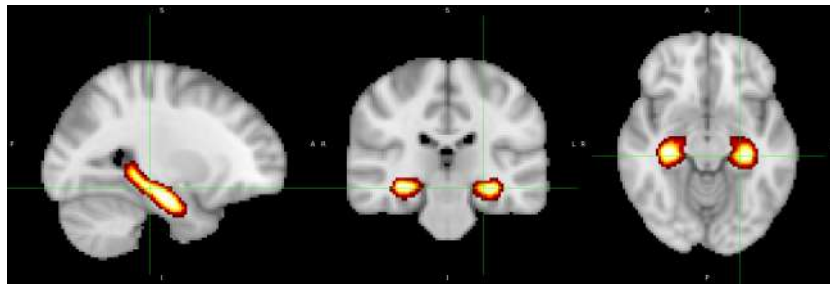


Figure 76: MNI152 2mm brain template with the hippocampus highlighted.

14.28 The Pituitary Gland

Description: Often referred to as the "master gland," the pituitary is a pea-sized endocrine organ that translates the neural commands of the hypothalamus into chemical signals (hormones) that dictate the function of glands throughout the entire body. In BME, it is heavily studied in neuroendocrinology and targeted nanomedicine delivery systems.

Location: Resting securely within a small, protective bony cavity at the base of the skull called the sella turcica, situated just below the hypothalamus.

Core Functions:

- **Growth Regulation:** Secretes Human Growth Hormone (HGH) to regulate cellular reproduction and physical growth.
- **Metabolic Control:** Releases Thyroid-Stimulating Hormone (TSH) to control the body's baseline metabolic rate.
- **Reproductive Function:** Regulates the reproductive system via the secretion of luteinizing hormone (LH) and follicle-stimulating hormone (FSH).
- **Water Retention:** The posterior lobe releases antidiuretic hormone (ADH) to signal the kidneys to conserve water and regulate blood osmolarity.

- **Stress Hormone Command:** Secretes Adrenocorticotrophic hormone (ACTH), prompting the adrenal glands to release cortisol during severe physical or psychological stress.

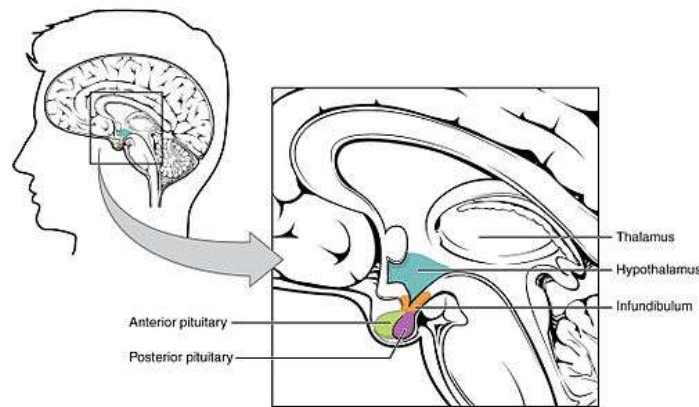


Figure 77: Anatomical illustration of the pituitary gland. *Source: Wikipedia*

14.29 The Sylvian Fissure (Lateral Sulcus)

14.30 The Corpus Callosum

Description: The corpus callosum is the largest white matter structure in the brain, consisting of over 200 million heavily myelinated axons. It is the primary neuroanatomical bridge that allows the left and right hemispheres to communicate. In clinical engineering, evaluating the structural integrity of these tracts via Diffusion Tensor Imaging (DTI) tractography is essential. Furthermore, it is pathologically relevant in severe epilepsy, where the corpus callosum may be surgically severed to prevent seizure electrical storms from spreading across the entire brain.

Location: Deep within the longitudinal fissure, forming the roof of the lateral ventricles and physically binding the two cerebral hemispheres together.

Core Functions:

- **Interhemispheric Communication:** Facilitates the rapid, bidirectional transfer of cognitive, motor, and sensory data between the left and right hemispheres.
- **Sensory Integration:** Merges independent sensory maps (e.g., the left and right visual fields) into a single, continuous perceptual experience.
- **Bilateral Motor Coordination:** Synchronizes complex, bimanual physical tasks (like typing or playing a piano) that require distinct but coordinated actions from both hands.
- **Cognitive Resource Allocation:** Allows lateralized functions (like language processing in the left hemisphere) to access data stored in the opposite hemisphere.

14.31 The Caudate Nucleus

Description: The caudate nucleus is a massive, C-shaped body of gray matter that forms a primary component of the basal ganglia (specifically the striatum). It is heavily involved in learning, memory, and the brain's reward system. Its progressive degeneration is the primary pathological cause of Huntington's Disease, making it a critical structure in neurodegenerative research and neuroimaging diagnostics.

Location: Deep within the brain, wrapping tightly around the lateral ventricles. It has a large "head" extending into the frontal lobe and a long "tail" that curves down into the temporal lobe.

Core Functions:

- **Procedural Learning:** Facilitates the acquisition and storage of automated, routine behaviors and habits.
- **Goal-Directed Action:** Integrates spatial information with motor execution to drive purposeful, directed physical movements.
- **Inhibitory Control:** Helps suppress unwanted or inappropriate motor actions and cognitive impulses.
- **Reward System Regulation:** Heavily populated with dopamine receptors, it processes motivation and reinforcement learning.



Figure 78: MNI152 2mm brain template with the caudate nucleus highlighted.

14.32 The Putamen

Description: The putamen is the outermost nucleus of the basal ganglia, physically fused with the caudate nucleus to form the striatum. While the caudate leans more toward cognitive control, the putamen is overwhelmingly dedicated to motor execution. The severe loss of dopamine input to the putamen is the hallmark driver of the rigid, freezing motor symptoms seen in Parkinson's disease.

Location: Situated deep within the cerebral hemisphere, lateral to (outside of) the globus pallidus and medial to (inside of) the insular cortex.

Core Functions:

- **Motor Preparation:** Prepares and organizes the specific sequence of muscle contractions required for a complex physical movement.
- **Movement Amplitude/Velocity:** Calculates and scales the speed and size of a voluntary movement.
- **Motor Sequence Learning:** Encodes repetitive motor patterns, shifting them from conscious, effortful execution to subconscious, automated execution.
- **Reinforcement Motor Learning:** Uses dopaminergic feedback to adjust and optimize motor performance over multiple trials.



Figure 79: MNI152 2mm brain template with the putamen highlighted.

14.33 The Globus Pallidus

Description: The globus pallidus is the primary output nucleus of the basal ganglia. It acts as a massive inhibitory brake on the brain's motor system. To initiate movement, the cortex must first signal the basal ganglia to temporarily "release the brake" applied by the globus pallidus. In neuroengineering, the Globus Pallidus internus (GPi) is one of the primary targets for surgically implanted Deep Brain Stimulation (DBS) pacemakers to treat severe movement disorders.

Location: Located medially (inward) to the putamen, nestled between the putamen and the internal capsule.

Core Functions:

- **Tonic Inhibition:** Continuously fires inhibitory signals (GABA) to the thalamus, preventing unwanted, spontaneous muscle twitches and movements at rest.
- **Voluntary Movement Gating:** Selectively pauses its inhibitory firing to allow specific, cortex-commanded motor plans to pass through the thalamus.
- **Postural Maintenance:** Modulates the baseline tone of core axial muscles to maintain stable posture and balance.
- **Subconscious Motor Regulation:** Smooths out voluntary movements, preventing jerky, hyperkinetic actions.

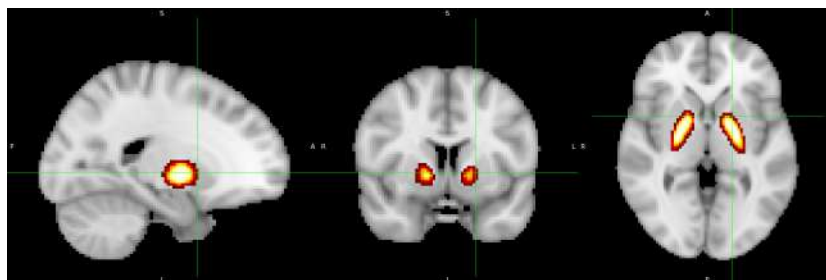


Figure 80: MNI152 2mm brain template with the globus pallidus highlighted.

14.34 The Internal Capsule

Description: The internal capsule is a dense, V-shaped bottleneck of white matter (myelinated axons) where nearly all of the brain's descending motor pathways and ascending sensory pathways converge into a tight bundle before spreading out. Because so much critical wiring is packed into such a tiny space, even microscopic strokes in the internal capsule can cause massive, devastating, full-body paralysis.

Location: Deep within the cerebral hemisphere, tightly sandwiched between the thalamus/caudate nucleus medially, and the lentiform nucleus (putamen/globus pallidus) laterally.

Core Functions:

- **Motor Output Pathway:** Carries the massive corticospinal tract, which delivers voluntary movement commands from the motor cortex down to the spinal cord.
- **Sensory Input Pathway:** Carries ascending touch, pain, and proprioceptive data from the thalamus up to the somatosensory cortex.
- **Cranial Nerve Routing:** Houses the corticobulbar tract, which controls the voluntary motor functions of the face, jaw, and throat.
- **Cortical-Thalamic Integration:** Facilitates massive, continuous bi-directional communication loops between the cerebral cortex and the thalamus.

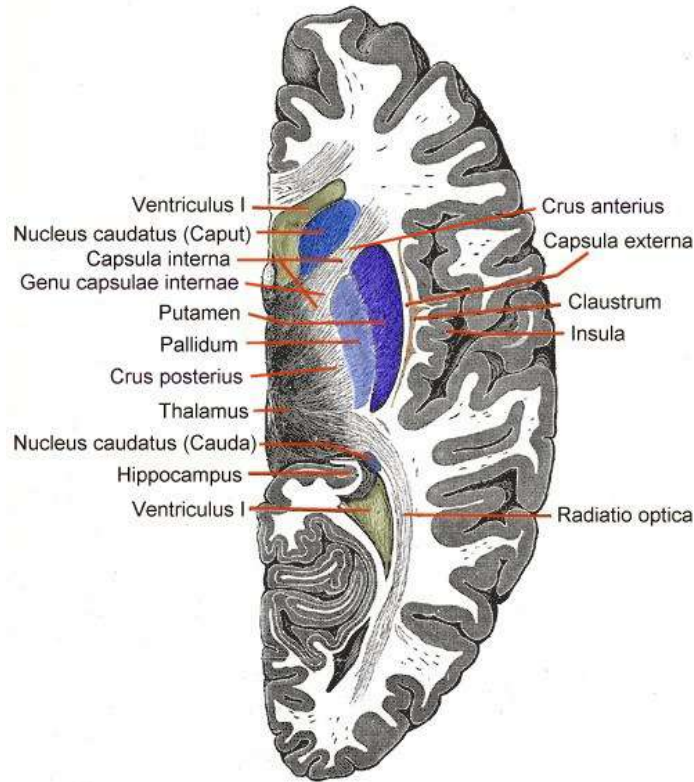


Figure 81: Anatomical illustration of the internal capsule. *Source: Wikipedia*

14.35 The Corona Radiata

Description: Translating to "radiating crown," the corona radiata is a sweeping, fan-like array of white matter tracts. It represents the structural continuation of the internal capsule as its massive bundles of nerve fibers spread outward to connect with all the varying regions of the expansive cerebral cortex. It is heavily mapped via DTI tractography in surgical planning to avoid severing connections between different brain regions.

Location: Extends superiorly (upward) and laterally (outward) from the dense internal capsule, filling the deep white matter space of the cerebral hemispheres just below the gray matter cortex.

Core Functions:

- **Signal Dispersion:** Takes the dense, concentrated neural traffic of the internal capsule and routes it geographically to the correct specialized regions of the cerebral cortex.
- **Signal Convergence:** Collects motor commands from across the expansive frontal lobe and funnels them downward into the internal capsule.
- **Structural Connectivity:** Provides the primary physical infrastructure linking the advanced computational centers of the neocortex with the lower brain and body.
- **Network Integration:** Facilitates the complex merging of independent cognitive, motor, and sensory networks required for cohesive human behavior.

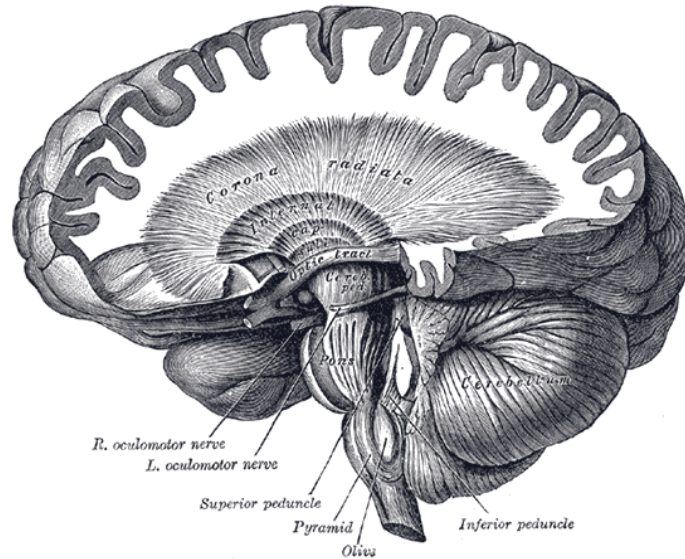


Figure 82: Anatomical illustration of the corona radiata. *Source: Wikipedia*

14.36 The Dura Mater

Description: Translating to "tough mother" in Latin, the dura mater is the outermost, thickest, and most fibrous layer of the meninges (the protective membranes surrounding the brain). For biomedical engineers, the dura is a highly significant physical and electrical barrier. It possesses high tensile strength, complicating surgical access, and it highly attenuates electrical signals, requiring engineers to design specialized electrodes (Electrocorticography, or ECoG) to place either above (epidural) or below (subdural) this layer for high-fidelity brain recordings.

Location: Adhered tightly to the inner surface of the skull bones, completely enveloping the brain and extending downward to sheath the spinal cord.

Core Functions:

- **Physical Protection:** Acts as a robust, leather-like shock absorber, shielding the fragile brain tissue from penetrating trauma and friction against the rough inner skull.
- **Electrical Insulation:** Acts as a major biological resistor, heavily filtering and diffusing the brain's electrical signals before they reach the scalp for EEG detection.
- **Structural Partitioning:** Infolds upon itself to create dense septa (partitions) that physically divide the cranial cavity, limiting brain movement during sudden head acceleration/deceleration.
- **Venous Infrastructure:** Splits into two layers (periosteal and meningeal) at certain points to create the dural venous sinuses, which act as the brain's primary plumbing system.

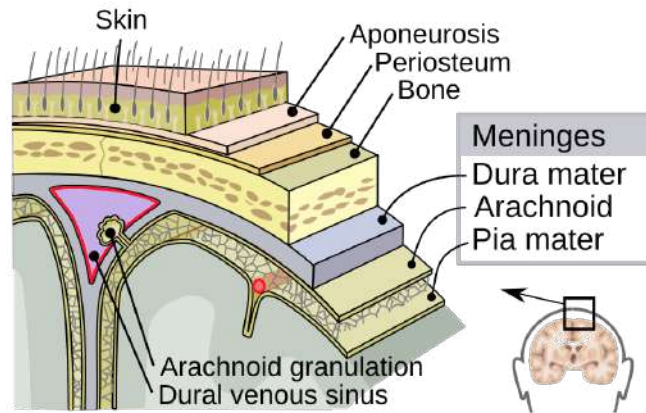


Figure 83: Anatomical illustration of the dura mater. *Source: Wikipedia*

14.37 The Falx Cerebri

Description: The falx cerebri is the largest of the dural septa, forming a dense, sickle-shaped biological wall. In the field of BME and biomechanics, the falx cerebri is a critical parameter in finite element modeling (FEM) of traumatic brain injuries (TBI) and concussions, as it dictates how shear stress propagates across the brain during violent rotational impacts.

Location: Descends vertically into the deep longitudinal fissure, sitting exactly on the midline between the left and right cerebral hemispheres.

Core Functions:

- **Hemispheric Compartmentalization:** Physically isolates the two cerebral hemispheres, preventing them from colliding or compressing one another.
- **Displacement Limitation:** Restricts excessive lateral (side-to-side) displacement of the brain within the skull.
- **Sinus Housing:** Structurally supports and envelopes the superior and inferior sagittal sinuses along its upper and lower margins.

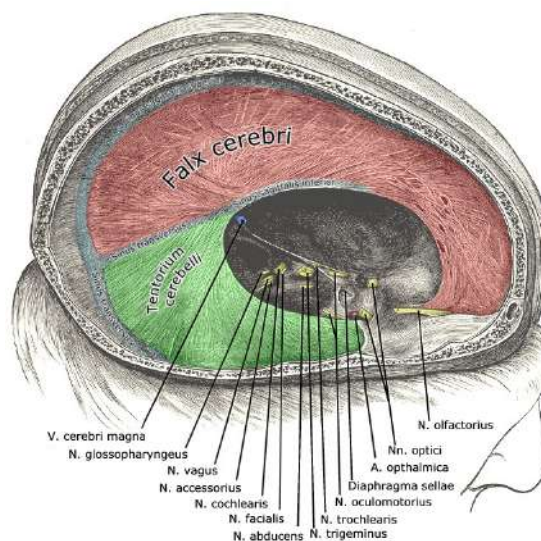


Figure 84: Anatomical illustration of the falx cerebri. *Source: Wikipedia*

14.38 The Tentorium Cerebelli

Description: The tentorium cerebelli is a horizontal, tent-like fold of dura mater. Clinically, it serves as the absolute dividing line for classifying brain structures into "supratentorial" (above) or "infratentorial" (below). In neuroengineering and critical care, monitoring intracranial pressure (ICP) here is vital, as increased pressure can force brain tissue through the narrow opening of the tentorium (tentorial herniation), leading to rapid, fatal brainstem compression.

Location: Stretched horizontally across the posterior cranial fossa, physically separating the occipital lobes of the cerebrum (above) from the cerebellum (below).

Core Functions:

- **Weight Distribution:** Acts as a structural hammock, supporting the heavy weight of the posterior cerebral hemispheres so they do not crush the underlying cerebellum.
- **Vertical Stabilization:** Prevents superior-inferior (up-and-down) displacement of the brain.
- **Vascular Routing:** Encloses the transverse sinuses along its attached outer margin.

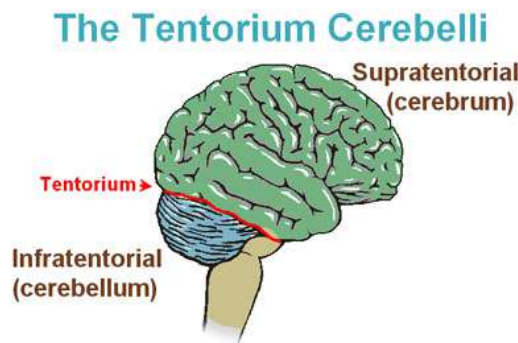


Figure 85: Anatomical illustration of the tentorium cerebelli. *Source: Wikipedia*

14.39 The Dural Venous Sinuses

Description: Unlike normal systemic veins, the dural venous sinuses lack muscular walls and functional valves; they are simply rigid, endothelium-lined channels formed by the splitting of the dural layers. For engineers modeling cerebral hemodynamics, computational fluid dynamics (CFD), and targeted drug delivery, these sinuses represent the primary clearance pathway for all cranial fluids.

Location: Distributed extensively throughout the dural folds across the entire inner vault of the skull.

Core Functions:

- **Venous Drainage:** Collect all oxygen-depleted blood from the brain tissue and channel it outward toward the jugular veins.
- **CSF Reabsorption:** House arachnoid granulations, tiny biological valves that continuously filter and reabsorb spent cerebrospinal fluid (CSF) back into the bloodstream.
- **Pressure Regulation:** Lack of valves allows for bi-directional blood flow depending on pressure gradients, helping to equilibrate sudden changes in intracranial pressure.
- **Thermal Regulation:** Facilitate the dissipation of heat generated by the highly metabolic brain tissue.

14.40 The Superior Sagittal Sinus

Description: The superior sagittal sinus is the largest and most prominent venous channel in the brain. Because it sits immediately below the skull and directly over the longitudinal fissure, it is the primary target for cutting-edge endovascular Brain-Computer Interfaces (like the Stentrode). Engineers can deploy stent-like electrode arrays directly into this sinus via the jugular vein, safely recording motor cortex signals without performing open brain surgery.

Location: Runs anteroposteriorly (front to back) along the exact midline of the skull vault, situated within the upper attached margin of the falx cerebri.

Core Functions:

- **Cortical Drainage:** Serves as the primary drainage route for the superior, medial, and lateral surfaces of the cerebral hemispheres.
- **Primary CSF Clearance:** Acts as the most heavily concentrated site for CSF reabsorption via arachnoid granulations.
- **Posterior Routing:** Channels massive volumes of deoxygenated blood backwards toward the confluence of sinuses.

14.41 The Confluence of Sinuses (Torcular Herophili)

Description: The confluence is the grand anatomical intersection where the major midline venous channels meet. It is a highly variable structure among individuals and serves as a critical junction in hemodynamic modeling, as flow dynamics here dictate how efficiently blood exits the brain.

Location: Located at the posterior pole of the cranial vault, pressing against the internal occipital protuberance of the skull.

Core Functions:

- **Venous Convergence:** Pools incoming blood from the superior sagittal sinus, the straight sinus, and the occipital sinus.
- **Flow Distribution:** Splits the massive pooled volume of blood laterally, distributing it into the left and right transverse sinuses.
- **Pressure Equilibration:** Acts as a central reservoir to balance venous blood pressure between the left and right hemispheres.

14.42 The Transverse Sinus

Description: The transverse sinuses are large, paired venous channels that route blood horizontally away from the midline. Due to turbulent flow and anatomical bottlenecks in this region, it is a common site for venous sinus thrombosis (blood clots) and idiopathic intracranial hypertension, conditions routinely diagnosed via heavily engineered Magnetic Resonance Venography (MRV) protocols.

Location: Runs laterally from the confluence of sinuses, tracking along the attached outer margin of the tentorium cerebelli toward the base of the skull.

Core Functions:

- **Horizontal Transport:** Carries pooled blood outward from the confluence of sinuses.
- **Regional Drainage:** Receives additional venous drainage from the inferior cerebrum and parts of the cerebellum.
- **Pathway Transition:** Serves as the structural bridge connecting the superior midline sinuses to the lower descending sinuses.

14.43 The Sigmoid Sinus

Description: The sigmoid sinus is the final S-shaped continuation of the transverse sinus. It acts as the ultimate cranial gateway, funnelling blood out of the skull. Altered fluid dynamics or turbulent flow within the sigmoid sinus can create acoustic vibrations that conduct through the skull directly to the inner ear, a phenomenon (pulsatile tinnitus) that biomedical acousticians and otolaryngologists aim to model and correct.

Location: Curves deeply downward in an "S" shape along the temporal and occipital bones, terminating at the jugular foramen at the base of the skull.

Core Functions:

- **Systemic Re-entry:** Channels all collected cranial venous blood directly into the internal jugular vein, returning it to the heart.
- **Final CSF Clearance:** Sweeps all reabsorbed cerebrospinal fluid out of the cranial vault and back into systemic circulation.
- **Pressure Buffering:** Its curved, S-like shape helps buffer the sheer force and velocity of blood as it rapidly descends out of the cranium.

14.44 The Cerebral Arteries and Neurovascular Engineering

Description: The human brain represents only 2% of the body's total mass but consumes roughly 20% of its oxygen and glucose. To meet this massive metabolic demand, the brain is fed by a highly pressurized, complex arterial network. For biomedical engineers, understanding this vascular architecture is the foundation for designing neurovascular stents, modeling stroke pathology via Computational Fluid Dynamics (CFD), and developing the very neuroimaging pipelines (like the fMRI BOLD signal) detailed earlier in this report. The brain's blood supply is distinctly divided into the anterior circulation (fed by the internal carotid arteries) and the posterior circulation (fed by the vertebral arteries), which meet at the base of the brain to form the Circle of Willis.

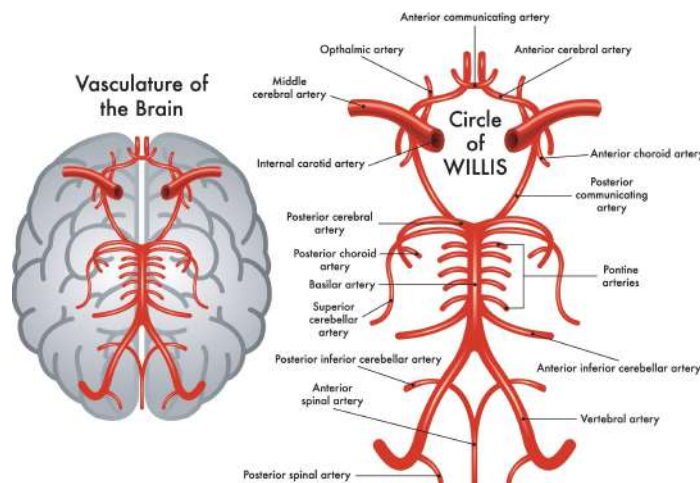


Figure 86: Anatomical illustration of the cerebral arteries and the Circle of Willis. *Source: Shutterstock*

14.45 The Internal Carotid Artery (ICA)

Description: The internal carotid arteries are massive, high-pressure vessels that provide roughly 80% of the brain's total blood supply, driving the entire anterior circulation. In BME, carotid artery stenosis (narrowing due to plaque) is heavily studied using Doppler ultrasound and hemodynamics to predict the shear stress that leads to plaque rupture and subsequent strokes.

Location: Bifurcates from the common carotid artery in the neck, ascends vertically, and enters the base of the skull through the carotid canal before terminating into the middle and anterior cerebral arteries.

Core Functions:

- **Anterior Circulation Supply:** Acts as the primary hemodynamic pipeline delivering oxygenated blood to the frontal, parietal, and temporal lobes.
- **Ophthalmic Perfusion:** Branches off to form the ophthalmic artery, providing the sole blood supply to the eyes and retinas.
- **Circle of Willis Input:** Feeds high-pressure blood into the anterior half of the Circle of Willis, ensuring collateral blood flow if other vessels fail.
- **Hemodynamic Buffering:** Its winding, S-shaped path through the skull base (the carotid siphon) helps dampen the pulsatile pressure of the heartbeat before the blood hits fragile brain tissue.

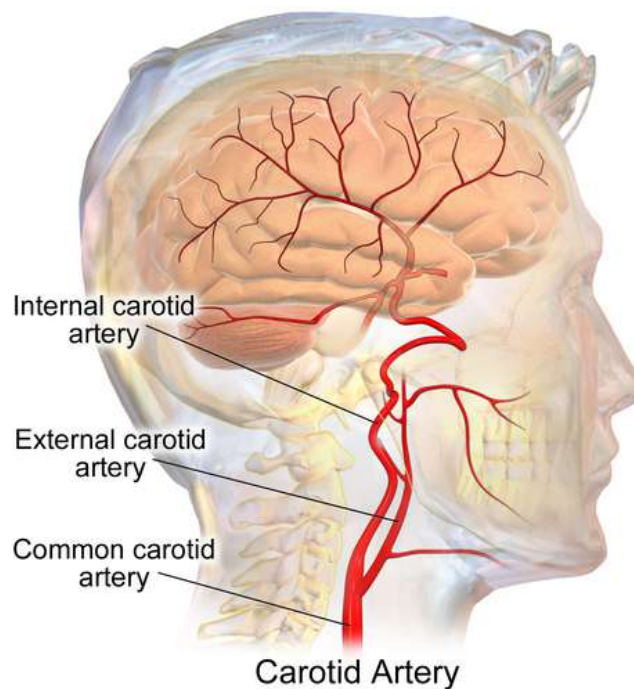


Figure 87: Anatomical illustration of the internal carotid artery (ICA). *Source: Wikipedia*

14.46 The Middle Cerebral Artery (MCA)

Description: The MCA is the largest and most direct continuation of the internal carotid artery. Because of its wide diameter and direct trajectory, it is the most common site for embolic strokes (where a blood clot from the heart shoots into the brain). Consequently, it is the primary target for engineers designing mechanical thrombectomy devices (clot retrievers).

Location: Branches from the ICA and runs laterally deep through the Sylvian fissure, fanning out across the entire lateral surface of the cerebral hemisphere.

Core Functions:

- **Lateral Cortex Perfusion:** Supplies blood to the vast majority of the lateral frontal, parietal, and temporal lobes.

- **Sensorimotor Supply (Upper Body):** Perfuses the lateral portions of the pre-central and post-central gyri, controlling the motor and sensory functions of the face, hands, and arms.
- **Language Network Supply:** Delivers blood directly to Broca's and Wernicke's areas; an MCA stroke almost always results in severe language deficits (aphasia).
- **Deep Brain Perfusion:** Sends tiny, fragile branches (lenticulostriate arteries) deep into the brain to supply the basal ganglia and internal capsule.

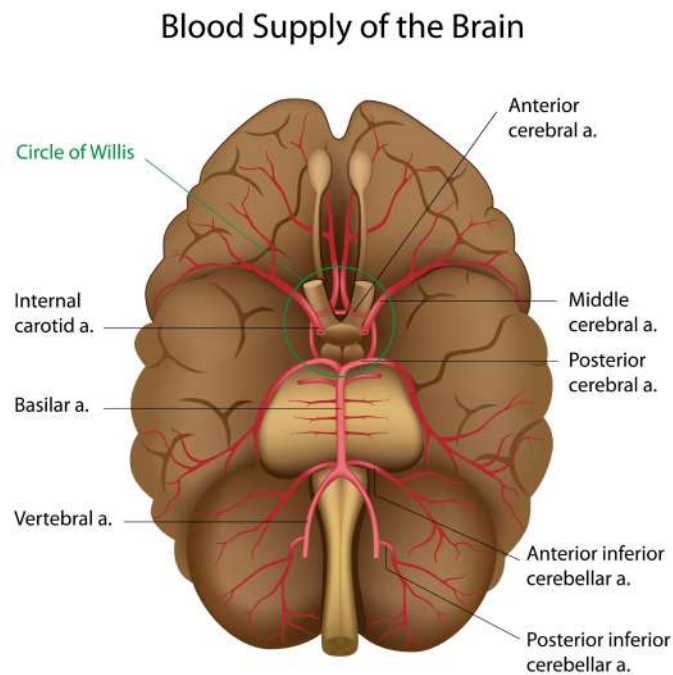


Figure 88: Anatomical illustration of the Middle Cerebral Artery (MCA). *Source: Shutterstock*

14.47 The Anterior Cerebral Artery (ACA)

Description: The ACA is the smaller of the two terminal branches of the internal carotid artery. It supplies the innermost structures of the brain. While strokes here are less common than in the MCA, they produce highly specific deficits that engineers modeling brain-machine interfaces must account for when mapping the motor cortex.

Location: Branches from the ICA, runs forward and upward into the deep longitudinal fissure, and arches backwards over the corpus callosum.

Core Functions:

- **Medial Cortex Perfusion:** Supplies the medial (inner) surfaces of the frontal and parietal lobes.
- **Sensorimotor Supply (Lower Body):** Perfuses the medial portions of the motor and sensory cortices, controlling the movement and sensation of the legs and feet.
- **Executive Function Support:** Supplies the prefrontal cortex, supporting high-level decision making and personality regulation.
- **Corpus Callosum Supply:** Delivers oxygenated blood to the anterior structural bridge connecting the two cerebral hemispheres.

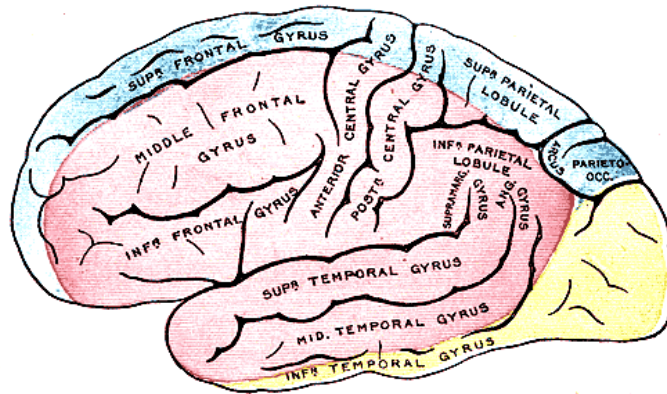


Figure 89: Anatomical illustration of the anterior cerebral artery (ACA). *Source: Wikipedia*

14.48 The Posterior Cerebral Artery (PCA)

Description: The PCA is the primary vessel of the posterior circulation. Because it supplies the visual cortex, disruption of this artery causes cortical blindness. For engineers developing visual prostheses, understanding the perfusion mapping of the PCA is critical to ensuring the viability of the target cortical tissue.

Location: Arises from the terminal bifurcation of the basilar artery, wrapping backward around the midbrain to supply the posterior cerebral hemispheres.

Core Functions:

- **Visual Cortex Perfusion:** Acts as the primary blood supply to the occipital lobe, driving the metabolic processes required for vision.
- **Inferior Temporal Supply:** Perfuses the bottom surface of the temporal lobe, aiding in complex visual recognition and memory processing.
- **Thalamic Supply:** Sends deep penetrating branches to supply the thalamus, the brain's massive sensory relay station.
- **Memory Network Support:** Supplies the medial temporal lobe, including the hippocampus, making it critical for the formation of new memories.

14.49 The Vertebral Arteries

Description: The vertebral arteries are paired vessels that form the foundational supply for the brain's posterior circulation, delivering the remaining 20% of the brain's blood. Because they travel through the mobile vertebrae of the neck, they are uniquely susceptible to mechanical shear and dissection from severe whiplash or physical trauma—a key consideration in biomechanical impact engineering.

Location: Ascend vertically through tiny holes (foramina) in the cervical vertebrae of the neck, entering the skull through the foramen magnum and running along the medulla.

Core Functions:

- **Posterior Circulation Initiation:** Provides the initial upward blood flow required to sustain the brainstem, cerebellum, and posterior cerebrum.
- **Spinal Cord Supply:** Gives off branches that fuse to form the anterior spinal artery, supplying the upper spinal cord.
- **Medulla Perfusion:** Delivers direct blood supply to the medulla oblongata, supporting vital cardiovascular and respiratory pacemaker cells.

- **Inferior Cerebellar Perfusion:** Branches into the Posterior Inferior Cerebellar Artery (PICA) to supply the lower portions of the cerebellum.

14.50 The Basilar Artery

Description: The basilar artery is a massive, unpaired vessel formed by the merging of the two vertebral arteries. It is the absolute lifeline for the brainstem. In clinical engineering, a "basilar occlusion" (clot in this artery) is considered one of the most catastrophic medical emergencies, often resulting in "locked-in syndrome" (total body paralysis with preserved consciousness) or immediate death, requiring rapid endovascular intervention.

Location: Runs vertically straight up the midline of the anterior surface of the pons, terminating by splitting into the left and right posterior cerebral arteries.

Core Functions:

- **Pontine Supply:** Sends numerous small pontine branches directly into the pons to sustain massive motor/sensory relay tracts and cranial nerve nuclei.
- **Cerebellar Supply:** Branches into the Anterior Inferior Cerebellar Artery (AICA) and Superior Cerebellar Artery (SCA) to perfuse the middle and upper cerebellum.
- **Midbrain Supply:** Provides terminal blood flow to the upper brainstem structures before splitting.
- **Circle of Willis Input:** Feeds high-pressure posterior blood upward into the Circle of Willis via the posterior communicating arteries, completing the brain's vascular safety loop.

14.51 The Ventricular System and Cerebrospinal Fluid (CSF)

Description: The ventricular system is a series of interconnected, fluid-filled cavities deep within the brain. These cavities continuously produce, circulate, and resorb Cerebrospinal Fluid (CSF). For biomedical engineers, the ventricular system is a profound study in biological fluid dynamics. When this plumbing system fails—often due to anatomical bottlenecks—it causes hydrocephalus (water on the brain), creating a massive buildup of intracranial pressure (ICP). Consequently, engineers dedicate significant resources to developing programmable Ventriculoperitoneal (VP) shunts, External Ventricular Drains (EVDs), and MRI phase-contrast sequences to measure CSF flow velocity.

Location: A continuous internal network spanning from the deep cerebral hemispheres down through the brainstem, eventually merging with the central canal of the spinal cord and the subarachnoid space.

Core Functions:

- **Buoyancy and Suspension:** CSF reduces the net weight of the brain from roughly 1,400 grams to just 50 grams, preventing the brain from crushing its own blood vessels against the base of the skull.
- **Shock Absorption:** Acts as a fluid buffer to protect the delicate neural tissue from sudden physical impacts or shear forces.
- **Chemical Homeostasis:** Continuously flushes metabolic waste (including misfolded proteins) away from the brain and regulates the precise extracellular chemical environment needed for neural firing.

14.52 The Lateral Ventricles

Description: The lateral ventricles are the two largest cavities in the ventricular system, forming a massive, symmetrical "C" shape in each cerebral hemisphere. They house the highest concentration of the choroid plexus, the specialized vascular tissue that produces the vast majority of the brain's CSF. In clinical neuroengineering, the right lateral ventricle is the primary anatomical target for surgically placing EVD catheters to rapidly relieve life-threatening intracranial pressure.

Location: Deep within each cerebral hemisphere, wrapping around the thalamus and basal ganglia.

Core Functions (and Structural Subdivisions):

- **CSF Production:** Acts as the primary biological manufacturing site for fresh cerebrospinal fluid.
- **The Anterior Horn:** Extends forward into the frontal lobe. Because it lacks choroid plexus, it is the safest, most common target for neurosurgical catheter insertion.
- **The Occipital Horn:** Extends backward into the occipital lobe, varying significantly in size and shape among individuals.
- **The Temporal Horn:** Curves downward and forward into the temporal lobe, running adjacent to the hippocampus. In neuroimaging pipelines, an enlarged temporal horn is a key structural biomarker used by diagnostic algorithms to track Alzheimer's disease progression.

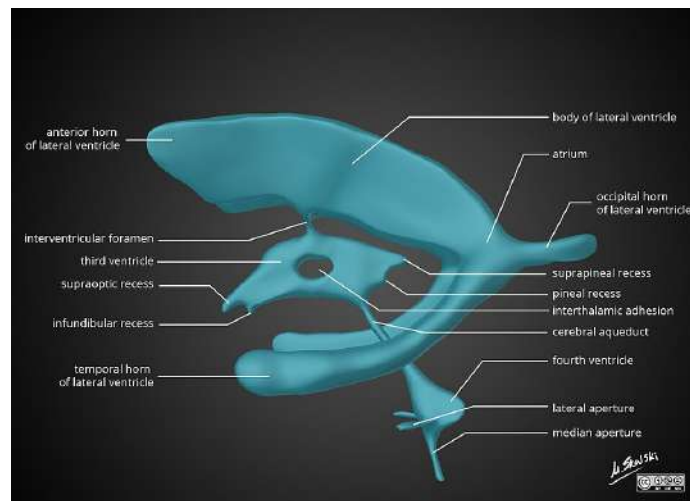


Figure 90: Anatomical illustration of the Ventricle. *Source: Wikipedia*

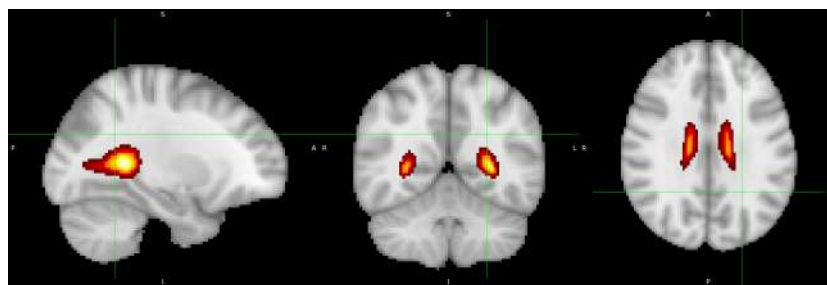


Figure 91: MNI152 2mm brain template with the lateral ventricles highlighted.

14.53 The Interventricular Foramen (of Monro)

Description: The foramen of Monro consists of two short, narrow, biological pipes. Because they are the sole exit routes for the massive volume of CSF produced in the lateral ventricles, they are critical structural bottlenecks. A blockage here (e.g., from a colloid cyst) instantly causes asymmetric, unilateral hydrocephalus.

Location: Situated at the midline, connecting the anterior portion of the lateral ventricles down to the third ventricle.

Core Functions:

- **Fluid Funneling:** Channels all CSF generated in the left and right lateral ventricles into the single, central third ventricle.

- **Pressure Equalization:** Maintains an equal hydrostatic pressure gradient between the massive upper ventricular chambers.

14.54 The Third Ventricle

Description: The third ventricle is a narrow, vertically oriented, slit-like cavity. Because it is tightly sandwiched between the brain's major relay stations (the thalami), tumors or fluid buildup here can place severe mechanical pressure on surrounding endocrine and sensory structures, requiring precise neurosurgical navigation to bypass or decompress.

Location: Located exactly on the midline of the diencephalon, compressed tightly between the left and right thalami and resting directly above the hypothalamus.

Core Functions:

- **CSF Pooling and Routing:** Receives fluid from the lateral ventricles, adds a small amount of its own newly produced CSF, and directs the total volume downward.
- **Structural Cushioning:** Provides a fluid buffer that protects the critical diencephalic structures (thalamus and hypothalamus) from internal mechanical stress.

14.55 The Cerebral Aqueduct (of Sylvius)

Description: The cerebral aqueduct is a long, thin, tube-like channel and the absolute narrowest point in the entire ventricular system. In the context of biomechanics and fluid dynamics, it is highly susceptible to stenosis (narrowing). Aqueductal stenosis is the leading cause of congenital obstructive hydrocephalus. Engineers utilize specialized MRI protocols (like phase-contrast MRI) specifically to quantify the turbulent flow and stroke volume of CSF pulsing through this tiny channel.

Location: Runs vertically down through the center of the midbrain, physically connecting the third ventricle (above) to the fourth ventricle (below).

Core Functions:

- **Midbrain Transit:** Acts as the sole conduit for all supratentorial CSF to exit the upper brain and reach the lower brainstem.
- **Pulsatile Dampening:** Modulates the hydraulic pressure wave of the CSF driven by the cardiac cycle before it reaches the fragile fourth ventricle.

14.56 The Fourth Ventricle

Description: The fourth ventricle is a pyramid-shaped cavity that acts as the final internal collection chamber for CSF before it is distributed to the outside surface of the brain and the spinal cord. It is clinically highly significant; tumors in the cerebellum (like medulloblastomas) frequently compress this ventricle, rapidly causing fatal increases in intracranial pressure.

Location: Situated in the posterior cranial fossa, nestled tightly between the brainstem (pons and medulla) anteriorly and the cerebellum posteriorly.

Core Functions:

- **Final Internal Pooling:** Collects all CSF arriving from the cerebral aqueduct.
- **Fluid Distribution:** Serves as the central hydraulic junction that finally pushes CSF out of the internal brain cavities and into the external subarachnoid space.

14.57 The Lateral Apertures (of Luschka) and Median Aperture (of Magendie)

Description: These three microscopic escape valves are the only biological doorways that allow CSF to leave the ventricular system and enter the subarachnoid space to bathe the outer surface of the brain. If these apertures are blocked (often by scarring from meningitis or hemorrhage), it traps the fluid internally, causing communicating hydrocephalus.

Location: Piercing the roof and lateral walls of the fourth ventricle.

Core Functions:

- **The Lateral Apertures (of Luschka):** A paired set of openings (left and right) that direct CSF laterally out of the fourth ventricle to wrap around the brainstem and flow upward over the cerebral hemispheres.
- **The Median Aperture (of Magendie):** A single, large opening at the midline base of the fourth ventricle that directs CSF straight downward into the cisterna magna, supplying the fluid that bathes the entire length of the spinal cord.
- **Subarachnoid Flushing:** By facilitating this exit, they enable the continuous, necessary flow of CSF into the venous sinuses for eventual reabsorption into the blood.

14.58 The Hippocampus: Anatomy, Function, Pathology, and Neuroengineering

Taking its name from the Greek word for "seahorse" due to its distinct, curved shape, the hippocampus is arguably one of the most intensely studied structures in all of neuroscience and neuroengineering. It is the core biological engine for memory consolidation and spatial navigation. For biomedical engineers, the hippocampus represents both a major vulnerability and a frontier of innovation. Because of its dense, highly excitable circuitry, it is the primary epicenter for severe seizure disorders. Consequently, it is the central target for advanced neuroimaging diagnostics, laser ablation surgeries, closed-loop neural pacemakers, and cutting-edge research into neural prosthetics aimed at digitally restoring lost memory function.

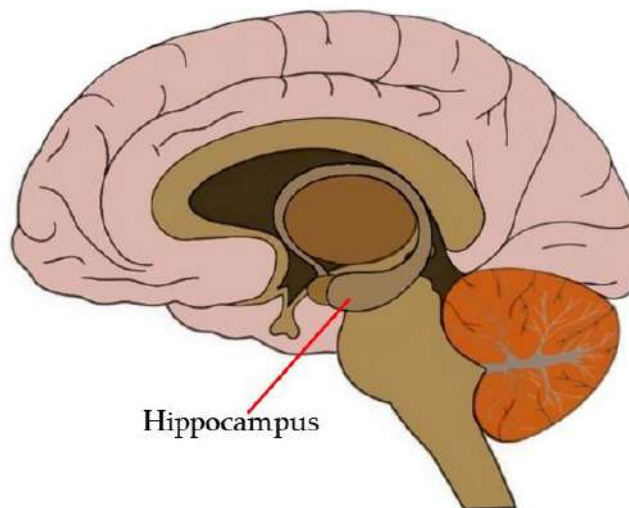


Figure 92: Anatomical illustration of the hippocampus. *Source: NEUROSCIENTIFICALLY CHALLENGED*

14.58.1 Anatomical Location and Topography

Location: The hippocampus is a paired structure (one in each hemisphere) buried deeply within the medial temporal lobe. It forms the medial floor of the temporal horn of the lateral ventricle.

The Hippocampal Formation: The hippocampus is not a single homogenous mass, but rather a complex, multi-layered assembly of distinct sub-regions collectively called the hippocampal formation. These include:

- **The Dentate Gyrus (DG):** A highly specialized, V-shaped band of tissue. It is one of only two confirmed regions in the adult human brain capable of neurogenesis (the continuous birth of brand-new neurons throughout life).
- **Cornu Ammonis (CA1, CA2, CA3, CA4):** Translating to "Amun's Horn," this is the main body of the hippocampus, divided into four functionally distinct zones of pyramidal neurons. CA1 is highly sensitive to hypoxia (lack of oxygen) and is usually the first area to die during a stroke or cardiac arrest. CA3 forms a massive auto-associative network, allowing the brain to recall an entire memory from just a fraction of a cue (e.g., a specific smell triggering a complex childhood memory).
- **The Subiculum:** The primary output gate of the hippocampus, funnelling processed memory signals out through a massive white matter tract called the fornix, which carries the data to the mammillary bodies and the thalamus for long-term storage.

14.58.2 The Trisynaptic Circuit (Micro-Architecture)

To understand the hippocampus from a biological engineering perspective, one must understand its precise electrical wiring. Memory formation occurs via a highly directional, one-way electrical loop known as the trisynaptic circuit:

1. **Input (The Perforant Path):** Raw sensory and cognitive data from the cortex enters the entorhinal cortex, which acts as the data portal. This data is fired via the perforant pathway directly into the Dentate Gyrus.
2. **Relay 1 (Mossy Fibers):** The Dentate Gyrus filters and pattern-separates this data, then fires it via mossy fibers into the CA3 region.
3. **Relay 2 (Schaffer Collaterals):** CA3 processes the associations and fires the signal via Schaffer collaterals into the CA1 region.
4. **Output:** CA1 completes the processing and routes the consolidated memory out through the Subiculum to be permanently stored in the cerebral cortex.

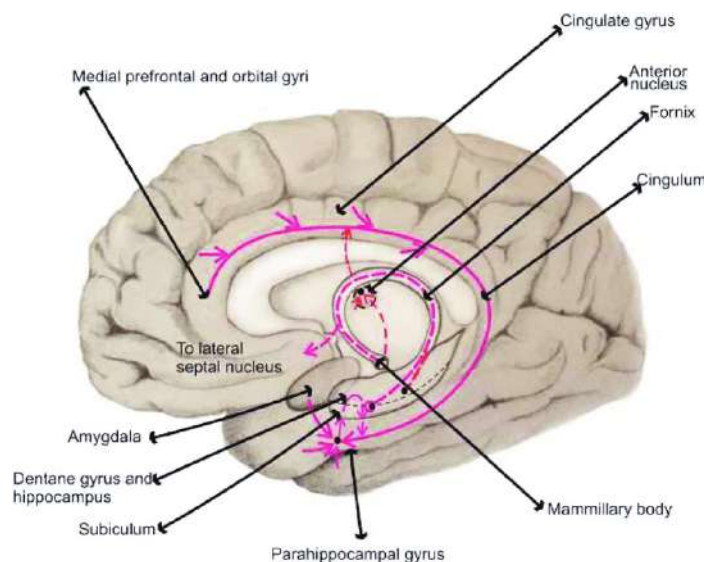


Figure 93: Illustration of the Trisynaptic Circuit. *Source: Cognitive problems related to vertebrobasilar circulation*

Note: Engineers developing "memory prostheses" map this exact circuit, creating microchip arrays that artificially record CA3 firing patterns and electrically stimulate CA1 to bypass damaged tissue.

14.58.3 Core Physiological Functions

- **Memory Consolidation:** The hippocampus does not store long-term memories. Instead, it acts like a biological RAM (Random Access Memory). It holds temporary short-term declarative memories (facts and events) and, primarily during deep sleep, "downloads" or consolidates them into the neocortex for permanent long-term storage.
- **Spatial Navigation and Mapping:** The hippocampus contains highly specialized neurons called Place Cells. These cells fire electrically only when an individual is in a specific physical location in their environment, effectively creating an internal, dynamic GPS map of the physical world.
- **Contextual Emotional Processing:** Working directly adjacent to the amygdala, the hippocampus provides spatial and temporal context to emotions (e.g., the amygdala generates the feeling of fear, while the hippocampus remembers where and when the threatening event occurred).

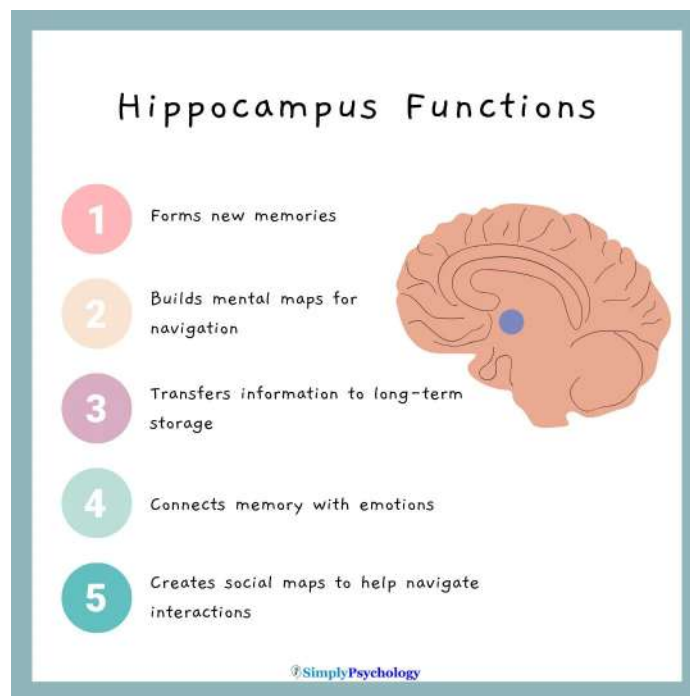


Figure 94: Illustration of the hippocampus functions. *Source: Simply Psychology*

14.58.4 Pathologies and Diseases

Due to its high metabolic demand and complex electrical circuitry, the hippocampus is highly vulnerable to several devastating pathologies:

- **Alzheimer's Disease (AD):**
 - *Pathology:* The hippocampus is the very first brain structure attacked by Alzheimer's disease. Toxic amyloid-beta plaques and tau protein tangles accumulate here, causing massive cellular death and structural atrophy.
 - *Symptoms:* Profound anterograde amnesia (the inability to form new memories), spatial disorientation, and getting lost in familiar places.
- **Mesial Temporal Lobe Epilepsy (MTLE):**

- *Pathology*: The excitatory nature of the trisynaptic circuit makes it prone to runaway electrical feedback loops. Recurrent seizures cause the hippocampus to scar, harden, and shrink—a condition known as Hippocampal Sclerosis.
- *Symptoms*: Severe, drug-resistant focal seizures, often preceded by auras involving sudden feelings of déjà vu, intense fear, or olfactory hallucinations.

- **Post-Traumatic Stress Disorder (PTSD) and Chronic Stress:**

- *Pathology*: The hippocampus is densely packed with glucocorticoid receptors, making it highly sensitive to cortisol (the stress hormone). Chronic, severe stress leads to cortisol toxicity, physically shrinking the volume of the hippocampus and halting adult neurogenesis in the Dentate Gyrus.
- *Symptoms*: Intrusive traumatic memories, impaired spatial reasoning, and severe anxiety.

- **Anterograde Amnesia (The "Patient H.M." Paradigm):**

- *Pathology*: Physical destruction of the hippocampus bilaterally (due to hypoxia, encephalitis, or surgical removal).
- *Symptoms*: The permanent inability to form any new declarative memories, trapping the patient perpetually in the present moment, though past memories and procedural motor skills remain intact.

14.58.5 Clinical Treatments and Neuroengineering Interventions

Treating hippocampal pathology involves a multi-disciplinary approach spanning pharmacology, neurosurgery, and advanced biomedical engineering:

- **Pharmacological Treatments:**

- **Cholinesterase Inhibitors (e.g., Donepezil)**: Used in early Alzheimer's to artificially boost levels of acetylcholine, a neurotransmitter critical for hippocampal memory encoding.
- **Anti-Epileptic Drugs (AEDs)**: Medications (like Levetiracetam or Lamotrigine) that suppress sodium or calcium channels to calm hyper-excitable hippocampal circuits and prevent seizures.
- **SSRIs (Antidepressants)**: Found to promote neurogenesis in the Dentate Gyrus, helping to slowly rebuild hippocampal volume in patients with chronic depression or PTSD.

- **Surgical and Engineering Interventions (For Epilepsy):**

- **Anterior Temporal Lobectomy**: If epilepsy is drug-resistant and strictly unilateral (on one side), neurosurgeons will physically amputate the scarred hippocampus and surrounding temporal lobe tissue to cure the seizures.
- **Laser Interstitial Thermal Therapy (LITT)**: A heavily engineered, minimally invasive alternative to open surgery. A fiber-optic laser catheter is stereotactically guided deep into the hippocampus. Under real-time MRI thermometry monitoring, the engineer/surgeon safely ablates the seizure-causing hippocampal tissue from the inside out.
- **Responsive Neurostimulation (RNS)**: A highly advanced Brain-Computer Interface (BCI). Engineers implant depth electrodes directly into the hippocampus, connected to a pacemaker in the skull. The device continuously monitors the hippocampal EEG for the specific electrical frequency of a starting seizure. When detected, it instantly fires a counter-electrical pulse to disrupt the seizure before the patient feels it.

- **The Frontier: Hippocampal Memory Prosthetics:**

- Still primarily in the clinical trial phase, neuroengineers are developing microelectrode arrays based on MIMO (Multi-Input Multi-Output) mathematical models. By bypassing the damaged CA3 region, these devices read the input code from the entorhinal cortex, run it through an artificial neural network algorithm, and electrically stimulate the CA1 region with the correct output code—essentially functioning as an artificial, silicon-based hippocampus to restore memory capacity in patients with severe dementia or brain trauma.

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